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On a Method for Large-Scale Constrained Stochastic Optimization

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Abstract

We study stochastic convex optimization problems subject to almost sure inequality constraints:

$$\min_{x \in \mathbb{R}^n} \{ f(x) := \mathbb{E}(F_\xi(x)) + r(x) \}$$
$$\text{subject to} \quad g(x, \xi) \leq 0 \quad \text{almost surely},$$

allowing for infinitely many constraints and composite objectives. To handle the constraints without projections, we develop a stochastic single-loop penalty-based method inspired by smoothed exact penalties, combined with proximal operators to accommodate nonsmooth regularizers and more general objective structures. Under reasonable assumptions, we prove convergence in both the convex and strongly convex settings. In particular, our method attains rates of order $\mathcal{O}(\log^{1/2+\epsilon} K/\sqrt{K})$ and $\mathcal{O}(\log^{\epsilon} K/K)$, for any $\epsilon \in (0, \infty)$, respectively, matching the optimal stochastic rates up to arbitrarily small logarithmic factors, while remaining projection-free. Unlike several related methods, our analysis does not require a global bound on the variance of stochastic gradients, and allows for infinitely many functional constraints. Numerical experiments support the theoretical findings and illustrate the practical performance of the method.

Contents

1 Introduction **1**

2 Mathematical Background **7**

2.1 Notation 7

2.2 Basic inequalities 8

2.3 Probability theory 8

2.3.1 Measure and integration 8

2.3.2 Random variables, expected value, and variance 10

2.3.3 Densities 13

2.3.4 Conditioning 14

2.3.5 Martingales 16

2.4 Optimization theory 17

2.4.1 Lipschitz functions 17

2.4.2 Convex optimization 18

2.4.3 Stochastic optimization 21

2.4.4 Multifunctions and metric regularity 23

3 Stochastic Inexact Penalty Method **27**

3.1 Technical preliminaries 28

3.2 Almost sure convergence 37

3.3 Convergence rates in expectation 54

3.4 High-probability guarantees 71

4 Numerical Experiments **79**

4.1 Classification with support vector machines 79

4.2 Inventory control 81

5 Summary and Outlook **85**

List of Figures

4.1 Wisconsin Diagnostic Breast Cancer dataset: training and test misclassification rates over iterations. The shaded regions show ±1 standard deviations. 81
4.2 Inventory control: relative distance $\|\bar{x}_k - x^\star\|/\|x^\star\|$ over iterations. The shaded region shows ±3 standard deviations, and the dotted horizontal line indicates the level 0.2. 84

1

Introduction

Mathematical optimization is concerned with problems of the form

$$\min_{x \in \mathcal{X}} f(x),$$

where $f : \mathbb{R}^n \to \mathbb{R} \cup \{\infty\}$ is a function and $\mathcal{X} \subset \operatorname{dom}(f)$ is a set. Typically, the set $\mathcal{X}$ is called the *feasible set*, elements $x \in \mathcal{X}$ are called *feasible points*, and f is called the *objective function*, or simply *objective*. Oftentimes, one refers to $x \in \mathbb{R}^n$ as a *decision variable*. If $\mathcal{X}$ is nonempty, then the problem is called *feasible*. In that case, if there exists $x^\star \in \mathcal{X}$ such that $f(x^\star) \leq f(x)$ for all $x \in \mathcal{X}$, then $x^\star$ is called *solution*. The value $f(x^\star)$ is called *optimal value* or *minimal value*. In practice, the feasible set $\mathcal{X}$ is often defined implicitly through the use of auxilliary functions, which leads to the formulation

$$\begin{aligned} &\min_{x \in \mathbb{R}^n} f(x) \\ &\text{subject to (s. t.)} \quad g(x) \leq 0 \\ &\qquad\qquad\qquad\quad h(x) = 0 \end{aligned}$$

for functions $g : \mathbb{R}^n \to \mathbb{R}^m$, $h : \mathbb{R}^n \to \mathbb{R}^\ell$. The feasible set is then given by $\mathcal{X} = \{\, x \in \mathbb{R}^n \mid g(x) \leq 0 \text{ and } h(x) = 0 \,\}$. An important special case of optimization problems are *convex optimization problems*, where the objective f and the map g are convex functions, and h is affine. Convex problems have (among other things) the highly desirable property that every local minimum is a global minimum, which makes gradient-based optimization algorithms work effectively [1].

In this work, we will consider *stochastic* convex optimization problems, which incorporate uncertainty into the problem formulation through the use of random variables [2, 3]. Such problems appear in many areas of applied mathematics, including machine learning [4], supply chain management [5], and portfolio optimization

[6]. Specifically, the stochastic optimization problems that we will consider take the form

$$\begin{aligned} &\min_{x\in\mathbb{R}^n} \{ f(x) := \mathbb{E}(F_\xi(x)) + r(x) \} \\ &\text{subject to} \quad g(x,\xi) \le 0 \quad \text{almost surely}, \end{aligned} \tag{P}$$

where ξ is a random variable that captures uncertainty in the objective and the constraints. The objective f is composed of two functions: the expectation functional $\mathbb{E}(F_\xi)\colon \mathbb{R}^n \to \mathbb{R}$, which we assume to be smooth, and the potentially nonsmooth function $r\colon \mathbb{R}^n \to \mathbb{R}\cup\{\infty\}$, which is sometimes referred to as the *regularizer*. Notable instances of problems that can be formulated in this way include primal support vector machines [7, 8], and robust optimization by sampling [9, 10, 11]. We are particularly interested in the case where the number of constraints induced by the distribution of ξ is large, possibly even infinite. These *large-scale* constraints prevent the use of classical methods in deterministic optimization, which need to process all constraints at once. This motivates the development of a method that only needs to process a small amount of constraints at a time.

A popular approach to solve (P) is an iterative method called *stochastic gradient descent* (SGD), introduced by Robbins and Monro [2]. SGD is very similar to the classic *gradient descent* algorithm [1, 12]. The difference is that the gradients are replaced by *stochastic gradients*: A stochastic gradient of f at $x \in \mathbb{R}^n$ is a random variable g such that $\mathbb{E}(g) = \nabla f(x)$. For example, if $f(x) = \mathbb{E}(F_\xi(x))$ and F_ξ satisfies certain regularity conditions, then $\nabla f(x) = \mathbb{E}(\nabla F_\xi(x))$, which implies that $\nabla F_\xi(x)$ is a stochastic gradient of f at x for every $x \in \mathbb{R}^n$. The advantage is that the computation of the gradient $\nabla f(x)$ is replaced with the (potentially much) cheaper computation of $\nabla F_\xi(x)$.

Unfortunately, classic SGD is not generally applicable to problems such as (P), because in order to deal with the constraints, the algorithm needs to compute the *projection operator*, $x \mapsto \arg\min_{y\in\mathcal{X}} \|x-y\|$ with $\|\cdot\|$ denoting the euclidian norm on $\mathbb{R}^n$, in every iteration. This computation is well known to be tractable only for certain (simple) kinds of constraints [13]. In contrast, problem (P) may have a highly complex set of constraints, possibly even infinitely many.

A standard method for dealing with constraints in deterministic constrained optimization problems like (1.1) involves the use of *penalty functions* $\pi\colon \mathbb{R}^n \to [0,\infty)$ such that $\pi(x) = 0$ for all $x \in \mathcal{X}$ and $\pi(x) > 0$ otherwise. Instead of the constrained problem (1.1), one considers the unconstrained problem

$$\min_{x\in\mathbb{R}^n} f(x) + \gamma\pi(x),$$

where $\gamma \in (0, \infty)$ is a parameter that controls the degree to which infeasible points get penalized. In the case where the problem is purely inequality constrained, that is $h \equiv 0$, popular examples of penalty functions are

- the nonsmooth penalty $\pi_{\ell_1}(x) := \|(g(x))_+\|_1$ with $\|\cdot\|_1$ the ℓ_1-norm on $\mathbb{R}^m$, and
- the smooth penalty $\pi_{\ell_2}(x) := \|(g(x))_+\|_2^2$ with $\|\cdot\|_2$ the ℓ_2-norm on $\mathbb{R}^m$,

where we denote by $(y)_+$ the elementwise application of $x \mapsto \max(0, x)$ to the vector $y \in \mathbb{R}^m$. Unfortunately, these penalty functions suffer from well-known problems: The smooth penalty π_{ℓ_2} will, in general, not recover a feasible point of (1.1) — no matter how large γ is chosen. To get reasonably close to a feasible point, one is often forced to choose a very large γ, which introduces issues related to poor conditioning. On the other hand, the nonsmooth penalty π_{ℓ_1} is *exact*, which means that there does exist a penalty parameter $\gamma^\star \in (0, \infty)$ such that the minimizer of $f + \gamma^\star \pi_{\ell_1}$ is the same as that of the original constrained problem. However, it is not clear in general how to choose $\gamma^\star$, and the non-differentiability introduces problems for gradient-based methods [12, 14]. Lastly, as these penalties need to evaluate all m constraints at once, they are not applicable to the large-scale problems that we care about.

An alternative to the aforementioned penalty functions, which avoids many of their drawbacks, was introduced by Nedić et al. [15]. In their work, the authors analyzed a special case of our problem (P), namely

$$\begin{aligned} &\min_{x \in \mathbb{R}^n} f(x) \\ &\text{s.t.} \quad a_i^\top x - b_i \leq 0 \quad \text{for all } i \in \{1, \dots, m\}. \end{aligned} \tag{1.1}$$

Their method to solve (1.1) is a penalty method. However, instead of a fixed penalty function π, the authors introduced a sequence of smooth penalty functions $(\pi_k^{\text{hub}})_{k \in \mathbb{N}}$, defined as follows: Let $(\delta_k)_{k \in \mathbb{N}}$ be a sequence of positive real numbers. For $k \in \mathbb{N}$, we define

$$\pi_k^{\text{hub}}(x) := \frac{1}{m} \sum_{i=1}^{m} h_k^{\text{hub}}(x; a_i, b_i),$$

where

$$h_k^{\text{hub}}(x; a, b) := \begin{cases} \frac{\langle a, x \rangle - b}{\|a\|} & \text{if } \langle a, x \rangle - b > \delta_k, \\ \frac{(\langle a, x \rangle - b + \delta_k)^2}{4\delta_k \|a\|} & \text{if } -\delta_k \leq \langle a, x \rangle - b \leq \delta_k, \\ 0, & \text{if } \langle a, x \rangle - b < -\delta_k, \end{cases}$$

for $x, a \in \mathbb{R}^n$, and $b \in \mathbb{R}$. The crucial properties of the sequence $(\pi_k^{\text{hub}})_{k \in \mathbb{N}_0}$ are that (i) it is a sequence of smooth functions, (ii) the sequence converges to the exact

penalty function $x \mapsto \|\max(0, x)\|_1$ as $k \to \infty$, and (iii) the gradients of $(\pi_k^{\text{hub}})_{k \in \mathbb{N}_0}$ are uniformly bounded, in the sense that there exists a constant $G \in (0, \infty)$ such that $\nabla \pi_k^{\text{hub}}(x) \leq G$ for all $x \in \mathbb{R}^n$ and $k \in \mathbb{N}_0$. These properties are very desirable for mathematical analysis. Indeed, the authors designed a single-loop iterative method around the penalty sequence (π_k^{hub}), which only requires the evaluation of one constraint per iteration. They proved that — under reasonable assumptions — their method converges to the solution of problem (1.1). More specifically, for any $\epsilon \in (0, \infty)$, one can achieve convergence rates of $\mathcal{O}(\log^{(1+3\epsilon)/2}/\sqrt{K})$ in the convex case, and $\mathcal{O}(\log^{\epsilon}/K)$ in the case where the objective is *strongly convex*, as the number of iterations, $K \in \mathbb{N}$, goes to infinity. This almost matches the rates of $\mathcal{O}(1/\sqrt{K})$ and $\mathcal{O}(1/K)$ achievable by SGD in the convex and strongly convex settings, respectively, while being projection-free [16, 17]. It should also be highlighted that the method is single-loop, as opposed to the typical two-loop design of the previously mentioned penalty methods, which makes for a particularly simple implementation.

Apart from [15], among existing work on constrained stochastic optimization, the settings considered by Fercoq et al. [7], Li et al. [18], and Xu et al. [11] are the most closely related to ours. Of those, [7] and [15] consider the special case of affine constraints, and in [11], [15], and [18], the authors assume a finite constraint set. The setting we consider therefore generalizes those of prior works. Further, [18] focuses on strongly convex finite-sum objectives, in order to leverage accelerated stochastic gradient methods such as SVRG [19]. In contrast, we will consider objectives that contain general expected value functions, and provide analysis for both strongly convex, as well as general convex objectives. Finally, both [7] and [11] make assumptions to ensure that the stochastic gradients have uniformly bounded variance. However, this assumption does not apply to many realistic settings [20] and we will not rely on it for the analysis of our method.

Our contributions in this work are as follows: We build on the ideas of Nedić et al. [15] with a new algorithm that exploits the compositional structure in the objective and applies to a wider range of constraints. Specifically, our method and the analysis thereof apply to the general setting of problem (P), which allows for infinitely many nonlinear constraints and makes no finite-sum assumption on the objective. This clearly differentiates it from the settings considered in prior works. A notable feature of our proofs is that we do not make any global boundedness assumptions on the norm of the stochastic gradients, which stands in contrast to many related works in the stochastic optimization literature.

In the following, we will first begin with a review of the relevant background material in the fields of probability and convex optimization in chapter 2. The main

contributions of this work are then presented in chapter 3, where we formulate our algorithm and prove various convergence results under relatively mild assumptions on the problem setting beyond convexity. We support our theoretical analysis with numerical experiments in chapter 4. Finally, we summarize our findings and point to future research directions in chapter 5.

2

Mathematical Background

This chapter will serve as a review of relevant definitions and results that we will make use of all throughout this work. First, we will introduce our notational conventions. Then, we will highlight two inequalities, which will be very useful to us later on. Following this comes the main heart of this chapter, where we will give a review of concepts in probability and convex optimization, the two central fields that all of our analysis will rely upon.

2.1. Notation

Throughout, we let $\|\cdot\|_p$ denote the ℓ_p-norm on $\mathbb{R}^n$, defined as

$$\|x\|_p := \left(\sum_{i=1}^{n} |x_i|^p\right)^{\frac{1}{p}}$$

for $p \in [1, \infty)$. Since we will mostly be using the ℓ_2 ("euclidian") norm, we will omit its subscript, that is, we will write

$$\|\cdot\| := \|\cdot\|_2.$$

The euclidian inner product is denoted by $\langle \cdot, \cdot \rangle$.

For $x \in \mathbb{R}^n$, we will write $(x)_+$ to denote the elementwise application of $\max(\cdot, 0)$ to x, that is, $(x)_+$ is defined as the vector $y \in \mathbb{R}^n$ such that

$$y_i = \max(0, x_i),$$

where x_i and y_i denote the i-th component of x and y, respectively, for $i \in \{1, \ldots, n\}$.

For a function $f : \mathbb{R}^n \to \mathbb{R}$ that is differentiable at $x \in \mathbb{R}^n$, we denote the gradient of f at x by $\nabla f(x)$. If f is merely subdifferentiable at $x \in \mathbb{R}^n$ (see definition 2.44), we will use the notation $\tilde{\nabla} f(x)$ to refer to a subgradient of f at x, which will typically be arbitrary, unless otherwise specified.

2.2. Basic inequalities

We will make use of two classic inequalities in various places throughout this work. The first one is the **Cauchy-Schwarz inequality**:

$$|\langle x, y\rangle| \leq \|x\|\|y\|$$

for all $x, y \in \mathbb{R}^n$. The second one is **Young's inequality**, which says that

$$ab \leq \frac{\epsilon a^2}{2} + \frac{b^2}{2\epsilon}$$

for all $a, b \in \mathbb{R}$ and any $\epsilon \in (0, \infty)$. Combining the two inequalities, we get the useful bound

$$|\langle x, y\rangle| \leq \frac{\epsilon\|x\|^2}{2} + \frac{\|y\|^2}{2\epsilon}$$

for all $x, y \in \mathbb{R}^n$ and any $\epsilon \in (0, \infty)$.

2.3. Probability theory

Most of the contents of this section are standard and can be found in various texts such as [21, 22]. Some material is more specialized, in which case we will point to additional references.

2.3.1. Measure and integration

Definition 2.1. Let Ω be a set and let 2^Ω denote its power set. A subset $\mathcal{F} \subset 2^\Omega$ is called a σ-**algebra over** Ω if it satisfies the following three conditions:

1. $\emptyset \in \mathcal{F}$.

2. If $A, B \in \mathcal{F}$, then $B \backslash A \in \mathcal{F}$.

3. For any countable sequence $A_1, A_2, \cdots \in \mathcal{F}$, we have $\bigcup_{n=1}^{\infty} A_n \in \mathcal{F}$.

If $\mathcal{F}$ is a σ-algebra over Ω, then the tuple $(\Omega, \mathcal{F})$ is called a **measurable space**. For any subset $\mathcal{G} \subset 2^{\Omega}$, we define the σ-**algebra generated by** $\mathcal{G}$ as the intersection over all σ-algebras that contain $\mathcal{G}$ as an element, and we denote this σ-algebra by $\sigma(\mathcal{G})$.

Example 2.2. An important example of a σ-algebra over $\mathbb{R}^n$ is the **Borel** σ-**algebra** $\mathcal{B}(\mathbb{R}^n)$, which is defined to be the σ-algebra generated by the subset of all open sets on $\mathbb{R}^n$. Functions that are $\mathcal{B}(\mathbb{R}^n)$-measurable are called **Borel measurable**.

Unless otherwise stated, we will always consider $\mathbb{R}^n$ to be endowed with the Borel σ-algebra.

Definition 2.3. Let $(\Omega, \mathcal{F})$ and $(E, \mathcal{G})$ be measurable spaces. A map $f\colon \Omega \to E$ is called $\mathcal{F}, \mathcal{G}$-**measurable** if $f^{-1}(G) := \{\omega \in \Omega \mid f(\omega) \in G\} \in \mathcal{F}$ for all $G \in \mathcal{G}$. We may say f is $\mathcal{F}$-**measurable** or simply **measurable** if one or both of the σ-algebras are either clear from the context or not relevant.

Definition 2.4. Let $(\Omega, \mathcal{F})$ be a measurable space. A map $\mu : \mathcal{F} \to [0, \infty]$ is called a **measure on** $(\Omega, \mathcal{F})$ if it satisfies the following two conditions:

1. $\mu(\emptyset) = 0$.

2. For any countable sequence $A_1, A_2, \dots \in \mathcal{F}$, we have $\mu(\bigcup_{n=1}^{\infty} A_n) = \sum_{n=1}^{\infty} \mu(A_n)$.

If μ is a measure on $(\Omega, \mathcal{F})$, then the triplet $(\Omega, \mathcal{F}, \mu)$ is called a **measure space**.

Definition 2.5. Let $(\Omega, \mathcal{F}, \mu)$ be a measure space and let $f\colon \Omega \to \mathbb{R}$ be measurable. We define the (μ-)**integral** of f, denoted $\int f \, \mathrm{d}\mu$, in three steps:

1. If $f(\omega) = \sum_{i=1}^{n} c_i 1_{A_i}(\omega)$ for some $n \in \mathbb{N}$, $c_1, \dots, c_n \in (0, \infty)$, and disjoint measurable sets $A_1, \dots, A_n \in \mathcal{F}$, then we define

$$\int f \, \mathrm{d}\mu := \sum_{i=1}^{n} c_i \mu(A_i).$$

   In this case f is called a **simple function**. The set of all simple functions on Ω is denoted by $\mathcal{S}(\Omega)$.

2. If f is nonnegative, i. e. $f(\omega) \geq 0$ of all $\omega \in \Omega$, then we define

$$\int f \, \mathrm{d}\mu := \sup_{g \in \mathcal{S}(\Omega),\, g \leq f} \int g \, \mathrm{d}\mu.$$

3. If f is neither a simple function, nor nonnegative, but $\int |f| \, \mathrm{d}\mu < \infty$, then we define

$$\int f \, \mathrm{d}\mu := \int \max(0, f) \, \mathrm{d}\mu - \int \max(0, -f) \, \mathrm{d}\mu.$$

Otherwise, we say that the (μ-)integral of f does not exist. If any of these three conditions apply to f, we say that f is (μ-)**integrable**.

Apart from the notation used above, we will also sometimes use

$$\int f(x) \, \mu(\mathrm{d}x)$$

to denote the μ-integral of f.

Proposition 2.6. Let $(\Omega, \mathcal{F}, \mu)$ be a measure space and let $f \colon \Omega \to \mathbb{R}$ and $g \colon \Omega \to \mathbb{R}$ be integrable. Then the following hold:

(i) $\int f + g \, \mathrm{d}\mu = \int f \, \mathrm{d}\mu + \int g \, \mathrm{d}\mu$.

(ii) $\int cf \, \mathrm{d}\mu = c \int f \, \mathrm{d}\mu$ for all $c \in \mathbb{R}$.

(iii) If $f \leq g$, then $\int f \, \mathrm{d}\mu \leq \int g \, \mathrm{d}\mu$. If additionally $f < g$ on some set $A \in \mathcal{F}$ with $\mu(A) > 0$, then $\int f \, \mathrm{d}\mu < \int g \, \mathrm{d}\mu$.

Proposition 2.7. Let $(\Omega, \mathcal{F}, \mu)$ be a measure space and let $f \colon \Omega \to \mathbb{R}$ and $g \colon \mathbb{R} \to \mathbb{R}$ be integrable. Then it holds that

$$\int g \circ f \, \mathrm{d}\mu = \int g \, \mathrm{d}\mu^f,$$

where μ^f is the measure defined as

$$\mu^f(B) := \mu(f^{-1}(B))$$

for all $B \in \mathcal{B}(\mathbb{R})$.

2.3.2. Random variables, expected value, and variance

Definition 2.8. Let $(\Omega, \mathcal{F})$ be a measurable space. We say that a measure $\mathbb{P} : \mathcal{F} \to [0, \infty)$ is a **probability measure** measure on $(\Omega, \mathcal{F})$, if

1. $\mathbb{P}(A) \in [0, 1]$ for all $A \in \mathcal{F}$, and

2. $\mathbb{P}(\Omega) = 1$.

If $\mathbb{P}$ is a probability measure on $(\Omega, \mathcal{F})$, then we call the triplet $(\Omega, \mathcal{F}, \mathbb{P})$ a **probability space**. In this context, elements of $\mathcal{F}$ are called **events**.

Definition 2.9. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. An event $A \in \mathcal{F}$ is said to hold **almost surely** (a. s. for short) if $\mathbb{P}(A) = 1$.

Definition 2.10 (Random variable)**.** Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. A map $X \colon \Omega \to \mathbb{R}^n$ is called a **random variable** on $(\Omega, \mathcal{F}, \mathbb{P})$ if X is measurable. In the case $n > 1$, we may call X a **random vector**. Further, we define the notation $\mathbb{P}(X \in B) := \mathbb{P}(X^{-1}(B))$ for $B \in \mathcal{B}(\mathbb{R}^n)$. We define the **distribution of** X to be the probability measure $\mathbb{P}^X = \mathbb{P} \circ X^{-1}$ on $\mathbb{R}^n$. The **support of** X is the smallest closed set $S \subset \mathbb{R}^n$ such that $\mathbb{P}^X(S) = 1$. Finally, we define $\sigma(X) := \sigma(\{ X^{-1}(B) \mid B \in \mathcal{B}(\mathbb{R}^n) \})$ and call this the **σ-algebra generated by** X.

Definition 2.11. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. Two random variables X and Y on $(\Omega, \mathcal{F}, \mathbb{P})$ are called **independent** if $\mathbb{P}(X \in A,\, Y \in B) = \mathbb{P}(X \in A) \cdot \mathbb{P}(Y \in B)$ for all $A, B \in \mathcal{F}$. X and Y are called **identically distributed** if $\mathbb{P}^X = \mathbb{P}^Y$. We use the abbreviation **i. i. d.** as shorthand for "independent and identically distributed".

Definition 2.12. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $X \colon \Omega \to \mathbb{R}$ be a random variable on this space. If X is integrable, we define the **expected value of** X, denoted by $\mathbb{E}(X)$, as

$$\mathbb{E}(X) := \int X \, \mathrm{d}\mathbb{P}.$$

If $X \colon \Omega \to \mathbb{R}^n$ is a random vector with components $X_i \colon \Omega \to \mathbb{R}$ for $i \in \{1, \dots, n\}$, then we say that X **is integrable** if all components are integrable. In that case, we define **the expected value of** X as the vector

$$\mathbb{E}(X) := (\mathbb{E}(X_1), \dots, \mathbb{E}(X_n))^\top.$$

The following three properties will be used multiple times throughout this text without explicit mention. They follow directly from proposition 2.6.

Proposition 2.13. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $X \colon \Omega \to \mathbb{R}^n$ and $Y \colon \Omega \to \mathbb{R}^n$ be integrable random variables. Then the following hold:

(i) $\mathbb{E}(X + Y) = \mathbb{E}(X) + \mathbb{E}(Y)$.

(ii) $\mathbb{E}(cX) = c\,\mathbb{E}(X)$ for all $c \in \mathbb{R}$.

(iii) If $X \leq Y$, then $\mathbb{E}(X) \leq \mathbb{E}(Y)$. If, additionally, $X(\omega) < Y(\omega)$ for all ω in an event $A \in \mathcal{F}$ with $\mathbb{P}(A) > 0$, then $\mathbb{E}(X) < \mathbb{E}(Y)$.

(iv) $\|\mathbb{E}(X)\| \leq \mathbb{E}\|X\|$.

Proposition 2.14 (Jensen's inequality)**.** Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $X\colon \Omega \to \mathbb{R}^n$ be a random variable. Then we have

$$\mathbb{E}(X^2) \geq \mathbb{E}(X)^2.$$

In particular: If X^2 is integrable, then X must also be integrable.

Proposition 2.15 (Bounded convergence theorem)**.** Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $(X_k)_{k\in\mathbb{N}_0}$ be a sequence of uniformly bounded real-valued random variables on Ω such that $\lim_{k\to\infty} X_k = X$ almost surely for some random variable X. Then X is integrable and

$$\lim_{k\to\infty} \mathbb{E}(X_k) = \mathbb{E}(X).$$

Proposition 2.16 (Interchanging differentiation and expectation)**.** Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, let $f\colon \mathbb{R}^n \times \Omega \to \mathbb{R}$ be continuously differentiable in its first argument, and suppose $\mathbb{E}|f(x,\cdot)| < \infty$ for all $x \in \mathbb{R}^n$. For $x_0 \in \mathbb{R}^n$, assume that there exists a random variable $G\colon \Omega \to \mathbb{R}$ such that, for all $x \in \mathbb{R}^n$ in a neighborhood around x_0,

$$\|\nabla f(x,\cdot)\| \leq G$$

almost surely, where we consider the gradient with respect to the first argument. Then

$$\nabla\mathbb{E}(f(x_0,\cdot)) = \mathbb{E}(\nabla f(x_0,\cdot)).$$

Definition 2.17. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $X\colon \Omega \to \mathbb{R}^n$ be a random variable. If X^2 is integrable, we define the **variance of** X, denoted by $\mathrm{Var}(X)$, as $\mathrm{Var}(X) := \mathbb{E}\|X - \mathbb{E}(X)\|^2$.

By proposition 2.7, for any integrable random variable $X : \Omega \to \mathbb{R}$, it holds that $\int X \,\mathrm{d}\mathbb{P} = \int I \,\mathrm{d}\mathbb{P}^X$, where $I\colon \mathbb{R} \to \mathbb{R}$ is the identity operator. It is now easy to see that if two random variables X and Y are identically distributed, then they have the same expected value and variance.

Proposition 2.18. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $X_1, \ldots, X_N\colon \Omega \to \mathbb{R}^n$ be $N \in \mathbb{N}$ independent random variables, such that $\|X_i\|^2$ is integrable for all $i \in \{1, \ldots, N\}$. Then $\mathrm{Var}(\sum_{i=1}^N a_i X_i) = \sum_{i=1}^N a_i^2 \,\mathrm{Var}(X_i)$, for any $a_1, \ldots, a_N \in \mathbb{R}$. If, additionally, they are all identically distributed, it holds that $\mathrm{Var}(1/N \sum_{i=1}^N X_i) = \mathrm{Var}(X_1)/N$.

Proposition 2.19. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $X\colon \Omega \to \mathbb{R}^n$ be a random variable such that X^2 is integrable. Then $\mathrm{Var}(X) = \mathbb{E}(X^2) - \mathbb{E}(X)^2$.

2.3.3. Densities

Definition 2.20 (Density)**.** Let $(E, \mathcal{E})$ be a measurable space, and let μ and ν be measures on $(E, \mathcal{E})$. A ν-integrable function $f\colon E \to [0, \infty)$ is called a **ν-density of** μ, if

$$\mu(A) = \int_A f \, \mathrm{d}\nu$$

for all $A \in \mathcal{E}$. In the special case where $E = \mathbb{R}^n$ and ν is the Lebesgue-measure on $\mathbb{R}^n$, we refer to f as a **Lebesgue-density**, or simply **density**, of μ.

Definition 2.21 (Absolute continuity and equivalence)**.** Let $(\Omega, \mathcal{F})$ be a measurable space and let μ and ν be measures on $(\Omega, \mathcal{F})$. We say that μ is **absolutely continuous with respect to** ν, if

$$\nu(A) = 0 \quad \text{implies} \quad \mu(A) = 0$$

for all $A \in \mathcal{F}$. If μ is absolutely continuous with respect to ν, then we use the notation $\mu \ll \nu$. We say that μ and ν are **equivalent**, if $\mu \ll \nu$ and $\nu \ll \mu$. If μ and ν are equivalent, we use the notation $\mu \sim \nu$.

Proposition 2.22 (Radon-Nikodym)**.** Let $(\Omega, \mathcal{F})$ be a measurable space and let μ and ν be measures on $(\Omega, \mathcal{F})$. Then $\mu \ll \nu$ if and only if there exists a ν-density of μ. If μ and ν are equivalent and f is a ν-density of μ, then f is positive ν-almost everywhere and there exists a μ-density g of ν, such that $f \cdot g = 1$ ν-almost everywhere.

Proposition 2.23. Let $(E, \mathcal{E})$ be a measurable space and let μ, ν and τ be measures on $(E, \mathcal{E})$. Suppose that μ has a ν-density f and ν has a τ-density g. Then the function $h := f \cdot g$ is a τ-density of μ.

If f is a ν-density of μ, then f is often referred to as a **Radon-Nikodym derivative of μ with respect to ν**. Commonly used notations are

$$f = \frac{\mathrm{d}\mu}{\mathrm{d}\nu}$$

and

$$\mathrm{d}\mu = f \mathrm{d}\nu.$$

Putting the two together, we formally have

$$\mathrm{d}\mu = \frac{\mathrm{d}\mu}{\mathrm{d}\nu} \mathrm{d}\nu.$$

This notation suggest that, if $\mu \ll \nu$ and $\nu \ll \tau$ for some other measure τ, then we have

$$\mathrm{d}\mu = \frac{\mathrm{d}\mu}{\mathrm{d}\nu}\mathrm{d}\nu = \frac{\mathrm{d}\mu}{\mathrm{d}\nu}\frac{\mathrm{d}\nu}{\mathrm{d}\tau}\mathrm{d}\tau,$$

and hence $\mu \ll \tau$ with density $\frac{\mathrm{d}\mu}{\mathrm{d}\nu}\frac{\mathrm{d}\nu}{\mathrm{d}\tau}$, which is consistent with the above proposition.

We now move to back to the setting of probability spaces, where the concept of densities are of great help to work with random variables.

Definition 2.24 (Density of a random variable)**.** Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $X\colon \Omega \to \mathbb{R}^n$ be a random variable. Further, let μ be a measure on $(\Omega, \mathcal{F})$. A function $f\colon \mathbb{R} \to [0, \infty)$ is called a **μ-density of** X, if f is a μ-density of $\mathbb{P}^X$. That is,

$$\mathbb{P}(X \in B) = \int_B f \,\mathrm{d}\mu$$

for all $B \in \mathcal{B}(\mathbb{R}^n)$.

The following is an important special case of proposition 2.23.

Proposition 2.25. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $X\colon \Omega \to \mathbb{R}^n$ be a random variable. Further, let μ be a measure on $(\Omega, \mathcal{F})$ and suppose that X has a μ-density f such that $\int |x| f(x)\,\mathrm{d}x < \infty$. Then X is integrable and

$$\mathbb{E}(X) = \int x f(x)\,\mu(dx).$$

In particular, if μ is the Lebesgue-measure on $\mathbb{R}^n$, it holds that

$$\mathbb{E}(X) = \int x f(x)\,\mathrm{d}x.$$

If μ is instead the counting measure on a discrete subset $S \subset \mathbb{R}^n$, then

$$f(x) = \mathbb{P}(X = x)$$

for all $x \in S$, and

$$\mathbb{E}(X) = \sum_{x \in S} x\,\mathbb{P}(X = x).$$

2.3.4. Conditioning

Definition 2.26 (Conditional expectation)**.** Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $X\colon \Omega \to \mathbb{R}^n$ be a random variable. Further, let $\mathcal{C} \subset \mathcal{F}$ be a σ-algebra. We call a random variable $Z\colon \Omega \to \mathbb{R}^n$ a **conditional expectation of** X **given** $\mathcal{C}$, if

1. Z is $\mathcal{C}$-measurable, and
2. for all $C \in \mathcal{C}$, it holds that

$$\int_C Z \, \mathrm{d}\mathbb{P} = \int_C X \, \mathrm{d}\mathbb{P}.$$

If Z is a conditional expectation of X given $\mathcal{C}$ then we use the notation $\mathbb{E}(X \mid \mathcal{C}) := Z$.

If $Y \colon \Omega \to \mathbb{R}^m$ is a random variable such that $\mathcal{C} = \sigma(Y)$, then we use the notation $\mathbb{E}(X \mid Y) := \mathbb{E}(X \mid \sigma(Y))$. In that case, we call $\mathbb{E}(X \mid Y)$ the **conditional expectation of X given Y**. Further, for $\omega \in \Omega$ with $Y(\omega) = y \in \mathbb{R}^m$, the **conditional expectation of X given $Y = y$**, denoted by $\mathbb{E}(X \mid Y = y)$, is defined as $\mathbb{E}(X \mid Y = y) := \mathbb{E}(X \mid Y)(\omega)$.

Note that $E(X \mid Y)$ is not unique. However, if Z_1 and Z_2 are both conditional expectations of X given Y, then we always have $Z_1 = Z_2$ almost surely. For simplicity, we will keep the "almost surely" implicit.

Remark 2.27. If X and Y are integrable random variables on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, then the conditional expectation of X given $Y = y$ can be thought of as the expected value of X on a different probabiliy space $(\Omega, \mathcal{F}, \mathbb{P}_{Y=y})$, where $\mathbb{P}_{Y=y}(A) := \mathbb{P}(A \mid Y = y)$. More precisely, $\mathbb{E}(X \mid Y = y) = \int X \, \mathrm{d}\mathbb{P}_{Y=y}$. It follows that $\mathbb{E}(X \mid Y = \cdot)$ inherits all basic properties of $\mathbb{E}(\cdot)$ from proposition 2.13.

Below, we state some further properties of conditional expectations.

Proposition 2.28 (Properties of $\mathbb{E}(X \mid Y)$)**.** Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $X \colon \Omega \to \mathbb{R}^n$, $Y \colon \Omega \to \mathbb{R}^m$ be integrable random variables. Further, let $F \colon \mathbb{R}^n \times \Omega \to \mathbb{R}$ such that $\omega \mapsto F(x, \omega)$ is $\mathcal{F}$-measurable for all $x \in \mathbb{R}^n$ and let $G \colon \mathbb{R}^n \to \mathbb{R}$ be Borel measurable. Then the following hold:

(i) $\mathbb{E}(\mathbb{E}(X \mid Y)) = \mathbb{E}(X)$.

(ii) $\mathbb{E}(X \mid X) = X$.

(iii) If $F(x, \cdot) \leq G(x)$ a. s. for all $x \in \mathbb{R}^n$, it follows that $\mathbb{E}(F(X, \cdot) \mid X) \leq G(X)$.

Definition 2.29 (Conditional variance)**.** Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $X \colon \Omega \to \mathbb{R}^n$, $Y \colon \Omega \to \mathbb{R}^m$ be integrable random variables. The **conditional variance of X given Y**, denoted by $\mathrm{Var}(X \mid Y)$, is defined as

$$\mathrm{Var}(X \mid Y) := \mathbb{E}\big(||X - \mathbb{E}(X \mid Y)||^2 \mid Y\big).$$

For $y \in \mathbb{R}^n$, the **conditional variance of X given Y = y** is defined as

$$\mathrm{Var}(X \mid Y = y) := \mathbb{E}\big(||X - \mathbb{E}(X \mid Y = y)||^2 \mid Y = y\big).$$

Remark 2.30. Let X and Y be integrable random variables on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Similarly to the conditional expectation of X given $Y = y$, the conditional variance $\mathrm{Var}(X \mid Y = y)$ can be thought of as the variance of X on a different probability space $(\Omega, \mathcal{F}, \mathbb{P}_{Y=y})$ (see remark 2.27). Hence, all basic properties of $\mathrm{Var}(\cdot)$ are inherited by the conditional variance. In particular, if $X_1, \dots, X_N$ are i. i. d. random variables, it holds that $\mathrm{Var}(X_1 + \cdots + X_N \mid Y) = \mathrm{Var}(X_1 \mid Y)/N$ — a fact that will be used later.

Proposition 2.31 (Monotonicity)**.** Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $X : \Omega \to \mathbb{R}^n$ be an integrable random variable. Further, let $F : \mathbb{R}^n \times \Omega \to \mathbb{R}$ such that $\omega \mapsto F(x, \omega)$ is $\mathcal{F}$-measurable for all $x \in \mathbb{R}^n$ and $\mathbb{E}(F(X, \cdot)^2) < \infty$, and let $G : \mathbb{R}^n \to \mathbb{R}$ be Borel measurable. Then, if $\mathrm{Var}(F(x, \cdot)) \leq G(x)$ for all $x \in \mathbb{R}^n$, it holds that

$$\mathrm{Var}(F(X, \cdot) \mid X) \leq G(X).$$

2.3.5. Martingales

Definition 2.32 (Filtration)**.** Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. A sequence of σ-algebras $(\mathcal{F}_k)_{k \in \mathbb{N}_0}$ with $\mathcal{F}_k \subset \mathcal{F}\ \forall k \in \mathbb{N}_0$ is called a **filtration** if $\mathcal{F}_{k-1} \subset \mathcal{F}_k$ for all $k \in \mathbb{N}$. If $(\mathcal{F}_k)_{k \in \mathbb{N}_0}$ is a filtration on $(\Omega, \mathcal{F}, \mathbb{P})$, then we call $(\Omega, \mathcal{F}, \mathbb{P}, (\mathcal{F}_k)_{k \in \mathbb{N}_0})$ a **filtered probability space**.

Example 2.33 (Natural filtration)**.** Given a sequence of random variables $(X_k)_{k \in \mathbb{N}_0}$ on some common probabilty space $(\Omega, \mathcal{F}, \mathbb{P})$, one can define a filtration $(\mathcal{F}_k)_{k \in \mathbb{N}_0}$ via $\mathcal{F}_k := \sigma(X_0, \dots, X_k)$ for all $k \in \mathbb{N}_0$. This filtration is referred to as the **natural filtration** of the sequence $(X_k)_{k \in \mathbb{N}_0}$.

Definition 2.34 (Stochastic process)**.** Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. A sequence of random variables $(X_t)_{t \in T}$, where $T \subset \mathbb{R}$ is some set of indeces, is called a **stochastic process**. Let $(\mathcal{F}_k)_{k \in \mathbb{N}_0}$ be a filtration on $(\Omega, \mathcal{F}, \mathbb{P})$. A stochastic process $X := (X_k)_{k \in \mathbb{N}_0}$ is said to be **adapted to** $(\mathcal{F}_k)_{k \in \mathbb{N}_0}$, if X_k is $\mathcal{F}_k$-measurable for all $k \in \mathbb{N}_0$.

Clearly, any stochastic process is adapted to their natural filtration.

Definition 2.35 (Martingale)**.** Let $X := (X_k)_{k \in \mathbb{N}_0}$ be a stochastic process on a filtered probability space $(\Omega, \mathcal{F}, \mathbb{P}, (\mathcal{F}_k)_{k \in \mathbb{N}_0})$. Then we say that X is a **martingale**, if the following conditions are satisfied:

1. $\mathbb{E}|X_k| < \infty$ for all $k \in \mathbb{N}_0$.

2. X is adapted to $(\mathcal{F}_k)_{k \in \mathbb{N}_0}$, i. e. X_k is $\mathcal{F}_k$-measurable for all $k \in \mathbb{N}_0$.

3. $\mathbb{E}(X_{k+1} \mid X_k) = X_k$ for all $k \in \mathbb{N}_0$.

The following inequality has its origins in the work of Hoeffding [23], who proved it for sequences of independent random variables. It was subsequently extended to martingales by Azuma [24].

Proposition 2.36 (Azuma-Hoeffding inequality)**.** Let $(X_k)_{k\in\mathbb{N}_0}$ be a martingale on a filtered probability space $(\Omega, \mathcal{F}, \mathbb{P}, (\mathcal{F}_k)_{k\in\mathbb{N}_0})$ and let $(c_k)_{k\in\mathbb{N}}$ be a sequence such that

$$|X_k - X_{k-1}| \leq c_k$$

almost surely, for all $k \in \mathbb{N}$. Then, for all $K \in \mathbb{N}$ and all $t \in (0, \infty)$, it holds that

$$\mathbb{P}(X_K - X_0 \geq t) \leq \exp\left(\frac{-t^2}{2\sum_{k=1}^{K} c_k^2}\right).$$

2.4. Optimization theory

Most of the contents of this section are standard, and can be found in various works in optimization such as [25, 26]. For an introduction to convex optimization with a focus on modelling, we refer the reader to [1]. For more specialized content, we will point to additional references.

2.4.1. Lipschitz functions

Definition 2.37. Let $f : \mathbb{R}^n \to \mathbb{R}$ be differentiable and let $L \in (0, \infty)$. We say that f is **Lipschitz-smooth with constant** L, or simply L**-smooth**, if its gradient is L-Lipschitz continuous, i. e.

$$\|\nabla f(y) - \nabla f(x)\| \leq L\|y - x\|$$

for all $x, y \in \mathbb{R}^n$.

Proposition 2.38. Let $f : \mathbb{R}^n \to \mathbb{R}$ be L-smooth. Then it holds that

$$f(y) \leq f(x) + \langle \nabla f(x), y - x\rangle + \frac{L}{2}\|y - x\|^2$$

for all $x, y \in \mathbb{R}^n$.

Proposition 2.39. Let $f : \mathbb{R}^n \to \mathbb{R}$ be L-smooth. Then f is **locally Lipschitz continuous**, i. e. for all bounded subsets $B \subset \mathbb{R}^n$ there exists a constant $L_B \in (0, \infty)$ such that

$$|f(x) - f(y)| \leq L_B\|x - y\|$$

for all $x, y \in B$.

Proposition 2.40. A function $f\colon \mathbb{R}^n \to \mathbb{R}$ is locally Lipschitz continuous if and only if, for any bounded subset $B \subset \mathbb{R}^n$, there exists $L_B \in (0, \infty)$ such that

$$\sup_{x \in B} \|\nabla f(x)\| \leq L_B.$$

In particular, f is L-Lipschitz continuous if and only if

$$\sup_{x \in \mathbb{R}^n} \|\nabla f(x)\| \leq L.$$

2.4.2. Convex optimization

Definition 2.41 (Extended real-valued function)**.** The **extended real number line** is defined as $\bar{\mathbb{R}} := \mathbb{R} \cup \{\pm\infty\}$. A function $f\colon \mathbb{R}^n \to \bar{\mathbb{R}}$ is called **extended real-valued function**. An extended real-valued function f is called **proper**, if $f(x) > -\infty$ for all $x \in \mathbb{R}^n$ and there exists a point $x \in \mathbb{R}^n$ such that $f(x) < \infty$. The **(effective) domain** of $f\colon \mathbb{R}^n \to \bar{\mathbb{R}}$ is defined as

$$\operatorname{dom}(f) := \{\, x \in \mathbb{R}^n \mid f(x) < \infty \,\}.$$

Definition 2.42 (Convex function)**.** A set $C \subset \mathbb{R}^n$ is called **convex** if, for all $x, y \in C$ and $t \in [0, 1]$, it holds that $tx + (1-t)y \in C$. We say that a function $f\colon \mathbb{R}^n \to \bar{\mathbb{R}}$ is **convex** if $\operatorname{dom}(f)$ is convex and

$$f((1-t)x + ty) \leq (1-t)f(x) + tf(y)$$

for all $x, y \in \mathbb{R}^n$ and all $t \in (0, 1)$.

Proposition 2.43. Every extended real-valued convex function $f\colon \mathbb{R}^n \to \bar{\mathbb{R}}$ is continuous on the interior of its domain, $\operatorname{int}(\operatorname{dom}(f))$. In particular, if $\operatorname{dom}(f) = \mathbb{R}^n$, then f is continuous.

Definition 2.44 (Subgradient)**.** Let $f\colon \mathbb{R}^n \to \bar{\mathbb{R}}$ be convex. A vector $g \in \mathbb{R}^n$ is a **subgradient** of f at $x \in \operatorname{dom}(f)$ if

$$f(y) \geq f(x) + \langle g, y - x \rangle$$

for all $y \in \mathbb{R}^n$. The set of all subgradients of f at x is denoted by $\partial f(x)$ and we call this set the **subdifferential of f at x**. If $\partial f(x) \neq \emptyset$, then we call f **subdifferentiable at x**.

If f is subdifferentiable for all $x \in \mathbb{R}^n$, we call f **subdifferentiable**. For $x \notin \operatorname{dom}(f)$, we set $\partial f(x) = \emptyset$. If f is subdifferentiable at $x \in \mathbb{R}^n$, then we use the notation

$$\tilde{\nabla} f(x)$$

to refer to an arbitrary subgradient of f at x.

Proposition 2.45. Let $f : \mathbb{R}^n \to \bar{\mathbb{R}}$ be proper, convex, and continuous on $\operatorname{int}(\operatorname{dom}(f))$. Then f is subdifferentiable on $\operatorname{int}(\operatorname{dom}(f))$. If, additionally, f is L-Lipschitz continuous on $\operatorname{dom}(f)$, then the subgradients are uniformly bounded by L, i. e.

$$\sup_{\tilde{\nabla} f(x) \in \partial f(x)} \|\tilde{\nabla} f(x)\| \leq L$$

for all $x \in \operatorname{dom}(f)$.

Proposition 2.46. Let $f : \mathbb{R}^n \to \bar{\mathbb{R}}$ be proper, convex, and continuous on $\operatorname{int}(\operatorname{dom}(f))$. Then f is subdifferentiable on $\operatorname{int}(\operatorname{dom}(f))$. If, additionally, f is Lipschitz continuous on $\operatorname{dom}(f)$, then f is subdifferentiable on $\operatorname{dom}(f)$.

Proposition 2.47. Let $f : \mathbb{R}^n \to \mathbb{R}$ be differentiable. Then f is convex if and only if

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle$$

for all $x, y \in \mathbb{R}^n$. In particular, f is subdifferentiable. Moreover, the subgradients are unique, i. e. $\partial f(x) = \{\nabla f(x)\}$ for all $x \in \mathbb{R}^n$.

Proposition 2.48 (First-order condition for minimizers)**.** Let $f : \mathbb{R}^n \to \bar{\mathbb{R}}$ be convex and subdifferentiable, and let $C \subset \mathbb{R}^n$ be a convex set. Then, a point $x^\star \in \mathbb{R}^n$ minimizes f on C if and only if there exists a subgradient $g \in \partial f(x^\star)$, such that

$$\langle g, x - x^\star \rangle \geq 0$$

for all $x \in C$. In particular, if $C = \mathbb{R}^n$, then $x^\star$ is a (global) minimizer of f if and only if

$$0 \in \partial f(x^\star).$$

Definition 2.49 (Strongly convex function)**.** Let $f : \mathbb{R}^n \to \bar{\mathbb{R}}$ be subdifferentiable and let $\mu \in (0, \infty)$. We say that f is (μ-)**strongly convex** if $\operatorname{dom}(f)$ is convex and

$$f(y) \geq f(x) + \langle \tilde{\nabla} f(x), y - x \rangle + \frac{\mu}{2} \|x - y\|^2$$

for all $x, y \in \operatorname{dom}(f)$.

Clearly, strongly convex functions are convex.

Proposition 2.50. A function $f\colon \mathbb{R}^n \to \bar{\mathbb{R}}$ is μ-strongly if and only if there exists a convex function $g\colon \mathbb{R}^n \to \bar{\mathbb{R}}$ such that $f(x) = g(x) + \frac{\mu}{2}\|x\|^2$ for all $x \in \mathbb{R}^n$.

Proposition 2.51. Let $f\colon \mathbb{R}^n \to \bar{\mathbb{R}}$ be μ-strongly convex, and let $C \subset \mathbb{R}^n$ be a convex set. If $x^\star \in C$ minimizes f on C, it then holds that

$$f(x^\star) \leq f(x) - \frac{\mu}{2}\|x^\star - x\|^2$$

for all $x \in C$.

Proposition 2.52 (Convex functions have monotone gradients)**.** Let $f\colon \mathbb{R}^n \to \bar{\mathbb{R}}$ be convex and subdifferentiable. Then it holds that

$$\langle \tilde{\nabla} f(y) - \tilde{\nabla} f(x), y - x \rangle \geq 0$$

for all $x, y \in \operatorname{dom}(f)$. If, additionally, $f\colon \mathbb{R}^n \to \bar{\mathbb{R}}$ is μ-strongly convex for some $\mu \in (0, \infty)$, it holds that

$$\langle \tilde{\nabla} f(y) - \tilde{\nabla} f(x), y - x \rangle \geq \frac{\mu}{2}\|x - y\|^2$$

for all $x, y \in \operatorname{dom}(f)$.

Definition 2.53 (Level sets)**.** Let $f\colon \mathbb{R}^n \to \bar{\mathbb{R}}$ be an extended real-valued function. We say that f has **bounded lower-level sets** if the sets

$$\{\, x \in \mathbb{R}^n \mid f(x) \leq c \,\}$$

are bounded for all $c \in \mathbb{R}$.

Proposition 2.54. Let $f\colon \mathbb{R}^n \to \bar{\mathbb{R}}$ be proper, convex, and continuous, and let $C \subset \mathbb{R}^n$ be a closed convex set. If f has bounded lower level sets, then f attains a minimizer on C, and the set of minimizers of f on C is compact and convex.

Proposition 2.55. Strongly convex functions have bounded lower-level sets and attain a unique minimizer on any nonempty closed convex set. In particular, strongly convex functions defined on $\mathbb{R}^n$ attain a unique (global) minimizer on $\mathbb{R}^n$.

Definition 2.56 (Projection operator)**.** Let $C \subset \mathbb{R}^n$ be a nonempty closed convex set. The **(euclidian) projection operator onto C** is the map $\Pi_C\colon \mathbb{R}^n \to C$, defined as

$$\Pi_C(x) := \min_{y \in C} \|x - y\|$$

for $x \in \mathbb{R}^n$.

Proposition 2.57 (Projection is Lipschitz)**.** For a nonempty closed convex set $C \subset \mathbb{R}^n$, the projection operator Π_C is 1-Lipschitz continuous, that is

$$\|\Pi_C(x) - \Pi_C(y)\| \leq \|x - y\|$$

for all $x, y \in \mathbb{R}^n$.

Proposition 2.58 (Projection is convex)**.** For a nonempty closed convex set $C \subset \mathbb{R}^n$, the projection operator Π_C is convex.

Definition 2.59 (Distance functional)**.** For a nonempty set $C \subset \mathbb{R}^n$, the **(euclidian) distance functional** $\operatorname{dist}(\cdot, C)\colon \mathbb{R}^n \to [0, \infty)$ is defined as

$$\operatorname{dist}(x, C) := \inf_{y \in C} \|x - y\|.$$

Proposition 2.60 (Distance functional is convex)**.** For a nonempty convex set $C \subset \mathbb{R}^n$, the distance functional $\operatorname{dist}(\cdot, C)$ is convex. If, additonally, C is closed, it holds that $\operatorname{dist}(x, C) = \|x - \Pi_C(x)\|$ for all $x \in \mathbb{R}^n$.

2.4.3. Stochastic optimization

Throughout this subsection, we fix an arbitrary probability space $(\Omega, \mathcal{F}, \mathbb{P})$. The main reference for this section is [13].

Definition 2.61. Let $f\colon \mathbb{R}^n \to \bar{\mathbb{R}}$ be a proper and convex function. A random vector $g\colon \Omega \to \mathbb{R}^n$ is called a **stochastic (sub-)gradient of f at** $x \in \mathbb{R}^n$, if

$$\mathbb{E}(g) \in \partial f(x).$$

We note that our definition of stochastic gradients — in particular, the requirement that they are unbiased subgradient estimates — is not standard, and other literature on stochastic optimization might use a different definition.

Example 2.62 (Per-sample stochastic gradient)**.** Let $F\colon \mathbb{R}^n \times \Omega \to \mathbb{R}$ be continuously differentiable in its first argument and assume that

- $\mathbb{E}|F(x, \cdot)| < \infty$ for all $x \in \mathbb{R}^n$, and
- there exists an integrable random variable $G\colon \Omega \to \mathbb{R}^n$ such that $\|\nabla F(x, \cdot)\| \leq G$ almost surely, where the gradient is taken with respect to the first argument.

Algorithm 1 Stochastic Gradient Descent (SGD)

Require: Initial point $x_0 \in \mathbb{R}^n$, step sizes $(\eta_k)_{k\in\mathbb{N}_0}$, number of iterations $K \in \mathbb{N}$

```
for k = 0 to K − 1 do
    Sample ξ_k from distribution of ξ
    g ← ∇̃F_{ξ_k}(x_k)
    x_{k+1} ← Π_X(x_k − η_k g)
end for
return x_K
```

Further, let $f(x) := \mathbb{E}(F(x,\cdot))$ for all $x \in \mathbb{R}^n$. Then, for any $x \in \mathbb{R}^n$, the random vector $g_x \colon \Omega \to \mathbb{R}^n$, defined as

$$g_x(\omega) := \nabla F(x,\omega),$$

is a stochastic gradient of f at x. This follows from proposition 2.16, which lets us interchange the gradient and expectation operators.

A **convex stochastic optimization problem** has the form

$$\min_{x\in\mathcal{X}} \left\{ f(x) := \mathbb{E}(F_\xi(x)) \right\}, \tag{2.1}$$

where $\emptyset \neq \mathcal{X} \subset \mathbb{R}^n$, $\xi \colon \Omega \to \mathbb{R}^m$ is a random vector, and $F \colon \mathbb{R}^n \times \mathbb{R}^m \to \bar{\mathbb{R}}$ is a proper function that satisfies

- $x \to F_{\xi(\omega)}(x)$ is convex for almost every $\omega \in \Omega$, and
- $\omega \to F_{\xi(\omega)}(x)$ is measurable for all $x \in \mathbb{R}^n$.

Under those assumptions, the problem is well-defined and f is convex. If, additionally, f is subdifferentiable, then a standard method to solve such a problem is **stochastic gradient descent**, which is described in algorithm 1.

The convergence of the iterates $(x_k)_{k\in\mathbb{N}}$, generated by algorithm 1, to a solution of (2.1), depends heavily on the choice of step sizes $(\eta_k)_{k\in\mathbb{N}}$ and the variance of the stochastic gradients $\tilde{\nabla} F_\xi(x)$. A classical result by Robbins and Monro [2] states: If f is strongly convex and there exists a constant $M^2 \in (0,\infty)$ such that $\sup_{x\in\mathbb{R}^n} \mathbb{E}\|\tilde{\nabla} F_\xi(x)\|^2 \leq M^2$, then the step-size conditions

$$\sum_{k=0}^{\infty} \eta_k = \infty \quad \text{and} \quad \sum_{k=0}^{\infty} \eta_k^2 < \infty,$$

ensure that algorithm 1 converges to the minimum of f on $\mathcal{X}$.

In many applications (such as the one we care about, see (P)), the objective function in (2.1) exhibits special structure, which can be exploited in algorithm

design. This often leads to problems that can be formulated as

$$\min_{x \in \mathcal{X}} \left\{ f(x) := \mathbb{E}(F_\xi(x)) + r(x) \right\}, \tag{2.2}$$

where now the objective consists of a differentiable part $F\colon \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}$, and a potentially less well-behaved part $r\colon \mathbb{R}^n \to \bar{\mathbb{R}}$. For example, the function r can be the ℓ_1-regularizer $x \mapsto \lambda\|x\|_1$ to promote sparsity, which is a non-differentiable operation at $x = 0$. Another popular choice is to use r to encode some constraints of the form $x \in S \subset \mathbb{R}^n$ directly into f, by letting

$$r(x) := \begin{cases} 0, & \text{if } x \in S, \\ \infty, & \text{else.} \end{cases}$$

To deal with such objectives effectively, we use the following concept.

Definition 2.63 (Proximity operator)**.** Let $r\colon \mathbb{R}^n \to \bar{\mathbb{R}}$ be proper, convex, and continuous, and let $\eta \in (0, \infty)$. The **proximity operator** $\operatorname{prox}_{\eta r}\colon \mathbb{R}^n \to \mathbb{R}^n$ is defined as

$$\operatorname{prox}_{\eta r}(x) := \operatorname*{arg\,min}_{y \in \mathbb{R}^n} \left\{ r(x) + \frac{1}{2\eta}\|x - y\|^2 \right\}$$

for $x \in \mathbb{R}^n$.

Note that, under the stated assumptions, the minimization problem always has a unique solution, since the objective $x \mapsto r(x) + \frac{1}{2\eta}\|x - y\|^2$ is strongly convex (propositions 2.50 and 2.55). A property that will be useful for us is the following.

Proposition 2.64 (Prox is Lipschitz continuous)**.** Let $r\colon \mathbb{R}^n \to \bar{\mathbb{R}}$ be proper, convex, and continuous, and let $\eta \in (0, \infty)$. Then the proximity operator $\operatorname{prox}_{\eta r}\colon \mathbb{R}^n \to \mathbb{R}^n$ is 1-Lipschitz, i. e.

$$\|\operatorname{prox}_{\eta r}(x) - \operatorname{prox}_{\eta r}(y)\| \leq \|x - y\|$$

for all $x, y \in \mathbb{R}^n$.

We can now describe the **stochastic proximal gradient method** for solving problems of the form (2.2) in algorithm 2. Our algorithm, which we will analyze in chapter 3, is also a proximal gradient method.

2.4.4. Multifunctions and metric regularity

The following definitions and results will be used in the proof of theorem 3.6. The contents are based on [27]. For further reading, we refer the reader to [28].

Algorithm 2 Stochastic Proximal Gradient Descent (SPGD)

Require: Initial point $x_0 \in \mathbb{R}^n$, step sizes $(\eta_k)_{k\in\mathbb{N}_0}$, number of iterations $K \in \mathbb{N}$

```
for k = 0 to K − 1 do
    Sample ξ_k from distribution of ξ
    g ← ∇̃F_{ξ_k}(x_k)
    x_{k+1} ← prox_{η_k r}(x_k − η_k g)
end for
return x_K
```

Definition 2.65 (Multifunction)**.** Let X and Y be Banach spaces. A function $\Psi\colon X \to 2^Y$ is called **multifunction**. The **domain** (denoted $\operatorname{dom}(\Psi)$) and the **range** (denoted $\operatorname{range}(\Psi)$) of a multifunction $\Psi\colon X \to 2^Y$ are defined as

$$\begin{aligned} \operatorname{dom}(\Psi) &:= \{\, x \in X \mid \Psi(x) \neq \emptyset \,\} \quad \text{and} \\ \operatorname{range}(\Psi) &:= \{\, y \in Y \mid y \in \Psi(x) \text{ for some } x \in X \,\}. \end{aligned}$$

The **graph** of Ψ is

$$\operatorname{graph}(\Psi) := \{\, (x, y) \in X \times Y \mid y \in \Psi(x), x \in X \,\}.$$

The **(graph) inverse** $\Psi^{-1}\colon Y \to 2^X$ of Ψ is defined as

$$\Psi^{-1}(y) := \{\, x \in X \mid y \in \Psi(x) \,\}.$$

The multifunction Ψ is said to be **closed at a point** $x \in X$ if, for sequences $(x_k)_{k\in\mathbb{N}} \subset X$ and $(y_k)_{k\in\mathbb{N}} \subset Y$, the relations $\lim_{k\to\infty} x_k = x$, $y_k \in \Psi(x_k)$ $\forall k \in \mathbb{N}$, and $\lim_{k\to\infty} y_k = y$ imply that $y \in \Psi(x)$. If Ψ is closed at all points $x \in X$, then Ψ is said to be **closed**. Finally, it is said that Ψ is **convex** if, for any $x_1, x_2 \in X$ and $t \in [0, 1]$,

$$t\Psi(x_1) + (1 - t)\Psi(x_2) \subset \Psi(tx_1 + (1 - t)x_2).$$

Definition 2.66 (Metric regularity)**.** We say that the multifunction $\Psi\colon X \to 2^Y$ is **metric regular** at a point $(x_0, y_0) \in \operatorname{graph}(\Psi)$ at rate $c \in \mathbb{R}_{\geq 0}$, if there exists a neighborhood $U \subset X \times Y$ containing (x_0, y_0), such that

$$\operatorname{dist}(x, \Psi^{-1}(y)) \leq c \operatorname{dist}(y, \Psi(x))$$

for all $(x, y) \in U$.

Proposition 2.67 (Robinson-Ursescu stability theorem)**.** Let $\Psi\colon X \to 2^Y$ be a closed convex multifunction. Then Ψ is metric regular at $(x_0, y_0) \in \operatorname{graph}(\Psi)$ if and only if

the regularity condition $y_0 \in \operatorname{int}(\operatorname{range} \Psi)$ holds.

3

Stochastic Inexact Penalty Method

In this chapter we will analyze the convergence properties of our method, illustrated in algorithm 3, applied to the constrained stochastic optimization problem

$$\begin{aligned} &\min_{x \in \mathbb{R}^n} \{ f(x) := \mathbb{E}(F_\xi(x)) + r(x) \} \\ &\text{subject to} \quad g(x, \xi) \leq 0 \quad \text{almost surely.} \end{aligned} \tag{P}$$

The algorithm proceeds as follows. We start from an initial point $x_0 \in \operatorname{dom}(f)$. In the beginning of each iteration $k \in \mathbb{N}_0$, we sample from the distribution of ξ. Then, based on this sample, we calculate a stochastic gradient (see example 2.62), g_k, of the **penalized objective**

$$x \mapsto \mathbb{E}(F_\xi(x)) + \gamma_k \pi_k(x),$$

at the current iterate x_k. Here, $\gamma_k \in (0, \infty)$ and

$$\pi_k(x) := \mathbb{E}(h_k(g(x, \xi))) \quad \forall x \in \mathbb{R}^n$$

with $(h_k)_{k \in \mathbb{N}_0}$ a sequence of functions that we will specify the properties of later. Following that, we take a standard gradient step with step size $\eta_k \in (0, \infty)$ on the current iterate, followed by an application of the proximity operator:

$$x_{k+1} := \operatorname{prox}_{\eta_k r}(x_k - \eta_k g_k),$$

which yields the next iterate x_{k+1}. After a set amount of $K \in \mathbb{N}$ iterations of this proceedure, the algorithm computes a weigthed average of the iterates $x_1, \ldots, x_K$. This yields $\bar{x}_K$, the output of algorithm 3.

It should be noted that, instead of sampling a single realization of ξ in each iteration, one could instead sample a minibatch of them, that is, take multiple sam-

Algorithm 3 Stochastic Inexact Penalty Method

Require: Initial point $x_0 \in \operatorname{dom}(f)$, step sizes $(\eta_k)_{k\in\mathbb{N}_0}$, penalty weights $(\gamma_k)_{k\in\mathbb{N}_0}$, function sequence $(h_k)_{k\in\mathbb{N}_0}$, iterate weights $(w_k)_{k\in\mathbb{N}}$, maximum iterations $K \in \mathbb{N}$

```
for k = 0 to K − 1 do
    Sample ξ_k from the distribution of ξ
    g_k ← ∇F_{ξ_k}(x_k) + γ_k ∇(h_k ∘ g)(x, ξ)
    x_{k+1} ← prox_{η_k r}(x_k − η_k g_k)
end for
S_K ← Σ_{k=1}^{K} w_k
x̄_K ← S_K^{−1} Σ_{k=1}^{K} w_k x_k
return x̄_K
```

ples, and average over the resulting stochastic gradients. This would lead to lower variance in the gradient estimates, which is desirable. However, in order to reduce notational clutter we choose to omit minibatches from the presentation of the algorithm and its theoretical analysis.

We will start with some technical preliminaries and first assumptions to set the stage for the subsequent analysis. Then, in section 3.2, we will establish the very important result that we can choose the parameters of algorithm 3 in a way that guarantees that the sequence of iterates $(x_k)_{k\in\mathbb{N}_0}$ is bounded with probability 1. Based on this result, we can then move on to prove convergence of the iterates $(x_k)_{k\in\mathbb{N}_0}$ to the solution of problem (P) in sections 3.3 and 3.4.

3.1. Technical preliminaries

We let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space on which the random variable $\xi\colon \Omega \to \mathbb{R}^p$, as well as the expected value mapping, $\mathbb{E}(\cdot)$, are defined. Further, we endow the probability space with a filtration $(\mathcal{F}_k)_{k\in\mathbb{N}}$, defined by

$$\mathcal{F}_k := \sigma(\xi_0, \dots, \xi_{k-1}),$$

where $(\xi_k)_{k\in\mathbb{N}_0}$ is a sequences of random variables such that

$$\xi_0, \xi_1, \xi_2, \dots$$

are i. i. d. samples from $\mathbb{P}^\xi$. We denote the conditional expectation given $\mathcal{F}_k$ as

$$\mathbb{E}_k(X) := \mathbb{E}(X \mid \mathcal{F}_k)$$

for all $k \in \mathbb{N}$. Similarly, we write

$$\mathrm{Var}_k(X) := \mathbb{E}_k \|X - \mathbb{E}_k(X)\|^2$$

for the conditional variance given $\mathcal{F}_k$, for all $k \in \mathbb{N}$. Note that the sequence of iterates $(x_k)_{k\in\mathbb{N}}$, generated by algorithm 3, is adapted to $(\mathcal{F}_k)_{k\in\mathbb{N}}$ and thus $\mathbb{E}_k(x_k) = x_k$ for all $k \in \mathbb{N}$. We denote the feasible set for problem (P) as

$$\mathcal{X} := \{\, x \in \mathrm{dom}(f) \mid g(x, \xi) \leq 0 \ \text{ a.s.} \,\},$$

and the set of solutions as

$$\mathcal{X}^\star := \{\, x^\star \in \mathrm{dom}(f) \mid f(x^\star) \leq f(x) \ \ \forall x \in \mathcal{X} \,\}.$$

Algorithm 3 works with the function sequence $(h_k)_{k\in\mathbb{N}_0}$, where $h_k \colon \mathbb{R} \to \mathbb{R}$ for all $k \in \mathbb{N}_0$, which induces the sequence of penalty functions $(\pi_k)_{k\in\mathbb{N}_0}$ given by

$$\pi_k(x) = \mathbb{E}(h_k(g(x, \xi)))$$

for $x \in \mathbb{R}^n$ and $k \in \mathbb{N}_0$. The penalty functions $(\pi_k)_{k\in\mathbb{N}_0}$, in turn, induce the sequence of unconstrained (penalized) problems

$$\min_{x\in\mathbb{R}^n} \left\{ f_k(x) := f(x) + \gamma_k \pi_k(x) \right\} \tag{P^k}$$

for $k \in \mathbb{N}_0$, where $(\gamma_k)_{k\in\mathbb{N}_0}$ is a sequence of positive real numbers. The properties of $(h_k)_{k\in\mathbb{N}_0}$ and $(\pi_k)_{k\in\mathbb{N}_0}$ are specified below in assumption 1.

For $k \in \mathbb{N}_0$, we define the solution set of problem (P^k) as

$$\mathcal{X}_k^\star := \{\, x_k^\star \in \mathrm{dom}(f) \mid f_k(x_k^\star) \leq f(x) \ \ \forall x \in \mathbb{R}^n \,\}.$$

Since proximal methods separate the smooth part of the above objective, given by $x \mapsto \mathbb{E}(F_\xi(x)) + \gamma_k \pi_k(x)$, from the possibly nonsmooth part, $x \mapsto r(x)$, it is useful to also define the functions

$$\psi_k(x) := \mathbb{E}(F_\xi(x)) + \gamma_k \pi_k(x)$$

for all $k \in \mathbb{N}_0$.

We will denote the iterates of algorithm 3 by $(x_k)_{k\in\mathbb{N}_0}$. The parameters $(w_k)_{k\in\mathbb{N}_0}$ are used to compute the final output — a weighted average of the iterates of the first

$K \in \mathbb{N}$ iterations — given by

$$\bar{x}_K := S_K^{-1} \sum_{k=1}^{K} w_k x_k,$$

where

$$S_K := \sum_{k=1}^{K} w_k.$$

Since this is a convex combination,[1] it follows that $\bar{x}_K \in \operatorname{dom}(f)$ if f is convex and $(x_k)_{k\in\mathbb{N}} \subset \operatorname{dom}(f)$.

In the analysis of algorithm 3, we will need to control the variance of the error introduced by using stochastic gradients. We define the **stochastic error of the k-th stochastic gradient** of algorithm 3, as

$$z_k := \nabla\psi_k(x_k) - g_k \tag{3.1}$$

for $k \in \mathbb{N}_0$. Note that $\mathbb{E}_k(z_k) = 0$, by definition of g_k and the aforementioned fact that x_k is $\mathcal{F}_k$-measurable.

To ensure that the the problems (P) and (P^k) are well-behaved enough to analyze convergence of algorithm 3, we make the following basic assumptions.

Assumption 1 (Basic assumptions)**.** Problem (P) satisfies the following:

1. The function $x \mapsto F_\xi(x)$ is almost surely L-smooth for some $L \in (0, \infty)$, and there exists a point $x \in \mathbb{R}^n$ such that $\mathbb{E}\|\nabla F_\xi(x)\|^2 < \infty$. Further, the expectation $x \mapsto \mathbb{E}(F_\xi(x))$ is μ-strongly convex for some $\mu \in [0, \infty)$. If $\mu = 0$, we additionally assume that f has bounded lower-level sets.

2. The function $r\colon \mathbb{R}^n \to \bar{\mathbb{R}}$ is proper, convex, and Lipschitz continuous on $\operatorname{dom}(r)$ with constant $L_r \in (0, \infty)$.

3. The sequence $(\gamma_k)_{k\in\mathbb{N}_0}$ is nondecreasing and unbounded.

4. The iterate weights $(w_k)_{k\in\mathbb{N}}$ are nonnegative and $w_k > 0$ for at least one $k \in \{1, \ldots, K\}$, where $K \in \mathbb{N}$ is the number of iterations of algorithm 3.

5. There exists at least one feasible point.

Notice that **we allow the strong convexity constant to take the value zero**, because it allows for a unified treatment of both strongly convex and convex objectives.

[1] As long as $w_1, \ldots, w_K$ are nonnegative and at least one w_k is positive.

That is, the statement "f is μ-strongly convex with $\mu \in [0, \infty)$" means "f is either convex or μ-strongly convex with $\mu \in (0, \infty)$". For brevity, we will typically just say "f is μ-strongly convex", even if $\mu = 0$ is allowed. In cases where we need $\mu > 0$, we will make that clear. If $\mu = 0$, the additional assumption of bounded lower-level sets ensures that the family of sets $(\mathcal{X}_k^\star)_{k\in\mathbb{N}_0}$ is nonempty and uniformly bounded (see lemma 3.2). A simple way to guarantee bounded lower-level sets in the (non-strongly) convex case is to choose r such that $\mathrm{dom}(r)$ is compact.

Next, we make some assumptions about the constraints in problem (P).

Assumption 2. In the setting of problem (P), the constraint map $g\colon \mathbb{R}^n \times \mathbb{R}^p \to \mathbb{R}$ satisfies the following conditions:

- The function $\omega \mapsto g(x, \xi(\omega))$ is measurable for all $x \in \mathbb{R}^n$, and
- the function $x \mapsto g(x, \xi)$ is almost surely convex and continuously differentiable.

Further, we assume that the gradient $x \mapsto \nabla g(x, \xi)$ is almost surely bounded on $\mathrm{dom}(f)$. That is, we assume that there exists a constant $M \in (0, \infty)$ such that

$$\sup_{x\in\mathrm{dom}(f)} \|\nabla g(x, \xi)\| \leq M \text{ almost surely,}$$

where, for $z \in \mathbb{R}^p$, we let $\nabla g(x, z)$ denote the gradient of the function $x \mapsto g(x, z)$, evaluated at $x \in \mathbb{R}^n$.

A sufficient condition for the last assumption is that $(x, z) \mapsto \nabla g(x, z)$ is continuous, ξ has compact support, and $\mathrm{dom}(r)$ is a compact subset of $\mathbb{R}^n$. The third holds for many applications, for example, when x represents a spent amount of some limited budget.

Next, we make some assumptions about the penalty sequence of algorithm 3.

Assumption 3. The functions $(\pi_k)_{k\in\mathbb{N}_0}$ in (P^k) have the form

$$\pi_k(x) := \mathbb{E}(h_k(g(x, \xi)))$$

for all $k \in \mathbb{N}_0$, where, for all $k \in \mathbb{N}_0$, we let $(h_k)_{k\in\mathbb{N}_0}$ be any sequence of functions from $\mathbb{R}$ to $\mathbb{R}$ with the following properties:

1. For all $k \in \mathbb{N}_0$, h_k is convex, continuously differentiable, and

$$\mathbb{E}|h_k(g(x, \xi))| < \infty$$

   for all $x \in \mathbb{R}^m$.

2. There exists a constant $s \in (0, \infty)$ such that, for all $x \in \mathbb{R}^n$ and $k \in \mathbb{N}_0$, it holds that $\pi_k(x) \geq s\,\mathbb{E}((g(x,\xi))_+)$.

3. There exists a sequence $(\alpha_k)_{k\in\mathbb{N}_0}$ such that $\pi_k(\hat{x}) \leq \alpha_k$ and $\sup_{k\in\mathbb{N}_0} \gamma_k \alpha_k < \infty$ for all $k \in \mathbb{N}_0$ and all feasible points $\hat{x} \in \mathcal{X}$.

4. The gradients of $(h_k)_{k\in\mathbb{N}_0}$ have uniformly bounded derivatives:

$$\sup_{k\in\mathbb{N}_0} \sup_{x\in\mathbb{R}} |h_k'(x)| < \infty.$$

By the chain rule, assumptions 2 and 3 together imply the following: For all $k \in \mathbb{N}_0$, the concatenation $h_k \circ g$ is continuously differentiable in the first argument, and there exists a constant $G \in (0, \infty)$ such that

$$\sup_{x\in\mathrm{dom}(f)} \|\nabla(h_k \circ g)(x,\xi)\| \leq G \text{ almost surely,} \tag{3.2}$$

where, for $z \in \mathbb{R}^p$ and $k \in \mathbb{N}_0$, we let $\nabla(h_k \circ g)(x,z)$ denote the gradient of the function $x \mapsto (h_k \circ g)(x,z)$, evaluated at $x \in \mathbb{R}^n$.

Before we proceed, we will present an example of a penalty sequence that satisfies assumption 3. We will refer back to this example in our numerical experiments in chapter 4.

Example 3.1 (Huber-type penalty)**.** Let $(\delta_k)_{k\in\mathbb{N}_0}$ be a nonincreasing sequence of positive real numbers with $\lim_{k\to\infty} \delta_k = 0$. For $x \in \mathbb{R}$, we define

$$h_k^{\mathrm{hub}}(x) := \begin{cases} x & \text{if } x > \delta_k, \\ \frac{(x+\delta_k)^2}{4\delta_k} & \text{if } -\delta_k \leq x \leq \delta_k, \\ 0, & \text{if } x < -\delta_k, \end{cases}$$

for all $k \in \mathbb{N}_0$. The sequence $(h_k^{\mathrm{hub}})_{k\in\mathbb{N}_0}$ is a sequence of convex and continuously differentiable functions. For all $k \in \mathbb{N}_0$, it holds that $h_k^{\mathrm{hub}}(x) \geq (x)_+$ for all $x \in \mathbb{R}$ with $x \geq 0$, and

$$h_k^{\mathrm{hub}}(x) \leq \frac{\delta_k}{4}$$

for all $k \in \mathbb{N}_0$ and $x \in \mathbb{R}$ with $x \leq 0$. Further, it holds that

$$(h_k^{\mathrm{hub}})'(x) = \begin{cases} 1, & \text{if } x > \delta_k, \\ \frac{x+\delta_k}{2\delta_k}, & \text{if } -\delta_k \leq x \leq \delta_k, \\ 0, & \text{if } x < -\delta_k, \end{cases}$$

for all $k \in \mathbb{N}_0$ and $x \in \mathbb{R}$, so, in particular,

$$|(h_k^{\text{hub}})'(x)| \leq 1$$

for all $k \in \mathbb{N}_0$ and $x \in \mathbb{R}$. We define the sequence of functions $(\pi_k^{\text{hub}})_{k\in\mathbb{N}_0}$ by

$$\pi_k^{\text{hub}}(x) := \mathbb{E}(h_k(g(x,\xi)))$$

for all $k \in \mathbb{N}_0$ and $x \in \mathbb{R}^n$. It follows that $(\pi_k^{\text{hub}})_{k\in\mathbb{N}_0}$ and $(h_k^{\text{hub}})_{k\in\mathbb{N}_0}$ satisfy all conditions of assumption 3 with a majorizing sequence $(\alpha_k)_{k\in\mathbb{N}_0}$ given by

$$\alpha_k := \frac{\delta_k}{4}.$$

This concludes the example.

Assumptions 1 to 3 have multiple useful implications, which are captured by the following lemma.

Lemma 3.2 (Basic properties)**.** Assumptions 1 to 3 imply the following:

1. The gradients of $(\pi_k)_{k\in\mathbb{N}_0}$ are uniformly bounded on $\text{dom}(f)$: There exists $G \in (0,\infty)$ such that
$$\sup_{k\in\mathbb{N}_0} \sup_{x\in\text{dom}(f)} \|\nabla\pi_k(x)\| \leq G.$$
In particular, π_k is Lipschitz continuous for all $k \in \mathbb{N}$.

2. The objective f of (P) is μ-strongly convex with $\mu \in [0,\infty)$, locally Lipschitz continuous on $\text{dom}(f)$, and subdifferentiable on $\text{dom}(f)$. Further, f has bounded lower-level sets.

3. The objectives $(f_k)_{k\in\mathbb{N}_0}$ of (P^k) are μ-strongly convex with $\mu \in [0,\infty)$, locally Lipschitz continuous on $\text{dom}(f_k)$, and subdifferentiable on $\text{dom}(f_k)$. Further, f_k has bounded lower-level sets for all $k \in \mathbb{N}_0$.

4. The functions $(\psi_k)_{k\in\mathbb{N}_0}$ are continuously differentiable and μ-strongly convex with $\mu \in [0,\infty)$.

5. Let $B \subset \text{dom}(f)$ be a bounded subset. Then, for all $k \in \mathbb{N}_0$, the map $g_k\colon \mathbb{R}^n \times \Omega \to \mathbb{R}^n$, defined as
$$g_k(x,\omega) := \nabla F_{\xi(\omega)}(x) + \gamma_k \nabla(h_k \circ g)(x,\xi(\omega)),$$

for $x \in \mathbb{R}^n$, $\omega \in \Omega$, is a stochastic gradient of ψ_k at $x \in \mathbb{R}^n$. Further, g_k satisfies

$$\sup_{x \in B} \|g_k(x, \cdot)\| = \mathcal{O}(\gamma_k)$$

almost surely, for all $k \in \mathbb{N}_0$. More specifically, there exists a constant $C \in (0, \infty)$ such that

$$\limsup_{k \to \infty} \frac{\|g_k(x, \omega)\|}{\gamma_k} \leq C$$

for all $x \in B$ and $\mathbb{P}$-almost every $\omega \in \Omega$.

6. The set $\mathcal{X}^\star$ is nonempty, compact, and convex. For all $k \in \mathbb{N}_0$, the set $\mathcal{X}_k^\star$ is nonempty, compact, and convex, and the family $(\mathcal{X}_k^\star)_{k \in \mathbb{N}_0}$ is uniformly bounded. In particular, the map $x \mapsto \Pi_{\mathcal{X}}(x)$ is well-defined and bounded when restricted to the set $\cup_{k=0}^{\infty} \mathcal{X}_k^\star$.

7. The iterates $(x_k)_{k \in \mathbb{N}}$ of algorithm 3 are all included in $\mathrm{dom}(f)$. In particular, $\bar{x}_K \in \mathrm{dom}(f)$ for all $K \in \mathbb{N}$.

Proof. 1. Let $k \in \mathbb{N}_0$. By assumptions 2 and 3 and proposition 2.16, we are allowed to exchange differentiation and integration as follows:

$$\nabla \pi_k(x) = \nabla \mathbb{E}(h_k(g(x, \xi))) = \mathbb{E}(\nabla (h_k \circ g)(x, \xi))$$

for all $x \in \mathbb{R}^n$. Hence, proposition 2.13 implies

$$\|\nabla \pi_k(x)\| \leq \mathbb{E}\|\nabla (h_k \circ g)(x, \xi)\|.$$

The claim now follows from (3.2) with Lipschitz continuity following from proposition 2.40.

2. By L-smoothness of F_ξ, the expectation $\mathbb{E}(F_\xi)$ is also L-smooth, thus locally Lipschitz. By Lipschitz continuity of r on $\mathrm{dom}(r)$, it follows that $f = \mathbb{E}(F_\xi) + r$ is also locally Lipschitz continuous on $\mathrm{dom}(r) = \mathrm{dom}(f)$. Similarly, as the sum of a μ-strongly convex function and a convex function, f is μ-strongly convex. The function r is proper, convex, and Lipschitz continuous on $\mathrm{dom}(r)$, which implies that it is subdifferentiable on $\mathrm{dom}(r)$, by proposition 2.45. Since $x \mapsto \mathbb{E}(F_\xi(x))$ is differentiable, it follows that f is subdifferentiable on $\mathrm{dom}(r) = \mathrm{dom}(f)$, by proposition 2.47. Finally, f has bounded lower-level sets per part 1 of assumption 1, if $\mu = 0$, and by properties of strongly convex functions, if $\mu > 0$ (see proposition 2.55).

3. By the above, f is μ-strongly convex, Lipschitz continuous on $\mathrm{dom}(f)$, and subdifferentiable on $\mathrm{dom}(f)$. Combining this with the fact that π_k is convex, Lipschitz continuous, and differentiable, as well as $\gamma_k \in (0, \infty)$, it follows that $f_k = f + \gamma_k \pi_k$ is also μ-strongly convex, Lipschitz continuous on $\mathrm{dom}(f)$, and subdifferentiable on $\mathrm{dom}(f)$, for all $k \in \mathbb{N}_0$. The bounded lower-level sets property follows from the same argument as above. Finally, the claim follows from $\mathrm{dom}(f) = \mathrm{dom}(f_k) \ \ \forall k \in \mathbb{N}_0$.

4. For all $k \in \mathbb{N}_0$, ψ_k is the sum of two differentiable functions, one μ-strongly convex, the other convex. Hence, ψ_k is μ-strongly convex and differentiable for all $k \in \mathbb{N}_0$.

5. Let $k \in \mathbb{N}_0$. The claim that $g_k(x, \cdot)$ is a stochastic gradient of ψ_k at $x \in \mathbb{R}^n$ follows from example 2.62. Let $B \subset \mathrm{dom}(f)$ be bounded. By the triangle inequality, we have
$$\|g_k(x)\| \leq \|\nabla F_\xi(x)\| + \gamma_k \|\nabla(h_k \circ g)(x, \xi)\|$$
for all $x \in B$. The claim now follows from the fact that F_ξ is almost surely locally Lipschitz continuous (see proposition 2.40), and (3.2).

6. As we have already shown, f is convex and has bounded lower-level sets. The set $\mathcal{X}$ is the intersection of the closed convex sets $\{\, x \in \mathrm{dom}(f) \mid g(x, z) \leq 0 \,\}$ over z in the support of ξ. Thus, $\mathcal{X}$ is itself closed and convex. Combining with the bounded lower-level sets property of f, proposition 2.54 implies that f attains a minimizer on $\mathcal{X}$ and the set of minimizers, $\mathcal{X}^\star$, is compact and convex. For $k \in \mathbb{N}_0$, we have also shown that f_k is convex and has bounded lower-level sets, so the same proposition implies that f_k has a global minimizer, and the set $\mathcal{X}_k^\star$ is compact and convex for all $k \in \mathbb{N}_0$. To prove uniform boundedness of the family $(\mathcal{X}_k^\star)_{k \in \mathbb{N}_0}$, fix any feasible point $\hat{x} \in \mathcal{X}$ (which exists per part 5 of assumption 1). By part 3 of assumption 3, there exists $c \in (0, \infty)$ such that, for all $k \in \mathbb{N}_0$ and all minima $x_k^\star \in \mathcal{X}_k^\star$,
$$f(x_k^\star) \leq f_k(x_k^\star) \leq f_k(\hat{x}) = f(\hat{x}) + \gamma_k \pi_k(\hat{x}) \leq f(\hat{x}) + \gamma_k \alpha_k \leq f(\hat{x}) + c.$$
Hence,
$$\mathcal{X}_k^\star \subset \{\, x \in \mathbb{R}^n \mid f(x) \leq f(\hat{x}) + c \,\} =: f_{\leq f(\hat{x})+c}$$
for all $k \in \mathbb{N}_0$. Coupled with the fact that f has bounded lower-level sets, it follows that the family $(\mathcal{X}_k^\star)_{k \in \mathbb{N}_0}$ is uniformly bounded. In particular, since $f_{\leq f(\hat{x})+c}$ is also closed, by continuity of f, and thus compact, it follows that the

image of the set $f_{\leq f(\hat{x})+c}$ under the continuous map $\Pi_{\mathcal{X}}$ (proposition 2.57) is compact. Therefore, we obtain

$$\sup_{k\in\mathbb{N}} \sup_{x_k^\star \in \mathcal{X}_k^\star} \|\Pi_{\mathcal{X}}(x_k^\star)\| \leq \sup_{x\in f_{\leq f(\hat{x})+c}} \|\Pi_{\mathcal{X}}(x)\| < \infty,$$

as desired.

7. By definition of the proximal operator,

$$x_{k+1} = \arg\min_{x\in\mathbb{R}^n} \left\{ r(x) + \frac{1}{2\eta_k}\|x - (x_k - \eta_k g_k)\|^2 \right\}$$

for all $k \in \mathbb{N}_0$. Since r is proper, it must therefore hold that $x_k \in \mathrm{dom}(r) = \mathrm{dom}(f)$, for all $k \in \mathbb{N}_0$. By definition of $\bar{x}_K$ as a convex combination of the iterates $x_1, \ldots, x_K$, $K \in \mathbb{N}$, and convexity of f, it follows that $\bar{x}_K \in \mathrm{dom}(f)$.

This concludes the proof. □

A key challenge in the analysis of algorithm 3 is the fact that both the generated iterates $(x_k)_{k\in\mathbb{N}_0}$ and the true minima $(x_k^\star)_{k\in\mathbb{N}_0}$ need not be feasible, which prevents the use of standard arguments from the SGD literature to analyse convergence. Our proof methods combine approaches from recent works, mainly the already mentioned [15], as well as [29]. The latter work investigates stochastic optimization problems under *distributional drift*. While these kinds of problems are not exactly the same as the ones we are working with, the two settings do indeed exhibit striking resemblences. Namely, the authors investigate a sequence of time-dependent composite problems of the form

$$\min_{x\in\mathbb{R}^n} g_t(x) + r_t(x),$$

where, for all $t \in \mathbb{N}$, g_t is smooth and strongly convex, and r_t is convex. Comparing to our problem (P^k), rewritten as

$$\min_{x\in\mathbb{R}^n} \left\{ f_k(x) = \psi_k(x) + r(x) \right\}, \tag{P^k}$$

we see that the two settings almost match, except for smoothness properties. Namely, we do not need the family $(\psi_k)_{k\in\mathbb{N}}$ itself to be smooth, but only that it is composed of an L-smooth function and a differentiable one with uniformly bounded gradients. Even though it would be realistic to assume Lipschitz smoothness of the penalties π_k, for typical examples — including the Huber-type penalty from example 3.1 — the smoothness constants blow up for $k \to \infty$. A naive application of the techniques

in Cutler et al. [29] would then lead to a restriction on the step sizes $(\eta_k)_{k\in\mathbb{N}}$ of the form $\eta_k \in (0, 1/L_k)$ with L_k the smoothness constant of ψ_k. With some care however, we manage to circumvent this restriction, as well as the Lipschitz assumption on $\nabla\pi_k$, and only require $\eta_k \in (0, 1/L)$. This allows for a lot more freedom in the choice of step sizes and is ultimately key for deriving the final convergence rates.

3.2. Almost sure convergence

In this section we will establish conditions under which we can guarantee almost sure convergence of the sequence of iterates $(x_k)_{k\in\mathbb{N}_0}$. The proof will also yield convergence in expectation of $(x_k)_{k\in\mathbb{N}_0}$ to $x^\star$ along a subsequence. The main use of the almost sure convergence result will be that we will have conditions on the step sizes $(\eta_k)_{k\in\mathbb{N}_0}$ and the penalty parameters $(\gamma_k)_{k\in\mathbb{N}_0}$ to ensure that $(x_k)_{k\in\mathbb{N}_0}$ is bounded with probability one. This, together with some of the results we prove along the way, will come in very handy in the subsequent analysis of the quantitative convergence rates of our methods.

The proof for almost sure convergence hinges on two technical lemmata. The first is the well-known Robbins-Siegmund lemma, which provides a general sufficient condition to guarantee almost sure convergence of so-called *almost supermartingales*.

Lemma 3.3 (Robbins-Siegmund)**.** Let $(\mathcal{F}_k)_{k\in\mathbb{N}}$ be an increasing sequence of σ-algebras and v_k, a_k, b_k, c_k be nonnegative $\mathcal{F}_k$-measurable random variables. If, for all $k \in \mathbb{N}$,

$$\mathbb{E}(v_{k+1} \mid \mathcal{F}_k) \leq v_k(1 + a_k) + b_k - c_k, \tag{3.3}$$

and $\sum_{k=1}^{\infty} a_k < \infty$, $\sum_{k=1}^{\infty} b_k < \infty$ a. s., then with probability one, $(v_k)_{k\in\mathbb{N}}$ is convergent and it holds that $\sum_{k=1}^{\infty} c_k < \infty$.

Proof. See [16]. □

Our goal for the rest of this section is to derive a recursive inequality for the sequence $\|x_k - x^\star\|^2$, which resembles (3.3). As a first step, we will analyze the convergence of sequences of minimizers $(x_k^\star)_{k\in\mathbb{N}_0}$ with

$$x_k^\star \in \mathcal{X}_k^\star \quad \forall k \in \mathbb{N}_0.$$

In particular, we will establish that $\operatorname{dist}(x_k^\star, \mathcal{X})$ converges to zero for $k \to \infty$, and that this convergence is independent of the sequence $(\gamma_k)_{k\in\mathbb{N}_0}$, as long as one insures that γ_k is eventually large enough. To do this, we will first show that one can locally

bound the distance $\operatorname{dist}(x, \mathcal{X})$ by (a term proportional to) the penalty $\pi_k(x)$ for all $k \in \mathbb{N}_0$. We start by introducing the following assumptions.

Assumption 4. There exists a compact set $\Xi \in \mathbb{R}^p$ such that ξ is supported on Ξ and $\mathbb{P}^\xi$ has a Lebesgue-density q such that

1. q is continuous on Ξ and
2. $q(x) > 0$ for all $x \in \Xi$.

Further, we assume that

$$\int_\Xi |g(x, z)| \, \mathrm{d}z < \infty$$

for all $x \in \mathbb{R}^n$.

A simple sufficient condition for the last part of assumption 4 is continuity of g on Ξ. The other condtions in assumption 4 can always be satisfied if the set Ξ (or a superset thereof) is known, by sampling uniformly. In fact, lemma 3.4 will demonstrate that uniform sampling is optimal,[2] in the sense that it minimizes constants that depend on the distribution of ξ in the derived upper bound. This is very intuitive: since we are working in the setting of almost sure constraints, no realization of ξ is more important than any other. However, it should be noted that this principle only applies if the set Ξ is fixed and cannot be somehow reduced to a smaller set, as ideally one would only need to sample from the set of maximizers of $z \mapsto g(x, z)$ on Ξ. Nevertheless, in the case where such a reduction to a smaller set $\bar{\Xi}$ is possible, the optimal way to sample would be to sample uniformly from $\bar{\Xi}$.

The next assumption is a common one in the optimization literature.

Assumption 5. There exists a *Slater point*, i. e. a point $x \in \mathcal{X}$ and a constant $\rho \in (0, \infty)$ such that $g(x, \xi) \leq -\rho$ almost surely.

We will now formulate and prove the lemma.

Lemma 3.4 (Subfeasibility bound via penalty functions)**.** Let assumptions 2 to 5 hold and let $C \subset \mathbb{R}^n$ be a compact subset. Then there exists a constant $\tau \in (0, \infty)$ such that

$$\tau \operatorname{dist}(x, \mathcal{X}) \leq \pi_k(x) \tag{3.4}$$

for all $x \in C$ and $k \in \mathbb{N}_0$.

[2] Of course, one may not have control over the sampling distribution.

Proof. We consider the Banach space

$$Y := L_1(\Xi, \mathbb{R}) = \left\{ y \colon \Xi \to \mathbb{R} \;\middle|\; \int_\Xi |y(z)| \, \mathrm{d}z < \infty \right\},$$

equipped with the norm

$$\|y\|_Y := \int_\Xi |y(z)| \, dz.$$

The distance map on Y, denoted dist_Y, is then given by

$$\mathrm{dist}_Y(y, S) = \inf_{y' \in S} \|y - y'\|_Y$$

for $y \in Y$ and $S \subset Y$. We define the multifunction $\Psi \colon \mathbb{R}^n \to 2^Y$ as

$$\Psi(x) := \{ \, y \in Y \mid g(x, \xi) \leq y(\xi) \text{ a. s.} \, \}.$$

This multifunction is closed and convex. Indeed, let $(x_k)_{k \in \mathbb{N}} \in \mathbb{R}^n$ be a sequence that converges to some $x \in \mathbb{R}^n$, and assume that there exists a sequence $(y_k)_{k \in \mathbb{N}}$ with $y_k \in \Psi(x_k)$ for all $k \in \mathbb{N}$, such that $\lim_{k \to \infty} y_k = y \in Y$. Then, with probability one,

$$g(x, \xi) = \lim_{k \to \infty} g(x_k, \xi) \leq \lim_{k \to \infty} y_k(\xi) = y(\xi),$$

where we used continuity of $x \mapsto g(x, \xi)$ from assumption 2. Hence $y \in \Psi(x)$, which proves closedness. For convexity, let $x_1, x_2 \in \mathbb{R}^n$ and $t \in [0, 1]$. Then

$$\begin{aligned} t\Psi(x_1) + (1-t)\Psi(x_2) = \{ \, y \mid \; & y = t y_1 + (1-t) y_2 \text{ for } y_1, y_2 \in Y \\ & \text{such that } g(x_1, \xi) \leq y_1(\xi) \text{ a. s.} \\ & \text{and } g(x_2, \xi) \leq y_2(\xi) \text{ a. s.} \, \}. \end{aligned}$$

Let $x := t x_1 + (1-t) x_2$. Then, for any $y \in t\Psi(x_1) + (1-t)\Psi(x_2)$, there exist $y_1, y_2 \in Y$, such that

$$g(x, \xi) \leq t g(x_1, \xi) + (1-t) g(x_2, \xi) \leq t y_1(\xi) + (1-t) y_2(\xi) = y(\xi),$$

where we used convexity of $x \mapsto g(x, \xi)$ from assumption 2. Hence, we have

$$t\Psi(x_1) + (1-t)\Psi(x_2) \subset \Psi(t x_1 + (1-t) x_2),$$

proving convexity. Now let $x_0 \in C$ be some arbitrary point. Per definition of the

inverse Ψ^{-1}, it holds that

$$\Psi^{-1}(0) = \{\, x \in \mathbb{R}^n \mid 0 \in \Psi(x) \,\} = \{\, x \in \mathbb{R}^n \mid g(x,\xi) \leq 0 \text{ a.s.} \,\} = \mathcal{X}.$$

Hence we have $\operatorname{dist}(x,\mathcal{X}) = \operatorname{dist}(x,\Psi^{-1}(0))$. Assumption 5 implies $0 \in \operatorname{int}(\operatorname{range}\Psi)$. Indeed, the constant function $\tilde{y} \equiv -\rho$ is an element of $L_1(\Xi,\mathbb{R})$, thus $\tilde{y} \in \operatorname{range}(\Psi)$, by assumption 5. In particular, $(\tilde{y}/2,\infty) \subset \operatorname{range}(\Psi)$. Since 0 is contained in $(\tilde{y}/2,\infty)$, we can conclude $0 \in \operatorname{int}(\operatorname{range}\Psi)$. Therefore, we can apply proposition 2.67, which guarantees the existence of a constant $c \in \mathbb{R}_{\geq 0}$ such that

$$\operatorname{dist}(x,\mathcal{X}) = \operatorname{dist}(x,\Psi^{-1}(y)) \leq c \operatorname{dist}_Y(y,\Psi(x))$$

for all (x,y) in some open neighborhood $U \subset \mathbb{R}^n \times Y$ containing $(x_0,0)$. In particular,

$$\operatorname{dist}(x,\mathcal{X}) \leq c \operatorname{dist}_Y(0,\Psi(x)) \tag{3.5}$$

for all x in the open set $U \cap (\mathbb{R}^n \times \{0\})$. Since x_0 is arbitrary, we can derive a similar bound that holds around an open neighborhood $U_x \subset \mathbb{R}^n$ of a point $x \in C$, for any $x \in C$, yielding corresponding constants $(c_x)_{x\in C}$. By compactness of C, the open covering

$$C \subset \bigcup_{x\in C} U_x$$

has a finite subcovering

$$C \subset \bigcup_{i=1}^{\ell} U_{x_i}$$

with $(x_i)_{i\in\{1,\dots,\ell\}} \subset C$. The corresponding constants $(c_{x_i})_{i\in\{1,\dots,\ell\}}$ have a maximum $c := \max_{i\in\{1,\dots,\ell\}} c_{x_i}$. Thus, we have shown that there exists $c \in \mathbb{R}_{\geq 0}$ such that

$$\operatorname{dist}(x,\mathcal{X}) \leq c \operatorname{dist}_Y(0,\Psi(x))$$

for all $x \in C$. Next, we will show that

$$\operatorname{dist}_Y(0,\Psi(x)) \overset{\text{def.}}{=} \int_\Xi (g(x,z))_+ \,\mathrm{d}z = \|(g(x,\cdot))_+\|_Y \tag{3.6}$$

for any $x \in C$. Fix $x \in C$ and let $y \in \Psi(x)$. By definition of $\Psi(x)$ and positivity of q on Ξ (recall assumption 4), it holds that $y \in \Psi(x) \iff g(x,z) \leq y(z)$ for all $z \in \Xi$. Set

$$\phi(z) := (g(x,z))_+$$

for all $z \in \Xi$. Clearly, $\phi \in \Psi(x)$. We will show that $|\phi(z)| \leq |y(z)|$ for all $z \in \Xi$ and $y \in \Psi(x)$. There are two cases. If $z \in \Xi$ such that $g(x,z) \leq 0$, then $|\phi(z)| = 0$, and thus $|\phi(z)| \leq |y(z)|$. Otherwise, for $z \in \Xi$ such that $g(x,z) > 0$, we have $|\phi(z)| = g(x,z)$, and so it also holds that $|\phi(z)| \leq |y(z)|$, per definition of the set $\Psi(x)$. Since y is arbitrary, it follows that

$$\int_\Xi \phi(z)\,\mathrm{d}z \leq \inf_{y \in \Psi(x)} \int_\Xi |y(z)|\,\mathrm{d}z = \mathrm{dist}_Y(0, \Psi(x)),$$

proving (3.6). To finish the proof, we need to establish a relationship between (3.6) and $\pi_k(x)$ for all $k \in \mathbb{N}_0$. By compactness of Ξ and continuity of q, the image $q(\Xi)$ must be compact. In particular, since $q(x) > 0$ for all $x \in \Xi$, there must exist some uniform positive lower bound $c_q \in (0, \infty)$ such that $q(x) \geq c_q$ for all $x \in \Xi$. If we denote the Lebesgue-measure by λ, we see that

$$\mathbb{P}^\xi(A) = \int_A q(z)\,\mathrm{d}z \geq c_q \lambda(A)$$

for all measurable $A \subset \Xi$. Therefore, $\mathbb{P}^\xi$ and λ are equivalent on Ξ, hence proposition 2.22 implies that q^{-1} is a $\mathbb{P}^\xi$-density of λ. Using proposition 2.23 and proposition 2.7, we can conclude

$$\begin{aligned}
\|y\|_Y &= \int_{z \in \Xi} |y(z)|\,\mathrm{d}z \\
&= \int_{z \in \Xi} |y(z)| q^{-1}(z)\,\mathbb{P}^\xi(dz) \\
&\leq c_q^{-1} \int_{z \in \Xi} |y(z)|\,\mathbb{P}^\xi(dz) \\
&= c_q^{-1} \int |y(\xi)|\,\mathrm{d}\mathbb{P} \\
&= c_q^{-1}\,\mathbb{E}|y(\xi)|.
\end{aligned}$$

Combining with (3.5) and (3.6), we obtain

$$\mathrm{dist}(x, \mathcal{X}) \leq c c_q^{-1} \mathbb{E}((g(x,\xi))_+).$$

Therefore, using part 2 of assumption 3, there exists $s \in (0, \infty)$ such that

$$\mathrm{dist}(x, \mathcal{X}) \leq c c_q^{-1} s^{-1} \pi_k(x).$$

The claim follows after setting $\tau := c^{-1} c_q s$. □

Notation 3.5. For the rest of this work, we will reserve the letter τ to refer to the constant from the proof of lemma 3.4, which guarantees that the bound (3.4) holds in the given context.

Note that τ^{-1} is proportional to the inverse of the minimum of the density of ξ on its support. For a fixed (bounded) support Ξ, this value is minimized if the density is the uniform density on Ξ. Therefore, if the support Ξ is known, it is optimal to sample uniformly from Ξ for algorithm 3.

We can now prove that, under certain assumptions, we can guarantee that the sequence of minimizers $(x_k^\star)_{k\in\mathbb{N}_0}$ corresponding to the problems (P^k) converge to the feasible set $\mathcal{X}$ with a rate that is controllable via the majorizing sequence $(\alpha_k)_{k\in\mathbb{N}_0}$ (recall part 3 of assumption 3).

Theorem 3.6 (Convergence of $x_k^\star$)**.** Let assumptions 1 to 5 hold and let $x^\star \in \mathcal{X}^\star$. Then there exists a constant $M \in (0,\infty)$, such that

$$\frac{\mu}{2}\|x^\star - x_k^\star\|^2 + \frac{\mu}{2}\|x^\star - \Pi_{\mathcal{X}}(x_k^\star)\|^2 + (\tau\gamma_k - M)\ \mathrm{dist}(x_k^\star, \mathcal{X}) \le \gamma_k\alpha_k$$

for all $k \in \mathbb{N}_0$ and $x_k^\star \in \mathcal{X}_k^\star$. In particular,

$$\mathrm{dist}(x_k^\star, \mathcal{X}) = \mathcal{O}(\alpha_k)$$

and, if $\mu > 0$, we further have

$$\|x^\star - x_k^\star\|^2 = \mathcal{O}(\gamma_k\alpha_k).$$

Proof. Let $k \in \mathbb{N}_0$, $x_k^\star \in \mathcal{X}_k^\star$, and $x^\star \in \mathcal{X}^\star$. By optimality of $x_k^\star$ for f_k, there exists $0 \in \partial f_k(x_k^\star)$. Hence, by strong convexity, we obtain

$$\frac{\mu}{2}\|x^\star - x_k^\star\|^2 \le f_k(x^\star) - f_k(x_k^\star) = f(x^\star) - f(x_k^\star) + \gamma_k\pi_k(x^\star) - \gamma_k\pi_k(x_k^\star). \tag{3.7}$$

We can write

$$f(x^\star) - f(x_k^\star) = f(x^\star) - f(\Pi_{\mathcal{X}}(x_k^\star)) + f(\Pi_{\mathcal{X}}(x_k^\star)) - f(x_k^\star)$$

and use strong convexity to obtain

$$\begin{aligned} f(x^\star) - f(x_k^\star) &= f(x^\star) - f(\Pi_{\mathcal{X}}(x_k^\star)) + f(\Pi_{\mathcal{X}}(x_k^\star)) - f(x_k^\star) \\ &\le -\frac{\mu}{2}\|x^\star - \Pi_{\mathcal{X}}(x_k^\star)\|^2 - \langle \tilde{\nabla} f(x^\star), \Pi_{\mathcal{X}}(x_k^\star) - x^\star\rangle + f(\Pi_{\mathcal{X}}(x_k^\star)) - f(x_k^\star), \end{aligned}$$

for any $\tilde{\nabla} f(x^\star) \in \partial f(x^\star)$. Since $\Pi_{\mathcal{X}}(x_k^\star) \in \mathcal{X}$ and $x^\star$ is optimal for f on $\mathcal{X}$, there exists a subgradient $\bar{\nabla} f(x^\star) \in \partial f(x^\star)$ such that

$$\langle \tilde{\nabla} f(x^\star), \Pi_{\mathcal{X}}(x_k^\star) - x^\star \rangle \geq 0.$$

Thus, we obtain

$$f(x^\star) - f(x_k^\star) \leq -\frac{\mu}{2}\|x^\star - \Pi_{\mathcal{X}}(x_k^\star)\|^2 + f(\Pi_{\mathcal{X}}(x_k^\star)) - f(x_k^\star).$$

By part 6 of lemma 3.2, we know that the family $(\mathcal{X}_k^\star)_{k\in\mathbb{N}_0}$ is uniformly bounded and

$$\sup_{k\in\mathbb{N}_0} \sup_{x_k^\star \in \mathcal{X}_k^\star} \|\Pi_{\mathcal{X}}(x_k^\star)\| < \infty.$$

Hence, local Lipschitz continuity of f (lemma 3.2) and Cauchy-Schwarz imply that there exists a constant $M \in (0, \infty)$, such that

$$f(\Pi_{\mathcal{X}}(x_k^\star)) - f(x_k^\star) \leq M \operatorname{dist}(x_k^\star, \mathcal{X})$$

for all $k \in \mathbb{N}_0$ and $x_k^\star \in \mathcal{X}_k^\star$. We now obain

$$f(x^\star) - f(x_k^\star) \leq -\frac{\mu}{2}\|x^\star - \Pi_{\mathcal{X}}(x_k^\star)\|^2 + M \operatorname{dist}(x_k^\star, \mathcal{X}).$$

Plugging this into (3.7), we obtain

$$\frac{\mu}{2}\|x^\star - x_k^\star\|^2 \leq -\frac{\mu}{2}\|x^\star - \Pi_{\mathcal{X}}(x_k^\star)\|^2 + M \operatorname{dist}(x_k^\star, \mathcal{X}) + \gamma_k \pi_k(x^\star) - \gamma_k \pi_k(x_k^\star).$$

Now, using our lower bound on $\pi_k(x_k^\star)$ from lemma 3.4, and combining terms, we arrive at the inequality

$$\frac{\mu}{2}\|x^\star - x_k^\star\|^2 \leq -\frac{\mu}{2}\|x^\star - \Pi_{\mathcal{X}}(x_k^\star)\|^2 + (M - \gamma_k \tau) \operatorname{dist}(x_k^\star, \mathcal{X}) + \gamma_k \pi_k(x^\star).$$

Using $\pi_k(x^\star) \leq \alpha_k$ (see part 3 of assumption 3) and rearranging, we obtain

$$\frac{\mu}{2}\|x^\star - x_k^\star\|^2 + \frac{\mu}{2}\|x^\star - \Pi_{\mathcal{X}}(x_k^\star)\|^2 + (\gamma_k \tau - M) \operatorname{dist}(x_k^\star, \mathcal{X}) \leq \gamma_k \alpha_k.$$

The asymptotic rate for $\|x^\star - x_k^\star\|^2$ now follows in the case $\mu > 0$. For the bound on $\operatorname{dist}(x_k^\star, \mathcal{X})$, recall that $\lim_{k\to\infty} \gamma_k \uparrow \infty$ (see part 3 of assumption 1), so we can let K be large enough such that $\gamma_k \tau > M$ for all $k \geq K$. Dividing by γ_k on both sides and

using the nonnegativity of the other terms on the left-hand side, we get

$$c \cdot \operatorname{dist}(x_k^\star, \mathcal{X}) \leq \frac{\gamma_k \tau - M}{\gamma_k} \operatorname{dist}(x_k^\star, \mathcal{X}) \leq \alpha_k$$

for all $k \geq K$ and some constant $c \in (0, 1)$, as desired. □

Having established the convergence of the sequences $(x_k^\star)_{k \in \mathbb{N}_0}$ to the feasible set $\mathcal{X}$, we will now shift our attention to the iterates $(x_k)_{k \in \mathbb{N}_0}$ of algorithm 3. We begin with the following fundamental recursive inequality.

Lemma 3.7 (One-step inequality)**.** Let assumptions 1 to 3 hold and let $\rho \in (0, 1)$ and $\eta_k \in (0, \rho L^{-1}]$ for all $k \in \mathbb{N}$. Then the iterates $(x_k)_{k \in \mathbb{N}_0}$, generated by algorithm 3 with step size schedule $(\eta_k)_{k \in \mathbb{N}_0}$, satisfy

$$\begin{aligned} 2\eta_k(f_k(x_{k+1}) - f_k(x)) \leq (1 - \mu\eta_k)\|x - x_k\|^2 + 2\eta_k\langle z_k, x_k - x\rangle - \|x - x_{k+1}\|^2 \\ + \frac{8G^2\eta_k^2\gamma_k^2}{1-\rho} + \frac{2\eta_k^2}{1-\rho}\|z_k\|^2 \end{aligned} \tag{3.8}$$

almost surely, for all $x \in \mathbb{R}^n$ and $k \in \mathbb{N}_0$.

Proof. For $k \in \mathbb{N}$ we denote by g_k the stochastic gradient of ψ_k at x_k that is used in iteration k of algorithm 3. We also let $\psi(x) := \mathbb{E}(F_\xi(x))$ for $x \in \mathbb{R}^n$, so that $\psi_k = \psi + \gamma_k \pi_k$ and $\nabla\psi_k = \nabla\psi + \gamma_k \nabla\pi_k$. By L-smoothness of ψ (assumption 1), we have

$$\begin{aligned} f_k(x_{k+1}) &= \psi_k(x_{k+1}) + r(x_{k+1}) \\ &= \psi(x_{k+1}) + \gamma_k \pi_k(x_{k+1}) + r(x_{k+1}) \\ &\leq \psi(x_k) + \langle \nabla\psi(x_k), x_{k+1} - x_k\rangle + \frac{L}{2}\|x_{k+1} - x_k\|^2 + \gamma_k \pi_k(x_{k+1}) + r(x_{k+1}) \\ &= \psi(x_k) + \langle \nabla\psi_k(x_k), x_{k+1} - x_k\rangle + \frac{L}{2}\|x_{k+1} - x_k\|^2 + \gamma_k \pi_k(x_{k+1}) + r(x_{k+1}) \\ &\quad + \gamma_k\langle \nabla\pi_k(x_k), x_k - x_{k+1}\rangle. \end{aligned}$$

By convexity of π_k, we further have

$$\pi_k(x_{k+1}) \leq \pi_k(x_k) + \langle \nabla\pi_k(x_{k+1}), x_{k+1} - x_k\rangle,$$

and an application of Cauchy-Schwarz and Young's inequality yields

$$\pi_k(x_{k+1}) \leq \pi_k(x_k) + \frac{\epsilon_k^{-1}}{2}\|\nabla\pi_k(x_{k+1})\|^2 + \frac{\epsilon_k}{2}\|x_{k+1} - x_k\|^2,$$

for any $\epsilon_k \in (0, \infty)$. By lemma 3.2, $(x_k)_{k\in\mathbb{N}_0} \subset \mathrm{dom}(f)$ and the gradients $(\nabla\pi_k)_{k\in\mathbb{N}_0}$ are uniformly bounded on $\mathrm{dom}(f)$. Hence, there exists $G \in (0, \infty)$ such that

$$\pi_k(x_{k+1}) \leq \pi_k(x_k) + \frac{\epsilon_k}{2}G^2 + \frac{\epsilon_k^{-1}}{2}\|x_{k+1} - x_k\|^2,$$

for any $\epsilon_k \in (0, \infty)$. With this, we can further bound $f_k(x_{k+1})$ by

$$\begin{aligned} f_k(x_{k+1}) \leq\ & \psi(x_k) + \langle \nabla\psi_k(x_k), x_{k+1} - x_k\rangle + \frac{L}{2}\|x_{k+1} - x_k\|^2 + r(x_{k+1}) \\ & + \gamma_k\Big(\pi_k(x_k) + \frac{\epsilon_k}{2}G^2 + \frac{\epsilon_k^{-1}}{2}\|x_{k+1} - x_k\|^2\Big) + \gamma_k\langle \nabla\pi_k(x_k), x_k - x_{k+1}\rangle. \\ =\ & \psi_k(x_k) + \langle \nabla\psi_k(x_k), x_{k+1} - x_k\rangle + \frac{L + \gamma_k\epsilon_k^{-1}}{2}\|x_{k+1} - x_k\|^2 + r(x_{k+1}) \\ & + \frac{\gamma_k\epsilon_k G^2}{2} + \gamma_k\langle \nabla\pi_k(x_k), x_k - x_{k+1}\rangle. \end{aligned}$$

Note that $\psi(x_k)$ and $\gamma_k\pi_k(x_k)$ were combined into $\psi_k(x_k)$ in the last step. By another application of Cauchy-Schwarz and Young's inequality, we have, for any $\epsilon_k \in (0, \infty)$,

$$\langle \nabla\pi_k(x_k), x_k - x_{k+1}\rangle \leq \frac{\epsilon_k}{2}G^2 + \frac{\epsilon_k^{-1}}{2}\|x_{k+1} - x_k\|^2,$$

where we again used the bound $\sup_{k\in\mathbb{N}_0}\sup_{x\in\mathrm{dom}(f)}\|\nabla\pi_k(x)\| \leq G$. Therefore, we obtain

$$\begin{aligned} f_k(x_{k+1}) \leq\ & \psi_k(x_k) + \langle \nabla\psi_k(x_k), x_{k+1} - x_k\rangle + \frac{L + 2\gamma_k\epsilon_k^{-1}}{2}\|x_{k+1} - x_k\|^2 + r(x_{k+1}) \\ & + \gamma_k\epsilon_k G^2 \end{aligned}$$

for any $\epsilon_k \in (0, \infty)$. The rest of the proof follows the strategy of Cutler et al. [29], lemma 2. Recall the error in the k-th stochastic gradient, $z_k = \nabla\psi_k(x_k) - g_k$. By adding and subtracting $\langle z_k, x_{k+1} - x_k\rangle$ in the above inequality, we get

$$\begin{aligned} f_k(x_{k+1}) \leq\ & \psi_k(x_k) + \langle g_k, x_{k+1} - x_k\rangle + \frac{L + 2\gamma_k\epsilon_k^{-1}}{2}\|x_{k+1} - x_k\|^2 + r(x_{k+1}) \\ & + \gamma_k\epsilon_k G^2 + \langle z_k, x_{k+1} - x_k\rangle. \end{aligned}$$

By yet another application of Cauchy-Schwarz and Young's inequality, we have

$$\langle z_k, x_{k+1} - x_k\rangle \leq \frac{\delta_k}{2}\|z_k\|^2 + \frac{\delta_k^{-1}}{2}\|x_{k+1} - x_k\|^2$$

for all $\delta_k \in (0, \infty)$, and therefore

$$\begin{aligned} f_k(x_{k+1}) &\leq \psi_k(x_k) + \langle g_k, x_{k+1} - x_k \rangle + \frac{L + 2\gamma_k \epsilon_k^{-1} + \delta_k^{-1}}{2} \|x_{k+1} - x_k\|^2 + r(x_{k+1}) \\ &\quad + \gamma_k \epsilon_k G^2 + \frac{\delta_k}{2} \|z_k\|^2 \\ &= \psi_k(x_k) + r(x_{k+1}) + \langle g_k, x_{k+1} - x_k \rangle + \frac{1}{2\eta_k} \|x_{k+1} - x_k\|^2 \\ &\quad + \frac{L + 2\gamma_k \epsilon_k^{-1} + \delta_k^{-1} - \eta_k^{-1}}{2} \|x_{k+1} - x_k\|^2 + \gamma_k \epsilon_k G^2 + \frac{\delta_k}{2} \|z_k\|^2, \end{aligned}$$

where in the last step we added and subtracted $(2\eta_k)^{-1}\|x_{k+1} - x_k\|^2$ (and moved $r(x_{k+1})$ further forward). From the definition of the proximal operator, it follows that

$$\begin{aligned} x_{k+1} &= \operatorname{prox}_{\eta_k r}(x_k - \eta_k g_k) \\ &= \arg\min_{x \in \mathbb{R}^n} \left\{ r(x) + \frac{1}{2\eta_k} \|x - (x_k - \eta_k g_k)\|^2 \right\} \\ &= \arg\min_{x \in \mathbb{R}^n} \left\{ r(x) + \langle g_k, x - x_k \rangle + \frac{1}{2\eta_k} \|x - x_k\|^2 \right\}, \end{aligned}$$

where the last step follows from expanding the square and dropping the constant term $\eta_k^2 \|g_k\|^2$ from the minimization. The function

$$x \mapsto r(x) + \langle g_k, x - x_k \rangle + \frac{1}{2\eta_k} \|x - x_k\|^2$$

is η_k^{-1}-strongly convex and minimized by x_{k+1}. Thus, comparing with our previous bound on $f_k(x_{k+1})$, and using proposition 2.51, we can conclude

$$\begin{aligned} f_k(x_{k+1}) &\leq \psi_k(x_k) + r(x) + \langle g_k, x - x_k \rangle + \frac{1}{2\eta_k} \|x - x_k\|^2 - \frac{1}{2\eta_k} \|x - x_{k+1}\|^2 \\ &\quad + \frac{L + 2\gamma_k \epsilon_k^{-1} + \delta_k^{-1} - \eta_k^{-1}}{2} \|x_{k+1} - x_k\|^2 + \gamma_k \epsilon_k G^2 + \frac{\delta_k}{2} \|z_k\|^2 \end{aligned}$$

for all $x \in \mathbb{R}^n$. By μ-strong convexity of ψ_k, we have

$$\begin{aligned} \psi_k(x_k) + r(x) + \langle g_k, x - x_k \rangle &= \psi_k(x_k) + r(x) + \langle \nabla \psi_k(x_k), x - x_k \rangle + \langle z_k, x_k - x \rangle \\ &\leq \psi_k(x) - \frac{\mu}{2} \|x - x_k\|^2 + r(x) + \langle z_k, x_k - x \rangle \\ &= f_k(x) - \frac{\mu}{2} \|x - x_k\|^2 + \langle z_k, x_k - x \rangle \end{aligned}$$

for all $x \in \mathbb{R}^n$. Hence,

$$f_k(x_{k+1}) \leq f_k(x) + \left(\frac{1}{2\eta_k} - \frac{\mu}{2}\right)\|x - x_k\|^2 + \langle z_k, x_k - x\rangle - \frac{1}{2\eta_k}\|x - x_{k+1}\|^2 + \frac{L + 2\gamma_k\epsilon_k^{-1} + \delta_k^{-1} - \eta_k^{-1}}{2}\|x_{k+1} - x_k\|^2 + \gamma_k\epsilon_k G^2 + \frac{\delta_k}{2}\|z_k\|^2. \tag{3.9}$$

Fix $\alpha \in (0, 1-\rho)$ and define $\epsilon_k := 2\alpha^{-1}\eta_k\gamma_k$. Then

$$(L + 2\gamma_k\epsilon_k^{-1})\eta_k = \left(L + 2\gamma_k\frac{\alpha}{2\eta_k\gamma_k}\right)\eta_k = L\eta_k + \alpha \leq \rho + \alpha < 1,$$

where we used that $\eta_k \leq \rho L^{-1}$, which holds per assumption in the statement of this lemma. Choosing

$$\delta_k := \frac{\eta_k}{1 - (L\eta_k + \alpha)} \in (0, \infty)$$

therefore yields

$$L + 2\gamma_k\epsilon_k^{-1} + \delta_k^{-1} - \eta_k^{-1} = L + \frac{\alpha}{\eta_k} - \frac{1 - (L\eta_k + \alpha)}{\eta_k} - \frac{1}{\eta_k} = 0.$$

Hence, we can drop the $\|x_{k+1} - x_k\|^2$ term from (3.9) and get

$$f_k(x_{k+1}) \leq f_k(x) + \left(\frac{1}{2\eta_k} - \frac{\mu}{2}\right)\|x - x_k\|^2 + \langle z_k, x_k - x\rangle - \frac{1}{2\eta_k}\|x - x_{k+1}\|^2 + 2\alpha^{-1}\eta_k\gamma_k^2 G^2 + \frac{\eta_k}{2(1 - (L\eta_k + \alpha))}\|z_k\|^2.$$

Subtracting by $f_k(x)$ and multiplying both sides by $2\eta_k$ yields

$$2\eta_k(f_k(x_{k+1}) - f_k(x)) \leq (1 - \mu\eta_k)\|x - x_k\|^2 + 2\eta_k\langle z_k, x_k - x\rangle - \|x - x_{k+1}\|^2 + 4\alpha^{-1}\eta_k^2\gamma_k^2 G^2 + \frac{\eta_k^2}{1 - (L\eta_k + \alpha)}\|z_k\|^2.$$

For the specific choice $\alpha := (1-\rho)/2$, it holds that $L\eta_k + \alpha \leq \rho + (1-\rho)/2 = (1+\rho)/2$. Hence, we arrive at

$$2\eta_k(f_k(x_{k+1}) - f_k(x)) \leq (1 - \mu\eta_k)\|x - x_k\|^2 + 2\eta_k\langle z_k, x_k - x\rangle - \|x - x_{k+1}\|^2 + \frac{8G^2\eta_k^2\gamma_k^2}{1-\rho} + \frac{2\eta_k^2}{1-\rho}\|z_k\|^2,$$

as desired. □

After rearranging the inequality from lemma 3.7, setting $x = x^\star$, and dropping the $-\mu\eta_k$ term, we get

$$\begin{aligned}\|x^\star - x_{k+1}\|^2 \leq \|x^\star - x_k\|^2 + 2\eta_k\langle z_k, x_k - x^\star\rangle + 16G^2\eta_k^2\gamma_k^2 + 4\eta_k^2\|z_k\|^2 \\ - 2\eta_k(f_k(x_{k+1}) - f_k(x^\star)),\end{aligned}$$

provided $\eta_k \leq 1/(2L)$. By properties of conditional expectation and the definition of stochastic gradients, we know that $\mathbb{E}_k\langle z_k, x_k - x\rangle = \langle \mathbb{E}_k(z_k), x_k - x\rangle = 0$. Further note that, per definition of z_k and g_k, it holds that

$$\mathbb{E}_k\|z_k\|^2 = \mathbb{E}_k\|g_k - \nabla\psi_k(x_k)\|^2 = \mathbb{E}_k\|g_k - \mathbb{E}_k(g_k)\|^2 = \mathrm{Var}_k(g_k).$$

Therefore, after applying conditional expectations to both sides of the above inequality, we obtain

$$\begin{aligned}\mathbb{E}_k\|x^\star - x_{k+1}\|^2 \leq \|x^\star - x_k\|^2 + 16G^2\eta_k^2\gamma_k^2 + 4\eta_k^2\,\mathrm{Var}_k(g_k) \\ - 2\eta_k\mathbb{E}_k(f_k(x_{k+1}) - f_k(x^\star)).\end{aligned}$$

This is not yet quite in the form needed to apply lemma 3.3. For one, we need to bound the noise term $\mathrm{Var}_k(g_k)$. Second, we cannot in general guarantee that $\mathbb{E}_k(f_k(x_{k+1}) - f_k(x^\star)) \geq 0$. However, as we show in the next lemma, we can find a lower bound that involves a nonnegative term plus a small negative term.

Lemma 3.8. Let assumptions 1 to 5 hold, and let $\eta_k \in (0, \rho L^{-1}]$ for all $k \in \mathbb{N}_0$ and some $\rho \in (0,1)$. Then the iterates $(x_k)_{k\in\mathbb{N}_0}$, generated by algorithm 3 with step size schedule $(\eta_k)_{k\in\mathbb{N}_0}$, satisfy

$$\begin{aligned}\|x^\star - x_{k+1}\|^2 \leq (1-\mu\eta_k)\|x^\star - x_k\|^2 + \frac{2\eta_k^2}{1-\rho}\|z_k\|^2 + \frac{8G^2\eta_k^2\gamma_k^2}{1-\rho} \\ + 2\eta_k\langle z_k, x_k - x^\star\rangle + O(\eta_k\gamma_k\alpha_k) - 2\eta_k(f_k(x_{k+1}) - f_k(x_k^\star))\end{aligned}$$

almost surely, for all $k \in \mathbb{N}_0$.

Proof. Let $k \in \mathbb{N}_0$. From lemma 3.7, we have

$$\begin{aligned}2\eta_k(f_k(x_{k+1}) - f_k(x^\star)) \leq (1-\mu\eta_k)\|x^\star - x_k\|^2 + 2\eta_k\langle z_k, x_k - x^\star\rangle - \|x^\star - x_{k+1}\|^2 \\ + \frac{8G^2\eta_k^2\gamma_k^2}{1-\rho} + \frac{2\eta_k^2}{1-\rho}\|z_k\|^2\end{aligned}$$

almost surely. Write

$$f_k(x_{k+1}) - f_k(x^\star) = (f_k(x_{k+1}) - f_k(x_k^\star)) + (f_k(x_k^\star) - f_k(\Pi_{\mathcal{X}}(x_k^\star))) \\ + (f_k(\Pi_{\mathcal{X}}(x_k^\star)) - f_k(x^\star)). \tag{3.10}$$

By definition, it holds that

$$\begin{aligned} f_k(\Pi_{\mathcal{X}}(x_k^\star)) - f_k(x^\star) &= f(\Pi_{\mathcal{X}}(x_k^\star)) + \gamma_k \pi_k(\Pi_{\mathcal{X}}(x_k^\star)) - f(x^\star) - \gamma_k \pi_k(x^\star) \\ &\geq f(\Pi_{\mathcal{X}}(x_k^\star)) - f(x^\star) - \gamma_k \pi_k(x^\star) \\ &\geq -\gamma_k \pi_k(x^\star), \end{aligned}$$

where the last step follows from the fact that $x^\star$ minimizes f on $\mathcal{X}$ and $\Pi_{\mathcal{X}}(x_k^\star) \in \mathcal{X}$. Further, feasibility implies $\pi_k(x^\star) \leq \alpha_k$. Hence, combining with (3.10), we have

$$f_k(x_{k+1}) - f_k(x^\star) \geq (f_k(x_{k+1}) - f_k(x_k^\star)) + f_k(x_k^\star) - f_k(\Pi_{\mathcal{X}}(x_k^\star)) - \gamma_k \alpha_k.$$

For the next steps, for all $x \in \operatorname{dom}(f)$ and $k \in \mathbb{N}_0$, we let $\tilde{\nabla} f(x) \in \partial f_k(x)$ denote an arbitrary, but fixed subgradient of f_k at x. To analyze $f_k(x_k^\star) - f_k(\Pi_{\mathcal{X}}(x_k^\star))$, we first use convexity and Cauchy-Schwarz to get

$$f_k(x_k^\star) - f_k(\Pi_{\mathcal{X}}(x_k^\star)) \geq \langle \tilde{\nabla} f_k(\Pi_{\mathcal{X}}(x_k^\star)), x_k^\star - \Pi_{\mathcal{X}}(x_k^\star) \rangle \geq -\|\tilde{\nabla} f_k(\Pi_{\mathcal{X}}(x_k^\star))\| \operatorname{dist}(x_k^\star, \mathcal{X}).$$

The sequence $(x_k^\star)_{k \in \mathbb{N}_0}$ is bounded (lemma 3.2), which implies that $(\Pi_{\mathcal{X}}(x_k^\star))_{k \in \mathbb{N}_0}$ is bounded, by continuity of the projection map (proposition 2.57). Lemma 3.2 says that there exists $K_1 \in \mathbb{N}$ such that the stochastic gradient

$$g_k(x, \omega) := \nabla F_{\xi(\omega)}(x) + \gamma_k \nabla (h_k \circ g)(x, \xi(\omega))$$

satisfies, for any bounded set $B \subset \operatorname{dom}(f)$,

$$\|g_k(x, \cdot)\| = O(\gamma_k) \tag{3.11}$$

almost surely, for all $x \in B$. By L_r-Lipschitz-continuity of r, we know that $\|\tilde{\nabla} r(x)\| \leq L_r$ for all $x \in \operatorname{dom}(r)$ (see proposition 2.45). It follow that, for all $x \in \mathbb{R}^n$,

$$\|\tilde{\nabla} f_k(x)\| = \|\mathbb{E}(g_k(x, \cdot)) + \tilde{\nabla} r(x)\| \leq \mathbb{E}\|g_k(x, \cdot)\| + L_r.$$

Combining with (3.11), and noting that we have already established that $(\Pi_{\mathcal{X}}(x_k^\star))_{k \in \mathbb{N}_0}$

is bounded almost surely, we obtain

$$\sup_{r\in\mathbb{N}_0} \|\tilde{\nabla} f_k(\Pi_{\mathcal{X}}(x_r^\star))\| = \mathcal{O}(\gamma_k)$$

almost surely. Hence, there exist constants $K_1 \in \mathbb{N}$ and $c_1 \in (0,\infty)$, such that

$$\sup_{r\in\mathbb{N}_0} \|\tilde{\nabla} f_k(\Pi_{\mathcal{X}}(x_r^\star))\| \leq c_1\gamma_k$$

for all $k \geq K_1$. By theorem 3.6, it holds that there exist $K_2 \in \mathbb{N}$ and $c_2 \in (0,\infty)$ such that, for all $k \geq K_2$, $\operatorname{dist}(x_k^\star, \mathcal{X}) \leq c_2\,\alpha_k$. Setting $K := \max(K_1, K_2)$, it therefore holds that

$$f_k(x_k^\star) - f_k(\Pi_{\mathcal{X}}(x_k^\star)) \geq -c_1c_2 \cdot \gamma_k\alpha_k$$

and

$$\begin{aligned} f_k(x_{k+1}) - f_k(x^\star) &\geq f_k(x_{k+1}) - f_k(x_k^\star) - c_1c_2 \cdot \gamma_k\alpha_k - \gamma_k\alpha_k \\ &= f_k(x_{k+1}) - f_k(x_k^\star) - (1 + c_1c_2)\,\gamma_k\alpha_k \end{aligned}$$

for all $k \geq K$. Plugging this into our original estimate, we have

$$\begin{aligned} 2\eta_k\big(f_k(x_{k+1}) - f_k(x_k^\star) - \mathcal{O}(1)\gamma_k\alpha_k\big) \leq (1-\mu\eta_k)\|x^\star - x_k\|^2 - \|x^\star - x_{k+1}\|^2 \\ + \frac{8G^2\eta_k^2\gamma_k^2}{1-\rho} + \frac{2\eta_k^2}{1-\rho}\|z_k\|^2 + 2\eta_k\langle z_k, x_k - x^\star\rangle \end{aligned}$$

almost surely. Hence

$$\begin{aligned} 2\eta_k(f_k(x_{k+1}) - f_k(x_k^\star)) \leq (1-\mu\eta_k)\|x^\star - x_k\|^2 - \|x^\star - x_{k+1}\|^2 + \frac{8G^2\eta_k^2\gamma_k^2}{1-\rho} \\ + \frac{2\eta_k^2}{1-\rho}\|z_k\|^2 + 2\eta_k\langle z_k, x_k - x^\star\rangle + \mathcal{O}(1)\eta_k\gamma_k\alpha_k \end{aligned}$$

almost surely, for all $k \geq K$. The claim follows. □

From the above lemma, we can derive

$$\begin{aligned} \mathbb{E}_k\|x^\star - x_{k+1}\|^2 \leq \|x^\star - x_k\|^2 + \frac{2\eta_k^2}{1-\rho}\operatorname{Var}_k(g_k) + \frac{8G^2\eta_k^2\gamma_k^2}{1-\rho} + \mathcal{O}(\eta_k\gamma_k\alpha_k) \\ - 2\eta_k\mathbb{E}_k(f_k(x_{k+1}) - f_k(x_k^\star)). \end{aligned}$$

The term $2\eta_k\mathbb{E}_k(f_k(x_{k+1}) - f_k(x_k^\star))$ is indeed nonnegative by optimality of $x_k^\star$ for f_k.

Assuming that $\sum_{k=0}^{\infty} \eta_k \gamma_k \alpha_k < \infty$ and $\sum_{k=0}^{\infty} \eta_k^2 \gamma_k^2 < \infty$, we are almost ready to apply lemma 3.3. The following lemma will give a bound on the gradient noise $\mathrm{Var}_k(g_k)$.

Lemma 3.9 (Bound on gradient noise)**.** Let assumptions 1 to 3 hold. Then, in the situation of algorithm 3, there exists a constant $\sigma^2 \in (0, \infty)$ such that

$$\mathrm{Var}_k(g_k) \leq 4L^2 \|x_k - x^\star\|^2 + (1 + \gamma_k^2)\sigma^2$$

for all $k \in \mathbb{N}_0$.

Proof. We will use the inequality $(a + b)^2 \leq 2a^2 + 2b^2 \;\; \forall a, b \in \mathbb{R}$ multiple times. For $k \in \mathbb{N}_0$, we have

$$\mathrm{Var}_k(g_k) \leq \mathbb{E}_k \|g_k\|^2 \leq 2\mathbb{E}_k \|\nabla F_{\xi_k}(x_k)\|^2 + 2\gamma_k^2 \, \mathbb{E}_k \|\nabla (h_k \circ g)(x_k, \xi_k)\|^2.$$

Using the almost sure L-smoothness of $x \mapsto F_\xi(x)$ from part 1 of assumption 1, we have

$$\begin{aligned}
\mathbb{E}_k \|\nabla F_{\xi_k}(x_k)\|^2 &= \mathbb{E}_k \|\nabla F_{\xi_k}(x_k) - \nabla F_{\xi_k}(x^\star) + \nabla F_{\xi_k}(x^\star)\|^2 \\
&\leq 2\Big(\mathbb{E}_k \|\nabla F_{\xi_k}(x_k) - \nabla F_{\xi_k}(x^\star)\|^2 + \mathbb{E}_k \|\nabla F_{\xi_k}(x^\star)\|^2\Big) \\
&\leq 2L^2 \, \|x_k - x^\star\|^2 + 2 \, \mathbb{E}_k \|\nabla F_{\xi_k}(x^\star)\|^2,
\end{aligned}$$

where in the last step we also used that x_k is $\mathcal{F}_k$-measurable. Also by part 1 of assumption 1, we can find a point $x \in \mathbb{R}^n$ such that $\mathbb{E}\|F_\xi(x)\|^2 < \infty$. Since ξ_k is independent of $\xi_0, \xi_1, \ldots, \xi_{k-1}$, we also have

$$\mathbb{E}_k \|\nabla F_{\xi_k}(x)\|^2 = \mathbb{E}\|\nabla F_{\xi_k}(x)\|^2 = \mathbb{E}\|\nabla F_\xi(x)\|^2.$$

Hence, combining with almost sure L-smoothness of $x \mapsto F_\xi(x)$, there exists a constant $M^2 \in (0, \infty)$ such that

$$\begin{aligned}
\mathbb{E}_k \|\nabla F_{\xi_k}(x^\star)\|^2 &= \mathbb{E}_k \|\nabla F_{\xi_k}(x^\star) - \nabla F_{\xi_k}(x) + \nabla F_{\xi_k}(x)\|^2 \\
&\leq 2 \, \mathbb{E}_k \|\nabla F_{\xi_k}(x^\star) - \nabla F_{\xi_k}(x)\|^2 + 2 \, \mathbb{E}\|\nabla F_\xi(x)\|^2 \\
&\leq 2L^2 \, \|x^\star - x\|^2 + 2 \, \mathbb{E}\|\nabla F_\xi(x)\|^2 \\
&\leq \frac{1}{4} M^2.
\end{aligned}$$

Therefore, combining with the previous inequality, we get the bound

$$\mathbb{E}_k \|\nabla F_{\xi_k}(x_k)\|^2 \leq 2L^2 \, \|x_k - x^\star\|^2 + \frac{1}{2} M^2.$$

Using (3.2), there exists a constant $\tilde{M}^2 \in (0, \infty)$ such that

$$\mathbb{E}_k \|\nabla(h_k \circ g)(x_k, \xi_k)\|^2 \leq \frac{1}{2}\tilde{M}^2 \quad \forall k \in \mathbb{N}_0.$$

Putting everything together, we obtain

$$\mathrm{Var}_k(g_k) \leq 4L^2 \|x_k - x^\star\|^2 + M^2 + \gamma_k^2 \tilde{M}^2 = 4L^2 \|x_k - x^\star\|^2 + M^2 + \gamma_k^2 \tilde{M}^2$$

for all $k \in \mathbb{N}_0$. Setting $\sigma^2 := \max(M^2, \tilde{M}^2)$ yields the desired result. □

We are now ready to prove the main theorem of this section.

Theorem 3.10 (Almost sure convergence)**.** Let assumptions 1 to 5 hold. Let $(x_k)_{k\in\mathbb{N}_0}$ be a sequence generated by algorithm 3 with parameters $(\eta_k)_{k\in\mathbb{N}_0}$, $(\gamma_k)_{k\in\mathbb{N}_0}$, $(h_k)_{k\in\mathbb{N}_0}$ that satisfiy

1. $\sum_{k=0}^{\infty} \eta_k = \infty$ and there exists $\rho \in (0, 1)$ such that $\eta_k \leq \rho L^{-1}$ for all $k \in \mathbb{N}$.
2. $\sum_{k=0}^{\infty} \eta_k \gamma_k \alpha_k < \infty$.
3. $\sum_{k=0}^{\infty} \eta_k^2 \gamma_k^2 < \infty$.

Then $(x_k)_{k\in\mathbb{N}_0}$ converges almost surely and, in particular, $(x_k)_{k\in\mathbb{N}_0}$ is bounded almost surely. In the case $\mu > 0$, it further holds that $\liminf_{k\to\infty} \mathbb{E}\|x_k - x^\star\|^2 = 0$.

Proof. By lemma 3.8 (dropping the $-\mu\eta_k$ term), we have

$$\begin{aligned}\mathbb{E}_k \|x^\star - x_{k+1}\|^2 \leq \|x^\star - x_k\|^2 + \frac{2\eta_k^2}{1-\rho} \mathrm{Var}_k(g_k) + \frac{8G^2\eta_k^2\gamma_k^2}{1-\rho} \\ + O(\eta_k\gamma_k\alpha_k) - 2\eta_k \mathbb{E}_k(f_k(x_{k+1}) - f_k(x_k^\star)),\end{aligned}$$

Lemma 3.9 lets us bound the variance by

$$\mathrm{Var}_k(g_k) \leq 4L^2 \|x_k - x^\star\|^2 + (1 + \gamma_k^2)\sigma^2$$

for a constant $\sigma^2 \in (0, \infty)$ and all $k \in \mathbb{N}_0$. Thus, we have

$$\begin{aligned}\mathbb{E}_k \|x^\star - x_{k+1}\|^2 &\leq \|x^\star - x_k\|^2 + \frac{2\eta_k^2}{1-\rho}\Big(4L^2 \|x_k - x^\star\|^2 + (1 + \gamma_k^2)\sigma^2\Big) \\ &\quad + O(\eta_k\gamma_k\alpha_k) - 2\eta_k \mathbb{E}_k(f_k(x_{k+1}) - f_k(x_k^\star)) \\ &= \Big(1 + 8L^2(1-\rho)^{-1}\eta_k^2\Big)\|x^\star - x_k\|^2 + 2\eta_k^2(1-\rho)^{-1}(1 + \gamma_k^2)\sigma^2 \\ &\quad + O(\eta_k\gamma_k\alpha_k) - 2\eta_k \mathbb{E}_k(f_k(x_{k+1}) - f_k(x_k^\star))\end{aligned}$$

for all $k \in \mathbb{N}_0$. Define the nonnegative sequences $(a_k)_{k\in\mathbb{N}_0}, (b_k)_{k\in\mathbb{N}_0}, (c_k)_{k\in\mathbb{N}_0}$ by

$$\begin{aligned} a_k &:= 8L^2(1-\rho)^{-1}\eta_k^2 \\ b_k &:= 2\eta_k^2(1-\rho)^{-1}(1+\gamma_k^2)\sigma^2 + O(\eta_k\gamma_k\alpha_k) \\ c_k &:= 2\eta_k\mathbb{E}_k(f_k(x_{k+1}) - f_k(x_k^\star)). \end{aligned}$$

Note that c_k is indeed nonnegative, since $x_k^\star$ minimizes f_k. The above inequality now takes the form

$$\mathbb{E}_k\|x^\star - x_{k+1}\|^2 \leq (1+a_k)\|x^\star - x_k\|^2 + b_k - c_k.$$

Our assumptions imply that $\sum_{k=0}^\infty a_k < \infty$ and $\sum_{k=0}^\infty b_k < \infty$, so we can apply lemma 3.3, which implies that with probability one the sequence $\|x^\star - x_k\|^2$ converges and $\sum_{k=0}^\infty \eta_k\mathbb{E}_k(f_k(x_{k+1}) - f_k(x_k^\star)) < \infty$. By the bounded convergence theorem, proposition 2.15, it further holds that

$$\begin{aligned} \infty > \mathbb{E}\Big(\sum_{k=0}^\infty \eta_k\mathbb{E}_k(f_k(x_{k+1}) - f_k(x_k^\star))\Big) &= \sum_{k=0}^\infty \eta_k\mathbb{E}(\mathbb{E}_k(f_k(x_{k+1}) - f_k(x_k^\star))) \\ &= \sum_{k=0}^\infty \eta_k\mathbb{E}(f_k(x_{k+1}) - f_k(x_k^\star)). \end{aligned}$$

Since $\sum_{k=0}^\infty \eta_k = \infty$, it must therefore hold that

$$\liminf_{k\to\infty} \mathbb{E}(f_k(x_{k+1}) - f_k(x_k^\star)) = 0.$$

Strong convexity of f_k and optimality of $x_k^\star$ for f_k imply

$$f_k(x_{k+1}) - f_k(x_k^\star) \geq \frac{\mu}{2}\|x_{k+1} - x_k^\star\|^2.$$

If $\mu > 0$, we therefore have $\liminf_{k\to\infty} \mathbb{E}\|x_{k+1} - x_k^\star\|^2 = 0$, which implies that

$$\liminf_{k\to\infty} \mathbb{E}\|x_k - x^\star\|^2 = 0,$$

since we also know that $x_k^\star$ converges to $x^\star$ in the case $\mu > 0$, by theorem 3.6. □

Having established conditions under which the sequence of iterates $(x_k)_{k\in\mathbb{N}_0}$ generated by algorithm 3 stay bounded with probability one, we can now move on to prove convergence of the algorithm.

3.3. Convergence rates in expectation

In this section, we will establish conditions under which the iterates $(x_k)_{k\in\mathbb{N}_0}$, generated by algorithm 3, converge to a solution $x^\star$ of (P). Concretely, we will show that the expected optimality gap, $\mathbb{E}|f(\bar{x}_k) - f(x^\star)|$, and the expected distance to the feasible set, $\mathbb{E}(\mathrm{dist}(\bar{x}_k, \mathcal{X}))$, both converge to zero. The attainable convergence rates are $\mathcal{O}(\log^{1/2+\epsilon} K/\sqrt{K})$ in the convex case and $\mathcal{O}(\log^{\epsilon} K/K)$ in the strongly convex case, respectively, for arbitrary $\epsilon \in (0, \infty)$.

In the previous section, we established that the iterates $(x_k)_{k\in\mathbb{N}_0}$ of algorithm 3 are bounded with probability one, provided that the parameters $(h_k)_{k\in\mathbb{N}_0}$, $(\eta_k)_{k\in\mathbb{N}_0}$, $(\gamma_k)_{k\in\mathbb{N}_0}$ satisfy certain conditions, which are captured in the following assumption.

Assumption 6. Like in assumption 3, we assume that the sequence $(h_k)_{k\in\mathbb{N}_0}$ is chosen such that there exists a sequence $(\alpha_k)_{k\in\mathbb{N}_0}$ that bounds $(\pi_k(x))_{k\in\mathbb{N}_0}$ for any feasible $\hat{x} \in \mathcal{X}$:

$$\pi_k(\hat{x}) \leq \alpha_k$$

for all $k \in \mathbb{N}_0$ and $\hat{x} \in \mathcal{X}$. Further, we assume that the parameters $(\eta_k)_{k\in\mathbb{N}_0}$, $(\gamma_k)_{k\in\mathbb{N}_0}$ of algorithm 3 satisfy

1. $\sum_{k=0}^{\infty} \eta_k = \infty$ and there exist constants $\rho \in (0, 1)$ and $K \in \mathbb{N}$ such that $\eta_k \leq \rho L^{-1}$ for all $k \in \mathbb{N}$ with $k \geq K$.
2. $\sum_{k=0}^{\infty} \eta_k \gamma_k \alpha_k < \infty$.
3. $\sum_{k=0}^{\infty} \eta_k^2 \gamma_k^2 < \infty$,

With the above assumption, it is now possible to prove the following crucial bound.

Lemma 3.11. Let assumptions 1 to 6 hold. Then, for all $k \in \mathbb{N}_0$, it holds that

$$\begin{aligned} 2\eta_k(f_k(x_k) - f_k(x)) \leq (1 - \mu\eta_k)\|x - x_k\|^2 + 2\eta_k\langle z_k, x_k - x\rangle - \|x - x_{k+1}\|^2 \\ + \frac{8G^2\eta_k^2\gamma_k^2}{1-\rho} + \frac{2\eta_k^2}{1-\rho}\|z_k\|^2 + \eta_k^2\mathcal{O}(\gamma_k^2) \end{aligned}$$

almost surely.

Proof. Let $k \in \mathbb{N}_0$. By lemma 3.7, it holds that

$$\begin{aligned} 2\eta_k(f_k(x_{k+1}) - f_k(x)) \leq (1 - \mu\eta_k)\|x - x_k\|^2 + 2\eta_k\langle z_k, x_k - x\rangle - \|x - x_{k+1}\|^2 \\ + \frac{8G^2\eta_k^2\gamma_k^2}{1-\rho} + \frac{2\eta_k^2}{1-\rho}\|z_k\|^2 \end{aligned} \tag{3.12}$$

almost surely, for all $x \in \mathbb{R}^n$. Let $\tilde{\nabla} f_k(x_k)$ denote a subgradient of f_k at x_k. By convexity of f_k and Cauchy-Schwarz, we have

$$f_k(x_{k+1}) \geq f_k(x_k) + \langle \tilde{\nabla} f_k(x_k), x_{k+1} - x_k \rangle \geq f_k(x_k) - \|\tilde{\nabla} f_k(x_k)\| \|x_{k+1} - x_k\|.$$

Therefore,

$$\begin{aligned} 2\eta_k(f_k(x_k) - f_k(x)) \leq (1 - \mu\eta_k)\|x - x_k\|^2 + 2\eta_k\langle z_k, x_k - x\rangle - \|x - x_{k+1}\|^2 \\ + \frac{8G^2\eta_k^2\gamma_k^2}{1-\rho} + \frac{2\eta_k^2}{1-\rho}\|z_k\|^2 + 2\eta_k\|\tilde{\nabla} f_k(x_k)\|\|x_{k+1} - x_k\| \end{aligned} \tag{3.13}$$

almost surely, for all $x \in \mathbb{R}^n$. The sequence $(x_k)_{k\in\mathbb{N}_0}$ is bounded almost surely (theorem 3.10), hence lemma 3.2 implies

$$\|\tilde{\nabla} f_k(x_k)\| = \mathcal{O}(\gamma_k)$$

almost surely. We will now analyze $\|x_{k+1} - x_k\|$. Let $y_k := \operatorname{prox}_{\eta_k r}(x_k)$. Then, using the triangle inequality and the Lipschitz property of the proximal operator (proposition 2.64), we have

$$\begin{aligned} \|x_{k+1} - x_k\| &= \|x_{k+1} - y_k + y_k - x_k\| \\ &\leq \|\operatorname{prox}_{\eta_k r}(x_k - \eta_k g_k) - \operatorname{prox}_{\eta_k r}(x_k)\| + \|y_k - x_k\| \\ &\leq \eta_k\|g_k\| + \|y_k - x_k\|. \end{aligned}$$

For the second term, we use the definition of y_k as the solution to

$$\min_{x\in\mathbb{R}^n} r(x) + \frac{1}{2\eta_k}\|x - x_k\|^2.$$

By the first-order optimality condition, there exists a subgradient $\tilde{\nabla} r(y_k) \in \partial r(y_k)$ such that

$$\tilde{\nabla} r(y_k) + \frac{y_k - x_k}{\eta_k} = 0 \iff \frac{x_k - y_k}{\eta_k} \in \partial r(y_k).$$

Since x_k is in the domain of r for all $k \in \mathbb{N}_0$ (part 7 of lemma 3.2), we can now use the above, together with convexity, to get

$$r(x_k) - r(y_k) \geq \langle \eta_k^{-1}(x_k - y_k), x_k - y_k \rangle = \eta_k^{-1}\|x_k - y_k\|^2$$

for all $k \in \mathbb{N}_0$. Finally, Lipschitz continuity of r on $\operatorname{dom}(r)$ (part 2 of assumption 1)

yields

$$L_r\|x_k - y_k\| \geq r(x_k) - r(y_k) \geq \eta_k^{-1}\|x_k - y_k\|^2 \quad \Longleftrightarrow \quad \|x_k - y_k\| \leq \eta_k L_r,$$

almost surely, where we used that, in addition to $(x_k)_{k\in\mathbb{N}_0}$, the sequence $(y_k)_{k\in\mathbb{N}_0}$ must also be contained in $\mathrm{dom}(r)$, by definition of $\mathrm{prox}_{\eta_k r}$ and our assumption that r is proper (part 2 of assumption 1). Combining with (3.13), we therefore obtain

$$\begin{aligned}2\eta_k(f_k(x_k) - f_k(x)) \leq (1-\mu\eta_k)\|x - x_k\|^2 + 2\eta_k\langle z_k, x_k - x\rangle - \|x - x_{k+1}\|^2 \\ + \frac{8G^2\eta_k^2\gamma_k^2}{1-\rho} + \frac{2\eta_k^2}{1-\rho}\|z_k\|^2 + \eta_k^2 O(\gamma_k)\big(\|g_k\| + L_r\big),\end{aligned}$$

almost surely, for all $k \in \mathbb{N}_0$. Finally, we can use part 5 of lemma 3.2, which tells us that $\|g_k\| = O(\gamma_k)$ almost surely (again using almost sure boundedness of $(x_k)_{k\in\mathbb{N}_0}$). Hence, we arrive at

$$\begin{aligned}2\eta_k(f_k(x_k) - f_k(x)) \leq (1-\mu\eta_k)\|x - x_k\|^2 + 2\eta_k\langle z_k, x_k - x\rangle - \|x - x_{k+1}\|^2 \\ + \frac{8G^2\eta_k^2\gamma_k^2}{1-\rho} + \frac{2\eta_k^2}{1-\rho}\|z_k\|^2 + \eta_k^2 O(\gamma_k^2),\end{aligned}$$

as desired. □

For the next lemma, recall the definition of z_k in (3.1).

Lemma 3.12. Let assumptions 1 to 6 hold. For all $k \in \mathbb{N}_0$, it holds that

$$\|z_k\|^2 \leq 8L_{\mathrm{loc}}^2 + 8\gamma_k^2 G^2$$

with probability one, where $L_{\mathrm{loc}} \in (0, \infty)$.

Proof. We have

$$\begin{aligned}\|g_k\| &= \|\nabla F_{\xi_k}(x_k) + \gamma_k \nabla(h_k \circ g)(x_k, \xi_k)\| \\ &\leq \|\nabla F_{\xi_k}(x_k)\| + \gamma_k\|\nabla(h_k \circ g)(x_k, \xi_k)\|.\end{aligned}$$

By almost sure boundedness of $(x_k)_{k\in\mathbb{N}_0}$ (theorem 3.10) and local Lipschitz smoothness of $x \mapsto F_\xi(x)$ (part 1 of assumption 1), proposition 2.40 implies

$$\|\nabla F_{\xi_k}(x_k)\| \leq L_{\mathrm{loc}},$$

for a constant $L_{\text{loc}} \in (0, \infty)$, almost surely. By (3.2), we have

$$\sup_{k \in \mathbb{N}_0} \sup_{x \in \text{dom}(f)} \|\nabla(h_k \circ g)(x, \xi_k)\| \leq G$$

almost surely. Using the inequality $(a+b)^2 \leq 2(a^2+b^2) \;\; \forall a, b \in \mathbb{R}$, it follows that

$$\|g_k\|^2 \leq 2L_{\text{loc}}^2 + 2\gamma_k^2 G^2$$

with probability one. The claim follows from $\nabla\psi_k(x_k) = \mathbb{E}_k(g_k)$ and

$$\|g_k - \nabla\psi_k(x_k)\|^2 \leq 2\|g_k\|^2 + 2\|\nabla\psi_k(x_k)\|^2 \leq 2\|g_k\|^2 + 2\mathbb{E}_k\|g_k\|^2,$$

where the last step uses Jensen's inequality. □

The following lemma will give us a parametrization of algorithm 3 that satisifes assumption 6.

Lemma 3.13. Let the parameters $(\eta_k)_{k\in\mathbb{N}_0}, (\gamma_k)_{k\in\mathbb{N}_0}, (h_k)_{k\in\mathbb{N}_0}$ of algorithm 3 satisfy

$$\eta_k = \mathcal{O}\Big(\frac{1}{k^c \log^{(1+3e)/2}(k+1)}\Big), \; \gamma_k = \mathcal{O}(\log^e(k+1)), \; \alpha_k = \mathcal{O}\Big(\frac{1}{k^d}\Big)$$

with $c \in [1/2, 1)$, $d \in (1/2, \infty)$, and $e \in (0, \infty)$. Then assumption 6 holds. In particular,

$$\sum_{k=0}^{\infty} \eta_k \gamma_k \alpha_k = \mathcal{O}(1) \;\text{ and }\; \sum_{k=0}^{\infty} \eta_k^2 \gamma_k^2 = \mathcal{O}(1).$$

Proof. See the proof of lemma 9 in [15]. □

Before the next theorem, we recall the definitions of S_K and $\bar{x}_K$, introduced in algorithm 3, as well as in the beginning of this chapter. We also introduce truncated versions of S_K and $\bar{x}_K$, defined as

$$S_{t,k} := \sum_{i=t}^{k} w_i$$

$$\bar{x}_{t,k} := S_{t,k}^{-1} \sum_{i=t}^{k} w_i x_i.$$

Note that, for any $k \in \mathbb{N}_0$, we have

$$\bar{x}_K = \frac{S_{1,k-1}}{S_K}\bar{x}_{1,k-1} + \frac{S_{k,K}}{S_K}\bar{x}_{k,K} \tag{3.14}$$

and

$$\frac{S_{1,k-1}}{S_K} + \frac{S_{k,K}}{S_K} = 1. \tag{3.15}$$

We will also prove the following lemma, which will be used in the proofs of both theorems that follow.

Lemma 3.14. Let assumptions 1 to 6 hold. Then there exists $k_0 \in \mathbb{N}$ such that

$$\begin{aligned} 2\eta_k(f(x_k) - f(x)) \leq (1 - \mu\eta_k)\|x - x_k\|^2 - \|x - x_{k+1}\|^2 + 2\eta_k\langle z_k, x_k - x\rangle + \eta_k^2 O(\gamma_k^2) \\ + 2\eta_k\gamma_k\alpha_k - 2\tau\eta_k\gamma_k \operatorname{dist}(x_k, \mathcal{X}) \end{aligned}$$

almost surely, for all $x \in \mathcal{X}$ and all $k \in \mathbb{N}_0$ with $k \geq k_0$.

Proof. Let $k_0 \in \mathbb{N}_0$ be large enough such that $\eta_{k_0} \leq 1/(2L)$ and fix some $k \in \mathbb{N}_0$ with $k \geq k_0$. We can apply lemma 3.11, which implies

$$\begin{aligned} 2\eta_k(f_k(x_k) - f_k(x)) \leq (1 - \mu\eta_k)\|x - x_k\|^2 + 2\eta_k\langle z_k, x_k - x\rangle - \|x - x_{k+1}\|^2 \\ + 16G^2\eta_k^2\gamma_k^2 + 4\eta_k^2\|z_k\|^2 + \eta_k^2 O(\gamma_k^2), \end{aligned}$$

almost surely, for all $x \in \mathbb{R}^n$. Combining with lemma 3.12, we find that

$$\begin{aligned} 2\eta_k(f_k(x_k) - f_k(x)) \leq (1 - \mu\eta_k)\|x - x_k\|^2 + 2\eta_k\langle z_k, x_k - x\rangle - \|x - x_{k+1}\|^2 \\ + \eta_k^2 O(\gamma_k^2) \end{aligned}$$

almost surely, for all $x \in \mathbb{R}^n$. Assume now that $x \in \mathcal{X}$. By a property of π_k, the fact that $(x_k)_{k\in\mathbb{N}_0}$ is bounded almost surely (theorem 3.10), and lemma 3.4, we can deduce that

$$\begin{aligned} f_k(x_k) - f_k(x) &= f(x_k) - f(x) + \gamma_k(\pi_k(x_k) - \pi_k(x)) \\ &\geq f(x_k) - f(x) + \gamma_k(\tau \operatorname{dist}(x_k, \mathcal{X}) - \alpha_k) \end{aligned}$$

with probability one. Hence, we have

$$\begin{aligned} 2\eta_k(f(x_k) - f(x)) \leq (1 - \mu\eta_k)\|x - x_k\|^2 - \|x - x_{k+1}\|^2 + 2\eta_k\langle z_k, x_k - x\rangle + \eta_k^2 O(\gamma_k^2) \\ + 2\eta_k\gamma_k\alpha_k - 2\tau\eta_k\gamma_k \operatorname{dist}(x_k, \mathcal{X}) \end{aligned}$$

almost surely, for all $x \in \mathcal{X}$, as desired. □

The following theorem will provide upper bounds on the optimality gap, $f(\bar{x}_K) - f(x^\star)$, and the feasibility gap, $\operatorname{dist}(\bar{x}_K)$, in the general convex case. Based on this result, we will then prove convergence rates of $\mathbb{E}|f(\bar{x}_K) - f(x^\star)|$ and $\mathbb{E}(\operatorname{dist}(\bar{x}_K))$.

Theorem 3.15 (Central bounds - general convex case)**.** Let assumptions 1 to 5 hold. Let the parameters $(\eta_k)_{k\in\mathbb{N}_0}, (\gamma_k)_{k\in\mathbb{N}_0}$, $(h_k)_{k\in\mathbb{N}_0}$ be as in lemma 3.13. Further, let the iterate weights $(w_k)_{k\in\mathbb{N}}$ be defined as $w_k := \eta_k$, for all $k \in \mathbb{N}$. Then there exists a constant $k_0 \in \mathbb{N}$ such that the averaged iterate $\bar{x}_K$, generated by algorithm 3 with the parameters $(\eta_k)_{k\in\mathbb{N}_0}$, $(\gamma_k)_{k\in\mathbb{N}_0}$, $(h_k)_{k\in\mathbb{N}_0}$, $(w_k)_{k\in\mathbb{N}}$, satisfies

$$f(\bar{x}_K) - f(x^\star) \leq \frac{S_{1,k_0-1}}{S_K}(f(\bar{x}_{1,k_0-1}) - f(x^\star)) + \frac{\|x^\star - x_{k_0}\|^2}{2S_K} + \frac{\sum_{k=k_0}^K \eta_k^2 \mathcal{O}(\gamma_k^2)}{2S_K} + \frac{\sum_{k=k_0}^K \eta_k\gamma_k\alpha_k}{S_K} + \frac{\sum_{k=k_0}^K \eta_k\langle z_k, x_k - x^\star\rangle}{S_K}$$

almost surely, for all $K \in \mathbb{N}$ with $K \geq k_0$. Furthermore, there exist constants $k_1 \in \mathbb{N}$ and $\tau \in (0, \infty)$ such that

$$\operatorname{dist}(\bar{x}_K, \mathcal{X}) \leq \frac{S_{1,k_1-1}}{S_K}\operatorname{dist}(\bar{x}_{1,k_1-1}, \mathcal{X}) + \frac{\gamma_{k_1}^{-1}\|x^\star - x_{k_1}\|^2}{\tau S_K} + \frac{\sum_{k=k_1}^K \eta_k^2 \mathcal{O}(\gamma_k)}{\tau S_K} + 2\frac{\sum_{k=k_1}^K \eta_k\alpha_k}{\tau S_K} + 2\frac{\sum_{k=k_1}^K \gamma_k^{-1}\eta_k\langle z_k, x_k - \Pi_{\mathcal{X}}(x_k)\rangle}{\tau S_K}$$

almost surely, for all $K \in \mathbb{N}$ with $K \geq k_1$.

Proof. Lemma 3.13 implies that the given parameters satisfy assumption 6. Hence, we can use lemma 3.14, which says that there exists $k_0 \in \mathbb{N}$ such that

$$2\eta_k(f(x_k) - f(x)) \leq (1 - \mu\eta_k)\|x - x_k\|^2 - \|x - x_{k+1}\|^2 + 2\eta_k\langle z_k, x_k - x\rangle + \eta_k^2\mathcal{O}(\gamma_k^2) + 2\eta_k\gamma_k\alpha_k - 2\tau\eta_k\gamma_k \operatorname{dist}(x_k, \mathcal{X}) \tag{3.16}$$

almost surely, for all $x \in \mathcal{X}$ and all $k \in \mathbb{N}$ with $k \geq k_0$. We can drop the $-\mu\eta_k$ and $-\operatorname{dist}(x_k, \mathcal{X})$ terms. Then, summing both sides from $k = k_0$ to $K \in \mathbb{N}$ with $K \geq k_0$,

we obtain

$$
\begin{aligned}
2\sum_{k=k_0}^{K}\eta_k(f(x_k)-f(x)) &\le \|x-x_{k_0}\|^2-\|x-x_{K+1}\|^2+2\sum_{k=k_0}^{K}\eta_k\langle z_k,x_k-x\rangle \\
&\quad +2\sum_{k=k_0}^{K}\eta_k\gamma_k\alpha_k+\sum_{k=k_0}^{K}\eta_k^2\mathcal{O}(\gamma_k^2) \\
&\le \|x-x_{k_0}\|^2+2\sum_{k=k_0}^{K}\eta_k\langle z_k,x_k-x\rangle \\
&\quad +2\sum_{k=k_0}^{K}\eta_k\gamma_k\alpha_k+\sum_{k=k_0}^{K}\eta_k^2\mathcal{O}(\gamma_k^2)
\end{aligned}
$$

almost surely, for all $x \in \mathcal{X}$. Setting $x := x^\star \in \mathcal{X}^\star$, dividing both sides by $2\sum_{k=k_0}^{K}\eta_k$, and using convexity of f, we obtain

$$
\begin{aligned}
f(\bar{x}_{k_0,K})-f(x^\star) \le \frac{\|x^\star-x_{k_0}\|^2}{2S_{k_0,K}}+\frac{\sum_{k=k_0}^{K}\eta_k\langle z_k,x_k-x^\star\rangle}{S_{k_0,K}} \\
+\frac{\sum_{k=k_0}^{K}\eta_k\gamma_k\alpha_k}{S_{k_0,K}}+\frac{\sum_{k=k_0}^{K}\eta_k^2\mathcal{O}(\gamma_k^2)}{2S_{k_0,K}}.
\end{aligned}
$$

Recall the relations (3.14) and (3.15). Using convexity of f, we can combine these with the latest bound on $(f(\bar{x}_{k_0,K})-f(x^\star))$, and arrive at

$$
\begin{aligned}
f(\bar{x}_K)-f(x^\star) \le \frac{S_{1,k_0-1}}{S_K}(f(\bar{x}_{1,k_0-1})-f(x^\star))+\frac{\|x^\star-x_{k_0}\|^2}{2S_K}+\frac{\sum_{k=k_0}^{K}\eta_k\langle z_k,x_k-x^\star\rangle}{S_K} \\
+\frac{\sum_{k=k_0}^{K}\eta_k\gamma_k\alpha_k}{S_K}+\frac{\sum_{k=k_0}^{K}\eta_k^2\mathcal{O}(\gamma_k^2)}{2S_K}
\end{aligned}
$$

almost surely. Next, we will derive the desired bound for $\operatorname{dist}(\bar{x}_K,\mathcal{X})$. Again, fix $k \ge k_0$. For the choice $x := \Pi_{\mathcal{X}}(x_k)$, inequality (3.16) gives us

$$
\begin{aligned}
2\eta_k(f(x_k)-f(\Pi_{\mathcal{X}}(x_k))) \le (1-\mu\eta_k)\operatorname{dist}(x_k,\mathcal{X})^2-\|\Pi_{\mathcal{X}}(x_k)-x_{k+1}\|^2+\eta_k^2\mathcal{O}(\gamma_k^2) \\
+2\eta_k\langle z_k,x_k-\Pi_{\mathcal{X}}(x_k)\rangle+2\eta_k\gamma_k\alpha_k-2\tau\eta_k\gamma_k\operatorname{dist}(x_k,\mathcal{X}).
\end{aligned}
$$

Note that $\|\Pi_{\mathcal{X}}(x_k)-x_{k+1}\| \ge \operatorname{dist}(x_{k+1},\mathcal{X})$, hence, after additionally rearranging, we

obtain

$$\begin{aligned}\operatorname{dist}(x_{k+1},\mathcal{X})^2 \leq (1-\mu\eta_k)\operatorname{dist}(x_k,\mathcal{X})^2 + \eta_k^2 O(\gamma_k^2) + 2\eta_k\langle z_k, x_k - \Pi_{\mathcal{X}}(x_k)\rangle \\ + 2\eta_k\gamma_k\alpha_k - 2\tau\eta_k\gamma_k\operatorname{dist}(x_k,\mathcal{X}) + 2\eta_k(f(\Pi_{\mathcal{X}}(x_k)) - f(x_k)).\end{aligned}$$

Almost sure boundedness of $(x_k)_{k\in\mathbb{N}_0}$ (theorem 3.10) implies that $(\Pi_{\mathcal{X}}(x_k))_{k\in\mathbb{N}_0}$ is also bounded almost surely, since continuous functions map compacta to compacta. Therefore, by local Lipschitz continuity of f (lemma 3.2), there exists a constant $M \in [0,\infty)$ such that

$$f(\Pi_{\mathcal{X}}(x_k)) - f(x_k) \leq M\operatorname{dist}(x_k,\mathcal{X})$$

almost surely. Combining with the previous inequality and gathering terms involving $\operatorname{dist}(x_k,\mathcal{X})$, we arrive at

$$\begin{aligned}\operatorname{dist}(x_{k+1},\mathcal{X})^2 \leq (1-\mu\eta_k)\operatorname{dist}(x_k,\mathcal{X})^2 + \eta_k^2 O(\gamma_k^2) + 2\eta_k\langle z_k, x_k - \Pi_{\mathcal{X}}(x_k)\rangle \\ + 2\eta_k\gamma_k\alpha_k - 2\eta_k(\tau\gamma_k - M)\operatorname{dist}(x_k,\mathcal{X}).\end{aligned}$$

Since $\gamma_k \uparrow \infty$, there exists $k_1 \in \mathbb{N}$ such that $\tau_k\gamma_k - M \geq \tau\gamma_k/2$ for all natural numbers $k \geq k_1$. Thus,

$$\begin{aligned}\operatorname{dist}(x_{k+1},\mathcal{X})^2 \leq (1-\mu\eta_k)\operatorname{dist}(x_k,\mathcal{X})^2 + \eta_k^2 O(\gamma_k^2) + 2\eta_k\langle z_k, x_k - \Pi_{\mathcal{X}}(x_k)\rangle \\ + 2\eta_k\gamma_k\alpha_k - \tau\eta_k\gamma_k\operatorname{dist}(x_k,\mathcal{X})\end{aligned} \tag{3.17}$$

for all $k \geq k_1$. Dropping the $-\mu\eta_k$ term and multiplying both sides by γ_k^{-1}, we get

$$\begin{aligned}\gamma_k^{-1}\operatorname{dist}(x_{k+1},\mathcal{X})^2 \leq \gamma_k^{-1}\operatorname{dist}(x_k,\mathcal{X})^2 + \eta_k^2 O(\gamma_k) \\ + 2\gamma_k^{-1}\eta_k\langle z_k, x_k - \Pi_{\mathcal{X}}(x_k)\rangle + 2\eta_k\alpha_k - \tau\eta_k\operatorname{dist}(x_k,\mathcal{X})\end{aligned}$$

for all $k \geq k_1$. Since γ_k is nondecreasing, we have $\gamma_k^{-1} \geq \gamma_{k+1}^{-1}$ for all $k \in \mathbb{N}$. Setting $d_j := \gamma_j^{-1}\operatorname{dist}(x_j,\mathcal{X})^2 \;\; \forall j \in \mathbb{N}$, we therefore have

$$d_{k+1} \leq d_k + \eta_k^2 O(\gamma_k) + 2\gamma_k^{-1}\eta_k\langle z_k, x_k - \Pi_{\mathcal{X}}(x_k)\rangle + 2\eta_k\alpha_k - \tau\eta_k\operatorname{dist}(x_k,\mathcal{X})$$

for all $k \geq k_1$. Rearranging again and summing from $k = k_1$ to $k = K$ for $K \in \mathbb{N}$ with

$K \geq k_1$, we have

$$\begin{aligned}
\tau \sum_{k=k_1}^{K} \eta_k \operatorname{dist}(x_k, \mathcal{X}) &\leq d_{k_1} - d_{K+1} + \sum_{k=k_1}^{K} \eta_k^2 O(\gamma_k) + 2 \sum_{k=k_1}^{K} \eta_k \alpha_k \\
&\quad + 2 \sum_{k=k_1}^{K} \gamma_k^{-1} \eta_k \langle z_k, x_k - \Pi_{\mathcal{X}}(x_k) \rangle \\
&\leq d_{k_1} + \sum_{k=k_1}^{K} \eta_k^2 O(\gamma_k) + 2 \sum_{k=k_1}^{K} \eta_k \alpha_k \\
&\quad + 2 \sum_{k=k_1}^{K} \gamma_k^{-1} \eta_k \langle z_k, x_k - \Pi_{\mathcal{X}}(x_k) \rangle,
\end{aligned}$$

where we used $d_{K+1} \geq 0$ in the last step. The distance functional $x \mapsto \operatorname{dist}(x, \mathcal{X})$ is convex (proposition 2.60), so if we divide both sides of the above inequality by $\tau \cdot S_{k_1,K}$, we obtain

$$\begin{aligned}
\operatorname{dist}(\bar{x}_{k_1,K}, \mathcal{X}) \leq \frac{d_{k_1}}{\tau S_{k_1,K}} + \frac{\sum_{k=k_1}^{K} \eta_k^2 O(\gamma_k)}{\tau S_{k_1,K}} + 2 \frac{\sum_{k=k_1}^{K} \eta_k \alpha_k}{\tau S_{k_1,K}} \\
+ 2 \frac{\sum_{k=k_1}^{K} \gamma_k^{-1} \eta_k \langle z_k, x_k - \Pi_{\mathcal{X}}(x_k) \rangle}{\tau S_{k_1,K}}.
\end{aligned}$$

To derive a bound for $\operatorname{dist}(\bar{x}_K, \mathcal{X})$, we proceed similarly as before. Recall again the relations (3.14) and (3.15). Using convexity of $\operatorname{dist}(\cdot, \mathcal{X})$, we can combine these with the latest bound on $\operatorname{dist}(\bar{x}_{k_1,K}, \mathcal{X})$, and arrive at

$$\begin{aligned}
\operatorname{dist}(\bar{x}_K, \mathcal{X}) &\leq \frac{S_{1,k_1-1}}{S_K} \operatorname{dist}(\bar{x}_{1,k_1-1}, \mathcal{X}) + \frac{S_{k_1,K}}{S_K} \operatorname{dist}(\bar{x}_{k_1,K}, \mathcal{X}) \\
&\leq \frac{S_{1,k_1-1}}{S_K} \operatorname{dist}(\bar{x}_{1,k_1-1}, \mathcal{X}) + \frac{d_{k_1}}{\tau S_K} + \frac{\sum_{k=k_1}^{K} \eta_k^2 O(\gamma_k)}{\tau S_K} + 2 \frac{\sum_{k=k_1}^{K} \eta_k \alpha_k}{\tau S_K} \\
&\quad + 2 \frac{\sum_{k=k_1}^{K} \gamma_k^{-1} \eta_k \langle z_k, x_k - \Pi_{\mathcal{X}}(x_k) \rangle}{\tau S_K}
\end{aligned}$$

almost surely, or all $K \geq k_1$. This concludes the proof. □

We can now give concrete convergence rates for the expected optimality gap and the expected distance to the feasible set of the output $\bar{x}_K$, generated by algorithm 3.

Corollary 3.16 (Convergence rates in expectation - convex case)**.** Let the assumptions of theorem 3.15 hold. Then the averaged iterate $\bar{x}_K$, generated by algorithm 3 with

the parameters $(\eta_k)_{k\in\mathbb{N}_0}$, $(\gamma_k)_{k\in\mathbb{N}_0}$, $(h_k)_{k\in\mathbb{N}_0}$, $(w_k)_{k\in\mathbb{N}}$, satisfies

$$\mathbb{E}|f(\bar{x}_K) - f(x^\star)| = \tau^{-1}\,\mathcal{O}\Big(\frac{\log^{(1+3e)/2}(K)}{K^{1-c}}\Big).$$

Further, it also holds that

$$\mathbb{E}(\mathrm{dist}(\bar{x}_K, \mathcal{X})) = \tau^{-1}\mathcal{O}\Big(\frac{\log^{(1+3e)/2}(K)}{K^{1-c}}\Big).$$

Proof. Theorem 3.15 gives us an upper bound for $f(\bar{x}_K) - f(x^\star)$,

$$\begin{aligned} f(\bar{x}_K) - f(x^\star) \leq \frac{S_{1,k_0-1}}{S_K}(f(\bar{x}_{1,k_0-1}) - f(x^\star)) + \frac{\|x^\star - x_{k_0}\|^2}{2S_K} + \frac{\sum_{k=k_0}^K \eta_k^2 \mathcal{O}(\gamma_k^2)}{2S_K} \\ + \frac{\sum_{k=k_0}^K \eta_k\gamma_k\alpha_k}{S_K} + \frac{\sum_{k=k_0}^K \eta_k\langle z_k, x_k - x^\star\rangle}{S_K} \end{aligned}$$

almost surely, for all $K \in \mathbb{N}$ with $K \geq k_0$, where $S_K := \sum_{k=1}^K \eta_k$. Note that

$$\mathbb{E}(\langle z_k, x_k - x^\star\rangle) = \mathbb{E}(\mathbb{E}_k(\langle z_k, x_k - x^\star\rangle)) = \mathbb{E}(\langle \mathbb{E}_k(z_k), x_k - x^\star\rangle) = 0$$

for all $k \in \mathbb{N}$. Hence, taking expectations on both sides of the above inequality, we get

$$\begin{aligned} \mathbb{E}(f(\bar{x}_K) - f(x^\star)) \leq \frac{S_{1,k_0-1}}{S_K}\mathbb{E}(f(\bar{x}_{1,k_0-1}) - f(x^\star)) + \frac{\mathbb{E}\|x^\star - x_{k_0}\|^2}{2S_K} + \frac{\sum_{k=k_0}^K \eta_k^2 \mathcal{O}(\gamma_k^2)}{2S_K} \\ + \frac{\sum_{k=k_0}^K \eta_k\gamma_k\alpha_k}{S_K}. \end{aligned} \tag{3.18}$$

By convexity, we have for all $k \in \mathbb{N}$

$$\begin{aligned} f(\bar{x}_k) - f(x^\star) &\geq \langle \tilde{\nabla} f(x^\star), \bar{x}_k - x^\star\rangle \\ &= \langle \tilde{\nabla} f(x^\star), \bar{x}_k - \Pi_{\mathcal{X}}(\bar{x}_k)\rangle + \langle \tilde{\nabla} f(x^\star), \Pi_{\mathcal{X}}(\bar{x}_k) - x^\star\rangle, \end{aligned}$$

for any subgradient $\tilde{\nabla} f(x^\star) \in \partial f(x^\star)$. Optimality of $x^\star$ for f on $\mathcal{X}$ implies that there exists a subgradient $\tilde{\nabla} f(x^\star) \in \partial f(x^\star)$ such that $\langle \tilde{\nabla} f(x^\star), \Pi_{\mathcal{X}}(\bar{x}_k) - x^\star\rangle \geq 0$. Combining these facts with the above inequality and applying Cauchy-Schwarz, we obtain

$$f(\bar{x}_k) - f(x^\star) \geq -\|\tilde{\nabla} f(x^\star)\|\,\mathrm{dist}(\bar{x}_k, \mathcal{X}) \tag{3.19}$$

for all $k \in \mathbb{N}$. Taking expectations in the bound for $\mathrm{dist}(\bar{x}_K, \mathcal{X})$ from theorem 3.18,

we know that there exists $k_1 \in \mathbb{N}$ such that for all $K \in \mathbb{N}$ with $K \geq k_1$, it holds that

$$\mathbb{E}(\mathrm{dist}(\bar{x}_K, \mathcal{X})) \leq \frac{S_{1,k_1-1}}{S_K}\mathbb{E}(\mathrm{dist}(\bar{x}_{1,k_1-1}, \mathcal{X})) + \frac{\gamma_{k_1}^{-1}\mathbb{E}\|x^\star - x_{k_1}\|^2}{\tau S_K} + \frac{\sum_{k=k_1}^K \eta_k^2 O(\gamma_k)}{\tau S_K} + 2\frac{\sum_{k=k_1}^K \eta_k \alpha_k}{\tau S_K}. \tag{3.20}$$

The first two terms converge to zero with rate $O(S_K^{-1})$ as $K \to \infty$. Since

$$\eta_k = O\left(\left(k^c \log^{(1+3e)/2}(k+1)\right)^{-1}\right),$$

it follows by a standard integral bound that

$$\begin{aligned} S_K &= \sum_{k=1}^K \eta_k \\ &= \sum_{k=1}^K \frac{1}{k^c \log^{(1+3e)/2}(k+1)} \\ &\geq \int_1^{K+1} \frac{1}{x^c \log^{(1+3e)/2}(x+1)}\,\mathrm{d}x \\ &\geq \frac{1}{\log^{(1+3e)/2}(K+2)} \int_1^{K+1} \frac{1}{x^c}\,\mathrm{d}x \\ &= \frac{K^{1-c}}{\log^{(1+3e)/2}(K+2)}. \end{aligned} \tag{3.21}$$

Hence, we have

$$S_K^{-1} = O\left(\frac{\log^{(1+3e)/2}(K)}{K^{1-c}}\right).$$

By lemma 3.13, it also holds that $\sum_{k=1}^K \gamma_k \eta_k \alpha_k = O(1)$ and $\sum_{k=1}^K \eta_k^2 \gamma_k^2 = O(1)$. Therefore, by combining with (3.20), we obtain

$$\mathbb{E}(\mathrm{dist}(\bar{x}_K, \mathcal{X})) = O\left(\frac{\log^{(1+3e)/2}(K)}{\tau K^{1-c}}\right).$$

Combining with (3.19), we obtain

$$\mathbb{E}(f(\bar{x}_k) - f(x^\star)) \geq -\|\nabla f(x^\star)\| \cdot O\left(\frac{\log^{(1+3e)/2}(K)}{\tau K^{1-c}}\right).$$

For the upper bound in (3.18), we similarly find that

$$\mathbb{E}(f(\bar{x}_K) - f(x^\star)) \leq \mathcal{O}\Big(\frac{\log^{(1+3e)/2}(K)}{K^{1-c}}\Big).$$

Putting the two bounds together, we finally arrive at

$$\mathbb{E}|f(\bar{x}_K) - f(x^\star)| = \tau^{-1}\mathcal{O}\Big(\frac{\log^{(1+3e)/2}(K)}{K^{1-c}}\Big),$$

as desired. □

It follows that we can achieve the convergence rate

$$\mathbb{E}|f(\bar{x}_K) - f(x^\star)| + \mathbb{E}(\mathrm{dist}(\bar{x}_K)) = \tau^{-1}\mathcal{O}\Big(\frac{\log^{(1+3e)/2}(K)}{\sqrt{K}}\Big)$$

in the general convex case.

We will now turn our attention to the special case where the objective is strongly convex, i. e. $\mu \in (0, \infty)$. In this setting, we can achieve a faster rate than in the general convex case, provided the parameters are set correctly.

Lemma 3.17. In the setting $\mu > 0$, let $\eta_k := 2/(\mu k)$ for all $k \in \mathbb{N}$ and $\eta_0 := 2/\mu$. Then, for any $e \in (0, \infty)$, and any sequences $(\gamma_k)_{k\in\mathbb{N}_0}$, $(h_k)_{k\in\mathbb{N}_0}$ with $\gamma_k = \mathcal{O}(\log^e(k+1))$ and $\alpha_k = \mathcal{O}(1/k)$, assumption 6 holds.

Proof. Fix $e \in (0, \infty)$ and let $\gamma_k = \mathcal{O}(\log^e(k))$, $\alpha_k = \mathcal{O}(1/k)$. Clearly, there exist $\rho \in (0, 1)$ and $K \in \mathbb{N}$ large enough such that $\eta_k \leq \rho L^{-1}$ for all $k \geq K$. It also holds that

1. $\sum_{k=0}^{\infty} \eta_k = \infty$,
2. $\sum_{k=0}^{\infty} \gamma_k^2 \eta_k^2 = \sum_{k=1}^{\infty} \mathcal{O}(\log^{2e}(k)/k^2) < \infty$,
3. $\sum_{k=0}^{\infty} \gamma_k \eta_k \alpha_k = \sum_{k=1}^{\infty} \mathcal{O}(\log^e(k)/k^2) < \infty$.

This proves the claim. □

With this choice of parameters, we can prove the following.

Theorem 3.18 (Central bounds - strongly convex case)**.** Let assumptions 1 to 5 hold, and assume that we are in the setting $\mu > 0$. Let the parameters $(\eta_k)_{k\in\mathbb{N}_0}$, $(\gamma_k)_{k\in\mathbb{N}_0}$, $(h_k)_{k\in\mathbb{N}_0}$ be as in lemma 3.17. Further, let the iterate weights $(w_k)_{k\in\mathbb{N}}$ be defined as $w_k := \eta_k^{-1}$, for all $k \in \mathbb{N}$. Then there exists a constant $k_0 \in \mathbb{N}$ such that the averaged

iterate $\bar{x}_K$, generated by algorithm 3 with the parameters $(\eta_k)_{k\in\mathbb{N}_0}$, $(\gamma_k)_{k\in\mathbb{N}_0}$, $(h_k)_{k\in\mathbb{N}_0}$, $(w_k)_{k\in\mathbb{N}}$, satisfies

$$f(\bar{x}_K) - f(x^\star) \leq \frac{S_{1,k_0-1}}{S_K}(f(\bar{x}_{1,k_0-1}) - f(x^\star)) + \frac{\eta_{k_0-1}^{-2}\|x^\star - x_{k_0}\|^2}{2S_K} + \frac{\sum_{k=k_0}^K \eta_k^{-1}\langle z_k, x_k - x^\star\rangle}{S_K} + \frac{\sum_{k=k_0}^K O(\gamma_k^2)}{S_K} + \frac{\sum_{k=k_0}^K \eta_k^{-1}\gamma_k\alpha_k}{S_K}.$$

almost surely, for all $K \in \mathbb{N}$ with $K \geq k_0$. Furthermore, there exist constants $k_1 \in \mathbb{N}$ and $\tau \in (0, \infty)$ such that

$$\operatorname{dist}(\bar{x}_K, \mathcal{X}) \leq \frac{S_{1,k_1-1}}{S_K}\operatorname{dist}(\bar{x}_{1,k_1-1}, \mathcal{X}) + \frac{\gamma_{k_1}^{-1}\eta_{k_1-1}^{-2}\operatorname{dist}(x_{k_1}, \mathcal{X})^2}{\tau S_K} + \frac{\sum_{k=k_1}^K O(\gamma_k)}{\tau S_K} + 2\frac{\sum_{k=k_1}^K \eta_k^{-1}\alpha_k}{\tau S_K} + 2\frac{\sum_{k=k_1}^K \gamma_k^{-1}\eta_k^{-1}\langle z_k, x_k - \Pi_{\mathcal{X}}(x_k)\rangle}{\tau S_K}$$

almost surely, for all $K \in \mathbb{N}$ with $K \geq k_1$.

Proof. Lemma 3.17 implies that the given parameters satisfy assumption 6. Hence, we can use lemma 3.14, which says that there exists $k_0 \in \mathbb{N}$ such that

$$2\eta_k(f(x_k) - f(x)) \leq (1 - \mu\eta_k)\|x - x_k\|^2 - \|x - x_{k+1}\|^2 + 2\eta_k\langle z_k, x_k - x\rangle + \eta_k^2 O(\gamma_k^2) + 2\eta_k\gamma_k\alpha_k - 2\tau\eta_k\gamma_k\operatorname{dist}(x_k, \mathcal{X})$$

almost surely, for all $x \in \mathcal{X}$ and all $k \in \mathbb{N}$ with $k \geq k_0$. From this starting point, we can closely follow the proof stategy of lemma 13 in [15] to arrive at our desired result. First, we will prove the claim for $f(\bar{x}_k) - f(x^\star)$. Dropping the $-\operatorname{dist}(x_k, \mathcal{X})$ term from the above inequality, setting $x := x^\star \in \mathcal{X}^\star$, and multiplying both sides by η_k^{-2}, we get

$$2\eta_k^{-1}(f(x_k) - f(x^\star)) \leq \eta_k^{-2}(1 - \mu\eta_k)\|x^\star - x_k\|^2 - \eta_k^{-2}\|x^\star - x_{k+1}\|^2 + 2\eta_k^{-1}\langle z_k, x_k - x^\star\rangle + O(\gamma_k^2) + 2\eta_k^{-1}\gamma_k\alpha_k.$$

For the choice of step size $\eta_k = 2/(\mu k)$, it holds that

$$\frac{1 - \mu\eta_k}{\eta_k^2} = \frac{\mu^2 k^2(1 - 2/k)}{4} = \frac{\mu^2(k^2 - 2k)}{4} = \frac{\mu^2((k-1)^2 - 1)}{4} \leq \frac{\mu^2(k-1)^2}{4} = \eta_{k-1}^{-2}.$$

Setting $e_j := \eta_{j-1}^{-2}\|x^\star - x_j\|^2$ for all $j \in \mathbb{N}$, we thus have

$$2\eta_k^{-1}(f(x_k) - f(x^\star)) \leq e_k - e_{k+1} + 2\eta_k^{-1}\langle z_k, x_k - x^\star\rangle + O(\gamma_k^2) + 2\eta_k^{-1}\gamma_k\alpha_k$$

almost surely, for all $k \in \mathbb{N}$. Summing both sides over $k = k_0, \dots, K$ for $K \in \mathbb{N}$ yields

$$\begin{aligned}
2\sum_{k=k_0}^{K} \eta_k^{-1}(f(x_k) - f(x^\star)) &\leq e_{k_0} - e_K + 2\sum_{k=k_0}^{K} \eta_k^{-1}\langle z_k, x_k - x^\star\rangle + 2\sum_{k=k_0}^{K} O(\gamma_k^2) \\
&\quad + 2\sum_{k=k_0}^{K} \eta_k^{-1}\gamma_k\alpha_k \\
&\leq e_{k_0} + 2\sum_{k=k_0}^{K} \eta_k^{-1}\langle z_k, x_k - x^\star\rangle + 2\sum_{k=k_0}^{K} O(\gamma_k^2) \\
&\quad + 2\sum_{k=k_0}^{K} \eta_k^{-1}\gamma_k\alpha_k,
\end{aligned}$$

where we used $e_K \geq 0$ in the second step. We define $S_{t,k} := \sum_{i=t}^{k} \eta_i^{-1}$, $S_k := S_{1,k}$, and $\bar{x}_{t,k} := S_{t,k}^{-1}\sum_{i=t}^{k} \eta_i^{-1} x_i$ for $t, k \in \mathbb{N}$. Using convexity of f, we get

$$\begin{aligned}
f(\bar{x}_{k_0,K}) - f(x^\star) &\leq S_K^{-1}\sum_{k=k_0}^{K} \eta_k^{-1} f(x_k) - f(x^\star) \\
&\leq \frac{e_{k_0}}{2S_{k_0,K}} + \frac{\sum_{k=k_0}^{K} \eta_k^{-1}\langle z_k, x_k - x^\star\rangle}{S_{k_0,K}} + \frac{\sum_{k=k_0}^{K} O(\gamma_k^2)}{S_{k_0,K}} \\
&\quad + \frac{\sum_{k=k_0}^{K} \eta_k^{-1}\gamma_k\alpha_k}{S_{k_0,K}},
\end{aligned}$$

for all $K \in \mathbb{N}$, as desired. Recall the relations (3.14) and (3.15). Using convexity of f, we can combine these with the latest bound on $(f(\bar{x}_{k_0,K}) - f(x^\star))$, and arrive at

$$\begin{aligned}
f(\bar{x}_K) - f(x^\star) \leq \frac{S_{1,k_0-1}}{S_K}(f(\bar{x}_{1,k_0-1}) - f(x^\star)) + \frac{e_{k_0}}{2S_K} + \frac{\sum_{k=k_0}^{K} \eta_k^{-1}\langle z_k, x_k - x^\star\rangle}{S_K} \\
+ \frac{\sum_{k=k_0}^{K} O(\gamma_k^2)}{S_K} + \frac{\sum_{k=k_0}^{K} \eta_k^{-1}\gamma_k\alpha_k}{S_K}.
\end{aligned}$$

Next, we will derive the desired bound for $\operatorname{dist}(\bar{x}_K, \mathcal{X})$. Our starting point is inequality (3.17) from the proof of theorem 3.15. Multiplying both sides by $\gamma_k^{-1}\eta_k^{-2}$ yields

$$\begin{aligned}
\gamma_k^{-1}\eta_k^{-2}\operatorname{dist}(x_{k+1}, \mathcal{X})^2 &\leq \gamma_k^{-1}\eta_k^{-2}(1 - \mu\eta_k)\operatorname{dist}(x_k, \mathcal{X})^2 + O(\gamma_k) \\
&\quad + 2\gamma_k^{-1}\eta_k^{-1}\langle z_k, x_k - \Pi_{\mathcal{X}}(x_k)\rangle + 2\eta_k^{-1}\alpha_k - \tau\eta_k^{-1}\operatorname{dist}(x_k, \mathcal{X}),
\end{aligned}$$

for all $k \geq k_1$. We have already shown that $\eta_k^{-2}(1 - \mu\eta_k) \leq \eta_{k-1}^2$ for all $k \in \mathbb{N}$. Also,

since γ_k is nondecreasing, we have $\gamma_k^{-1} \geq \gamma_{k+1}^{-1}$ for all $k \in \mathbb{N}$. Combining these two facts, we get

$$\gamma_{k+1}^{-1}\eta_k^{-2}\operatorname{dist}(x_{k+1},\mathcal{X})^2 \leq \gamma_k^{-1}\eta_{k-1}^2\operatorname{dist}(x_k,\mathcal{X})^2 + O(\gamma_k) + 2\gamma_k^{-1}\eta_k^{-1}\langle z_k, x_k - \Pi_{\mathcal{X}}(x_k)\rangle \\ + 2\eta_k^{-1}\alpha_k - \tau\eta_k^{-1}\operatorname{dist}(x_k,\mathcal{X}),$$

for all $k \geq k_1$. Setting $d_j := \gamma_j^{-1}\eta_{j-1}^{-2}\operatorname{dist}(x_j,\mathcal{X})^2 \ \forall j \in \mathbb{N}$ and rearranging again, we obtain

$$\tau\eta_k^{-1}\operatorname{dist}(x_k,\mathcal{X}) \leq d_k - d_{k+1} + O(\gamma_k) + 2\eta_k^{-1}\alpha_k + 2\gamma_k^{-1}\eta_k^{-1}\langle z_k, x_k - \Pi_{\mathcal{X}}(x_k)\rangle,$$

for all $k \geq k_1$. Summing over $k = k_1, \dots, K$ for $K \in \mathbb{N}$, we have

$$\begin{aligned}\tau\sum_{k=k_1}^{K}\eta_k^{-1}\operatorname{dist}(x_k,\mathcal{X}) &\leq d_{k_1} - d_{K+1} + \sum_{k=k_1}^{K} O(\gamma_k) + 2\sum_{k=k_1}^{K}\eta_k^{-1}\alpha_k \\ &\quad + 2\sum_{k=k_1}^{K}\gamma_k^{-1}\eta_k^{-1}\langle z_k, x_k - \Pi_{\mathcal{X}}(x_k)\rangle \\ &\leq d_{k_1} + \sum_{k=k_1}^{K} O(\gamma_k) + 2\sum_{k=k_1}^{K}\eta_k^{-1}\alpha_k \\ &\quad + 2\sum_{k=k_1}^{K}\gamma_k^{-1}\eta_k^{-1}\langle z_k, x_k - \Pi_{\mathcal{X}}(x_k)\rangle,\end{aligned}$$

where we used $d_{K+1} \geq 0$ in the last step. The distance functional $x \mapsto \operatorname{dist}(x,\mathcal{X})$ is convex (proposition 2.60),), so if we divide both sides of the above inequality by $\tau \cdot S_{k_1,K}$, we obtain

$$\begin{aligned}\operatorname{dist}(\bar{x}_{k_1,K},\mathcal{X}) \leq \frac{d_{k_1}}{\tau S_{k_1,K}} &+ \frac{\sum_{k=k_1}^{K} O(\gamma_k)}{\tau S_{k_1,K}} + 2\frac{\sum_{k=k_1}^{K}\eta_k^{-1}\alpha_k}{\tau S_{k_1,K}} \\ &+ 2\frac{\sum_{k=k_1}^{K}\gamma_k^{-1}\eta_k^{-1}\langle z_k, x_k - \Pi_{\mathcal{X}}(x_k)\rangle}{\tau S_{k_1,K}}.\end{aligned}$$

To derive a bound for $\operatorname{dist}(\bar{x}_K,\mathcal{X})$, we proceed similarly as before. Recall again the relations (3.14) and (3.15). Using convexity of $\operatorname{dist}(\cdot,\mathcal{X})$, we can combine these with

the latest bound on $\operatorname{dist}(\bar{x}_{k_1,K}, \mathcal{X})$, and arrive at

$$\begin{aligned}\operatorname{dist}(\bar{x}_K, \mathcal{X}) &\leq \frac{S_{1,k_1-1}}{S_K} \operatorname{dist}(\bar{x}_{1,k_1-1}, \mathcal{X}) + \frac{S_{k_1,K}}{S_K} \operatorname{dist}(\bar{x}_{k_1,K}, \mathcal{X}) \\ &\leq \frac{S_{1,k_1-1}}{S_K} \operatorname{dist}(\bar{x}_{1,k_1-1}, \mathcal{X}) + \frac{d_{k_1}}{\tau S_K} + \frac{\sum_{k=k_1}^K \mathcal{O}(\gamma_k)}{\tau S_K} + 2\frac{\sum_{k=k_1}^K \eta_k^{-1}\alpha_k}{\tau S_K} \\ &\quad + 2\frac{\sum_{k=k_1}^K \gamma_k^{-1}\eta_k^{-1}\langle z_k, x_k - \Pi_{\mathcal{X}}(x_k)\rangle}{\tau S_K}\end{aligned}$$

almost surely, for all $K \geq k_1$. This concludes the proof. □

From theorem 3.18, we can derive the following improved convergence rates in the strongly convex case.

Corollary 3.19 (Convergence rates in expectation - strongly convex case)**.** Let the assumptions of theorem 3.18 hold and assume that we are in the setting $\mu > 0$. Let the parameters $(\eta_k)_{k\in\mathbb{N}_0}, (\gamma_k)_{k\in\mathbb{N}_0}, (h_k)_{k\in\mathbb{N}_0}$ be the same as in lemma 3.17. Further, let the iterate weights $(w_k)_{k\in\mathbb{N}}$ be defined as $w_k := \eta_k^{-1}$ for all $k \in \mathbb{N}$. Then the averaged iterate $\bar{x}_K$, generated by algorithm 3 with the parameters $(\eta_k)_{k\in\mathbb{N}_0}$, $(\gamma_k)_{k\in\mathbb{N}_0}$, $(h_k)_{k\in\mathbb{N}_0}$, $(w_k)_{k\in\mathbb{N}}$, satisfies

$$\mathbb{E}|f(\bar{x}_K) - f(x^\star)| = \tau^{-1}\mathcal{O}\Big(\frac{\log^{2e}(K)}{K}\Big).$$

Further, it also holds that

$$\mathbb{E}(\operatorname{dist}(\bar{x}_K, \mathcal{X})) = \tau^{-1}\mathcal{O}\Big(\frac{\log^{e}(K)}{K}\Big).$$

Proof. Theorem 3.18 gives us an upper bound for $f(\bar{x}_K) - f(x^\star)$,

$$\begin{aligned}f(\bar{x}_K) - f(x^\star) \leq \frac{S_{1,k_0-1}}{S_K}(f(\bar{x}_{1,k_0-1}) - f(x^\star)) + \frac{e_{k_0}}{2S_K} + \frac{\sum_{k=k_0}^K \eta_k^{-1}\langle z_k, x_k - x^\star\rangle}{S_K} \\ + \frac{\sum_{k=k_0}^K \mathcal{O}(\gamma_k^2)}{S_K} + \frac{\sum_{k=k_0}^K \eta_k^{-1}\gamma_k\alpha_k}{S_K},\end{aligned}$$

almost surely, for all $K \in \mathbb{N}$ with $K \geq k_0$, where $e_{k_0} := \eta_{k_0-1}^{-2}\|x^\star - x_{k_0}\|^2$. Note that

$$\mathbb{E}(\langle z_k, x_k - x^\star\rangle) = \mathbb{E}(\mathbb{E}_k(\langle z_k, x_k - x^\star\rangle)) = \mathbb{E}(\langle \mathbb{E}_k(z_k), x_k - x^\star\rangle) = 0,$$

for all $k \in \mathbb{N}$. Hence, taking expectations on both sides of the above inequality, we

get

$$\mathbb{E}(f(\bar{x}_K) - f(x^\star)) \leq \frac{S_{1,k_0-1}}{S_K}\mathbb{E}(f(\bar{x}_{1,k_0-1}) - f(x^\star)) + \frac{\mathbb{E}(e_{k_0})}{2S_K} + \frac{\sum_{k=k_0}^K \mathcal{O}(\gamma_k^2)}{S_K} + \frac{\sum_{k=k_0}^K \eta_k^{-1}\gamma_k\alpha_k}{S_K}. \tag{3.22}$$

By convexity, we have, for all $k \in \mathbb{N}$,

$$\begin{aligned} f(\bar{x}_k) - f(x^\star) &\geq \langle \tilde{\nabla} f(x^\star), \bar{x}_k - x^\star \rangle \\ &= \langle \tilde{\nabla} f(x^\star), \bar{x}_k - \Pi_{\mathcal{X}}(\bar{x}_k) \rangle + \langle \tilde{\nabla} f(x^\star), \Pi_{\mathcal{X}}(\bar{x}_k) - x^\star \rangle, \end{aligned}$$

for any subgradient $\tilde{\nabla} f(x^\star) \in \partial f(x^\star)$. Optimality of $x^\star$ for f on $\mathcal{X}$ implies that there exists a subgradient $\tilde{\nabla} f(x^\star) \in \partial f(x^\star)$ such that $\langle \tilde{\nabla} f(x^\star), \Pi_{\mathcal{X}}(\bar{x}_k) - x^\star \rangle \geq 0$. Combining these facts with the above inequality, and applying Cauchy-Schwarz, we obtain

$$f(\bar{x}_k) - f(x^\star) \geq -\|\tilde{\nabla} f(x^\star)\| \operatorname{dist}(\bar{x}_k, \mathcal{X}) \tag{3.23}$$

for all $k \in \mathbb{N}$. Taking expectations in the bound for $\operatorname{dist}(\bar{x}_K, \mathcal{X})$ from theorem 3.18, we know that there exists $k_1 \in \mathbb{N}$ such that for all $K \in \mathbb{N}$ with $K \geq k_1$, it holds that

$$\mathbb{E}(\operatorname{dist}(\bar{x}_K, \mathcal{X})) \leq \frac{S_{1,k_1-1}}{S_K}\mathbb{E}(\operatorname{dist}(\bar{x}_{1,k_1-1}, \mathcal{X})) + \frac{\mathbb{E}(d_{k_1})}{\tau S_K} + \frac{\sum_{k=k_1}^K \mathcal{O}(\gamma_k)}{\tau S_K} + 2\frac{\sum_{k=k_1}^K \eta_k^{-1}\alpha_k}{\tau S_K},$$

where $d_{k_1} := \gamma_{k_1}^{-1}\eta_{k_1-1}^{-2}\operatorname{dist}(x_{k_1}, \mathcal{X})^2$. The first two terms are constants. Since $\eta_k^{-1} = (\mu k)/2$, it follows that $S_K = \mathcal{O}(K^2)$. Also, since $\alpha_k = \mathcal{O}(1/k)$, we have $\eta_k^{-1}\alpha_k = \mathcal{O}(1)$. Therefore,

$$\begin{aligned} \mathbb{E}(\operatorname{dist}(\bar{x}_K, \mathcal{X})) &= \mathcal{O}(K^{-2}) + \mathcal{O}\Big(K^{-2}\sum_{k=1}^K \log^e(k)\Big) + \mathcal{O}(K^{-1}) \\ &= \mathcal{O}(\tau^{-1}K^{-2}) + \mathcal{O}(\tau^{-1}\log^e(K)/K) + \mathcal{O}(\tau^{-1}K^{-1}) \\ &= \tau^{-1}\mathcal{O}\Big(\frac{\log^e(K)}{K}\Big). \end{aligned}$$

Combining with (3.23), we obtain

$$\mathbb{E}(f(\bar{x}_k) - f(x^\star)) \geq -\|\nabla f(x^\star)\| \cdot \mathcal{O}\Big(\frac{\log^{2e}(K)}{\tau K}\Big).$$

Note that $\mathcal{O}(\gamma_k^2) = \mathcal{O}(\log^{2e}(k))$ and $\eta_k^{-1}\gamma_k\alpha_k = \mathcal{O}(\log^e(k))$, so for the upper bound in

(3.22), we find that

$$\begin{aligned}\mathbb{E}(f(\bar{x}_K) - f(x^\star)) &\leq \mathcal{O}(K^{-2}) + \mathcal{O}\Big(K^{-2}\sum_{k=1}^{K}\log^{2e}(k)\Big) + \mathcal{O}\Big(K^{-2}\sum_{k=1}^{K}\log^{e}(k)\Big)\\ &= \mathcal{O}\Big(\frac{\log^{2e}(K)}{K}\Big).\end{aligned}$$

Putting the two bounds together, we obtain

$$\mathbb{E}|f(\bar{x}_K) - f(x^\star)| = \tau^{-1}\,\mathcal{O}\Big(\frac{\log^{2e}(K)}{K}\Big),$$

as desired. □

This concludes the section on convergence rates in expectation. Next, we will consider a different kind of notion for convergence.

3.4. High-probability guarantees

In the previous section, we proved that the expected values $\mathbb{E}|f(\bar{x}_k) - f(x^\star)|$ and $\operatorname{dist}(\bar{x}_k, \mathcal{X})$ converge to zero as $k \to \infty$, provided the parameters $(\eta_k)_{k\in\mathbb{N}_0}$, $(\gamma_k)_{k\in\mathbb{N}_0}$, $(h_k)_{k\in\mathbb{N}_0}$, and $(w_k)_{k\in\mathbb{N}}$ are suitably chosen. We will now turn our attention towards establishing *high-probability guarantees* for convergence, in contrast to guarantees that hold only in expectation. In particular, we will investigate under which conditions we can guarantee that an iterate $\bar{x}_k$ is both a) close to the feasible set and b) has function value close to the optimal value $f(x^\star)$ with high probability. This notion of "closeness with high probability" is formalized in the following definition.

Definition 3.20 ((ϵ, δ)-solution)**.** Let $\epsilon \in (0, \infty)$ and $\delta \in (0, 1)$. We call a random variable $x\colon \Omega \to \mathbb{R}^n$ an (ϵ, δ)**-solution** of (P), if

$$\mathbb{P}\Big(\max\big(|f(x) - f(x^\star)|, \operatorname{dist}(x, \mathcal{X})\big) \geq \epsilon\Big) \leq \delta.$$

In other words, an iterate x_k of algorithm 3 is an (ϵ, δ)-solution, if we can guarantee that $|f(x_k) - f(x^\star)| < \epsilon$ and $\operatorname{dist}(x_k, \mathcal{X}) < \epsilon$ with probability greater than $1 - \delta$. We can now state the central question we aim to answer in this section as follows:

For any given (ϵ, δ), how many iterations are needed to guarantee that the sequence $(\bar{x}_k)_{k\in\mathbb{N}}$ generated by algorithm 3 reaches an (ϵ, δ)-solution of (P)*?*

Our approach to answer this question is standard. We rely on the Azuma-Hoeffding

inequality (proposition 2.36) and the almost-sure bounds we derived in the previous sections.

For the following lemma, recall the definition of $z_k = \nabla\psi_k(x_k) - g_k$, where $k \in \mathbb{N}$, x_k is the k-th iterate of algorithm 3, and g_k is the stochastic gradient of ψ_k at x_k used in algorithm 3.

Lemma 3.21. Let $(\nu_k)_{k\in\mathbb{N}}$ be an arbitrary sequence of real numbers and let $(x_k)_{k\in\mathbb{N}_0}$ be the iterates produced by algorithm 3 for some arbitrary parameter choice. Further, let $(Y_k)_{k\in\mathbb{N}}$ be a bounded sequence of random variables such that $Y_k = Y(x_k)$ for some measurable function $Y\colon \mathbb{R}^n \to \mathbb{R}^n$, for all $k \in \mathbb{N}$. Then the stochastic process $(X_k)_{k\in\mathbb{N}_0}$, defined by $X_0 := 0$ and

$$X_k := \sum_{i=1}^{k} \nu_i \langle z_i, Y_i \rangle$$

for $k \in \mathbb{N}$, is a martingale with respect to the filtration $(\mathcal{G}_k)_{k\in\mathbb{N}_0}$, defined by $\mathcal{G}_k := \sigma(z_0, \dots, z_k)\ \forall k \in \mathbb{N}_0$.

Proof. First, note that Y_k is $\sigma(z_0, z_1 \dots, z_{k-1})$-measurable for all $k \in \mathbb{N}$. Indeed, for $k = 1$, we have

$$x_1 = x_0 + \eta_0 g_0 = x_0 + \eta_0(z_0 + \nabla\psi_0(x_0)),$$

and the quantities $x_0, \nabla\psi_0(x_0)$, and η_0 are deterministic. Assuming that x_k is a function of $z_0, z_1, \dots, z_{k-1}$, it follows inductively that

$$x_{k+1} = x_k + \eta_k g_k = x_k + \eta_k(z_k + \nabla\psi_k(x_k))$$

is a function of $z_1, \dots, z_k$. Per assumption of the lemma, Y_k is a (measurable) function of x_k, and hence Y_k is $\sigma(z_0, \dots, z_{k-1})$-measurable. Thus, the process $(X_k)_{k\in\mathbb{N}_0}$ is adapted to the filtration $(\mathcal{G}_k)_{k\in\mathbb{N}_0}$. With probability one we also have, for all $k \in \mathbb{N}_0$,

$$|X_k| \le \sum_{i=1}^{k} |\nu_i \langle z_i, Y_i \rangle| \le \sum_{i=1}^{k} \nu_i \|z_i\| \|Y_i\| = \sum_{i=1}^{k} \nu_i O(\gamma_i) O(1) \le \nu_k O(k\gamma_k),$$

where the third step follows from lemma 3.12 and the assumption that $(Y_k)_{k\in\mathbb{N}}$ is bounded. In particular,

$$\mathbb{E}|X_k| < \infty$$

for all $k \in \mathbb{N}_0$. Finally, we have

$$\begin{aligned}
\mathbb{E}(X_k \mid \mathcal{G}_{k-1}) &= \mathbb{E}\big(\nu_k\langle z_k, Y_k\rangle + X_{k-1} \mid \mathcal{G}_{k-1}\big) \\
&= \nu_k\mathbb{E}\big(\langle z_k, Y_k\rangle \mid \mathcal{G}_{k-1}\big) + X_{k-1} \\
&= \nu_k\big\langle \mathbb{E}(z_k \mid \mathcal{G}_{k-1}), Y_k\big\rangle + X_{k-1} \\
&= \nu_k\big\langle \mathbb{E}(z_k \mid x_k), Y_k\big\rangle + X_{k-1} \\
&= X_{k-1},
\end{aligned}$$

where we used that x_k is a function of $z_0, \dots, z_{k-1}$ and, per definition of stochastic gradients, $\mathbb{E}(z_k \mid x_k) = 0$. Therefore, $(X_k)_{k\in\mathbb{N}_0}$ is a martingale. □

We can now prove the main theorem of this section.

Theorem 3.22 (Convergence with high probability - convex case)**.** Let the assumptions of theorem 3.15 hold. Then, for arbitrary $\delta \in (0, \infty)$ and $K \in \mathbb{N}$, the averaged iterate $\bar{x}_K$, generated by algorithm 3 with the parameters $(\eta_k)_{k\in\mathbb{N}_0}$, $(\gamma_k)_{k\in\mathbb{N}_0}$, $(h_k)_{k\in\mathbb{N}_0}$, $(w_k)_{k\in\mathbb{N}}$, satisfies the bounds

$$|f(\bar{x}_K) - f(x^\star)| \le \delta\tau^{-1}\,\mathcal{O}\Big(\frac{\log^{(2+3e)/2}(K)}{\sqrt{K}}\Big)$$

and

$$\operatorname{dist}(\bar{x}_K, \mathcal{X}) \le \delta\tau^{-1}\,\mathcal{O}\Big(\frac{\log^{(2+3e)/2}(K)}{\sqrt{K}}\Big)$$

with probability at least $1 - \exp(-\delta^2/2)$.

Proof. Let $K \in \mathbb{N}$. From theorem 3.15, we have the almost-sure bound

$$\begin{aligned}
f(\bar{x}_K) - f(x^\star) \le \frac{S_{1,k_0-1}}{S_K}(f(\bar{x}_{1,k_0-1}) - f(x^\star)) + \frac{\|x^\star - x_{k_0}\|^2}{2S_K} + \frac{\sum_{k=k_0}^K \eta_k^2\mathcal{O}(\gamma_k^2)}{2S_K} \\
+ \frac{\sum_{k=k_0}^K \eta_k\gamma_k\alpha_k}{S_K} + \frac{\sum_{k=k_0}^K \eta_k\langle z_k, x_k - x^\star\rangle}{S_K}.
\end{aligned}$$

By theorem 3.10, the numerators in the first two terms are bounded almost surely, whereas the sum of the last two terms is on the order $\mathcal{O}\big(\log^{(1+3e)/2}(K)/\sqrt{K}\big)$, as argued in the proof of corollary 3.16. Thus,

$$f(\bar{x}_K) - f(x^\star) \le \mathcal{O}\Big(\frac{\log^{(1+3e)/2}(K)}{\sqrt{K}}\Big) + \frac{\sum_{k=k_0}^K \eta_k\langle z_k, x_k - x^\star\rangle}{S_K}$$

almost surely. To prove the claim, we will first bound the probability that the random

variable

$$\sum_{k=k_0}^{K} \nu_k \langle z_k, x_k - x^\star \rangle$$

is large, where

$$\nu_k := \frac{\eta_k}{S_K}.$$

The sequence $(x_k)_{k\in\mathbb{N}_0}$ is bounded almost surely by theorem 3.10. Therefore, lemma 3.21 implies that $(X_k)_{k\in\mathbb{N}_0}$ is a martingale, and we also have

$$|X_k - X_{k-1}| = \nu_k |\langle z_k, x_k - x^\star \rangle| \leq \nu_k \|z_k\| \|x_k - x^\star\| =: c_k$$

for all $k \in \mathbb{N}$. We can thus apply the Azuma-Hoeffding inequality (proposition 2.36) to obtain

$$\mathbb{P}\Big(\sum_{k=1}^{K} \nu_k \langle z_k, x_k - x^\star \rangle \geq t \Big) \leq \exp\Big(\frac{-t^2}{2\sum_{k=1}^{K} c_k^2} \Big)$$

for all $t \in (0, \infty)$. By lemma 3.12, it holds that $\|z_k\| \leq M \log^e(k)\, \forall k \in \mathbb{N}_0$, where $M := 4 \max(1, \min_{k\in\mathbb{N}_0} \gamma_k) \max(L_{\text{loc}}, G)$. Further, using (3.21), we have

$$\nu_k = \frac{\eta_k}{S_K} = \mathcal{O}\Big(\frac{1}{\log^{(1+3e)/2}(k)\sqrt{k} S_K} \Big) = \mathcal{O}\Big(\frac{\log^{(1+3e)/2}(K)}{\sqrt{k}\sqrt{K}} \Big)$$

for all $k \in \{1, \dots K\}$. Let $R \in (0, \infty)$ be large enough such that $\sup_{k\in\mathbb{N}} \|x_k - x^\star\| \leq R$ almost surely. We obtain

$$c_k = \nu_k \|z_k\| \|x_k - x^\star\| \leq \mathcal{O}\Big(\frac{MR \log^{(1+3e)/2}(K)}{\sqrt{k}\sqrt{K}} \Big),$$

which implies

$$\sum_{k=1}^{K} c_k^2 \leq \mathcal{O}\Big(\frac{M^2 R^2 \log^{1+3e}(K)}{K} \Big) \sum_{k=1}^{K} \frac{1}{k} = \mathcal{O}\Big(\frac{M^2 R^2 \log^{2+3e}(K)}{K} \Big)$$

almost surely, where the last step follows from

$$\sum_{k=1}^{K} \frac{1}{k} = \mathcal{O}(\log(K)).$$

Hence, it holds that

$$\mathbb{P}\Big(\sum_{k=1}^{K} \nu_k \langle z_k, x_k - x^\star\rangle \geq t\Big) \leq \exp\left(-\frac{1}{2}t^2 \mathcal{O}\Big(\frac{K}{M^2R^2\log^{2+3e}(K)}\Big)\right)$$

for all $t \in (0,\infty)$. Substituting $t := \delta\,\mathcal{O}(MR\log^{(2+3e)/2}(K)/\sqrt{K})$ for arbitrary $\delta \in (0,1)$, we obtain

$$\mathbb{P}\left(\sum_{k=1}^{K} \nu_k \langle z_k, x_k - x^\star\rangle \geq \delta\,\mathcal{O}\Big(\frac{MR\log^{(2+3e)/2}(K)}{\sqrt{K}}\Big)\right) \leq \exp\Big(-\frac{1}{2}\delta^2\Big).$$

Therefore,

$$f(\bar{x}_K) - f(x^\star) \leq \mathcal{O}\Big(\frac{\log^{2e}(K)}{K}\Big) + \delta\,\mathcal{O}\Big(\frac{MR\log^{(2+3e)/2}(K)}{\sqrt{K}}\Big) \tag{3.24}$$

with probability at least $1 - \exp(-\delta^2/2)$. To derive a lower bound, we will first prove the second claim, which follows from essentially the same argument. For $k \in \mathbb{N}$, define

$$\tilde{\nu}_k := \frac{\gamma_k^{-1}\eta_k}{\tau S_K}$$

and

$$\tilde{X}_k := \sum_{k=1}^{K} \tilde{\nu}_k \langle z_k, x_k - \Pi_{\mathcal{X}}(x_k)\rangle,$$

as well as $\tilde{X}_0 := 0$. The process $(x_k - \Pi_{\mathcal{X}}(x_k))_{k\in\mathbb{N}}$ is a function of x_k, and bounded by theorem 3.10 and continuity of $\Pi_{\mathcal{X}}$ (proposition 2.57). Hence, applying lemma 3.21, we find that $(\tilde{X}_k)_{k\in\mathbb{N}_0}$ is a martingale. Further, we have

$$|\tilde{X}_k - \tilde{X}_{k-1}| = \tilde{\nu}_k |\langle z_k, x_k - \Pi_{\mathcal{X}}(x_k)\rangle| \leq \tilde{\nu}_k \|z_k\| \operatorname{dist}(x_k, \mathcal{X}) =: d_k$$

for all $k \in \mathbb{N}$. It holds that

$$\tilde{\nu}_k = \frac{\gamma_k^{-1}\eta_k}{\tau S_K} \leq (\tau\gamma_0)^{-1}\frac{\eta_k}{S_K} = (\tau\gamma_0)^{-1}\nu_k \leq (\tau\gamma_0)^{-1}\mathcal{O}\Big(\frac{\log^{(1+3e)/2}(K)}{k^{1/2}K^{1/2}}\Big)$$

for all $k \in \mathbb{N}$. Letting $\tilde{R} \in (0,\infty)$ be large enough such that $\sup_{k\in\mathbb{N}} \operatorname{dist}(x_k, \mathcal{X}) \leq \tilde{R}$ a. s., we thus have

$$d_k \leq (\tau\gamma_0)^{-2}\,\mathcal{O}\Big(\frac{M\tilde{R}\log^{(1+3e)/2}(K)}{k^{1/2}K^{1/2}}\Big)$$

for all $k \in \mathbb{N}$. It follows that

$$\sum_{k=1}^{K} d_k^2 \leq (\tau\gamma_0)^{-2} \mathcal{O}\Big(\frac{M^2\tilde{R}^2 \log^{1+3e}(K)}{K}\Big) \sum_{k=1}^{K} \frac{1}{k} = (\tau\gamma_0)^{-2} \mathcal{O}\Big(\frac{M^2\tilde{R}^2 \log^{2+3e}(K)}{K}\Big).$$

Applying the Azuma-Hoeffding inequality (proposition 2.36), we obtain

$$\begin{aligned}\mathbb{P}\Big(\sum_{k=1}^{K} \tilde{v}_k \langle z_k, x_k - \Pi_{\mathcal{X}}(x_k)\rangle \geq t\Big) &\leq \exp\Big(\frac{-t^2}{2\sum_{k=1}^{K} d_k^2}\Big) \\ &\leq \exp\left(-\frac{1}{2} t^2 \mathcal{O}\Big(\frac{\tau_k^2 \gamma_0^2 K}{2M^2\tilde{R}^2 \log^{2+3e}(K)}\Big)\right)\end{aligned}$$

for all $t \in (0, \infty)$ and $K \in \mathbb{N}$. Substituting $t := \delta\, \mathcal{O}(M\tilde{R} \log^{(2+3e)/2}(K)/\tau\gamma_0\sqrt{K})$ for arbitrary $\delta \in (0, \infty)$, we obtain

$$\mathbb{P}\left(\sum_{k=1}^{K} \tilde{v}_k \langle z_k, x_k - \Pi_{\mathcal{X}}(x_k)\rangle \geq \delta(\tau\gamma_0)^{-1} \mathcal{O}\Big(\frac{M\tilde{R} \log^{(2+3e)/2}(K)}{\sqrt{K}}\Big)\right) \leq \exp\Big(-\frac{1}{2}\delta^2\Big)$$

for all $K \in \mathbb{N}$. We therefore have, for any $\delta \in (0, \infty)$ and $K \in \mathbb{N}$,

$$\begin{aligned}\operatorname{dist}(\bar{x}_K, \mathcal{X}) &\leq \tau^{-1} \mathcal{O}\Big(\frac{\log^{(1+3e)/2}(K)}{\sqrt{K}}\Big) + 2\delta(\tau\gamma_0)^{-1} \mathcal{O}\Big(\frac{M\tilde{R} \log^{(2+3e)/2}(K)}{\sqrt{K}}\Big) \\ &= \delta\tau^{-1} \mathcal{O}\Big(\frac{M\tilde{R} \log^{(2+3e)/2}(K)}{\sqrt{K}}\Big) \qquad (3.25)\end{aligned}$$

with probability at least $1 - \exp(-\delta^2/2)$. Note that for any valid choice of R, we can choose $\tilde{R}$ such that $\tilde{R} \leq R$, which brings us to the desired bound. We can derive a lower bound on $f(\bar{x}_K) - f(x^\star)$ via

$$\begin{aligned}f(\bar{x}_K) - f(x^\star) &\geq \langle \tilde{\nabla} f(x^\star), \bar{x}_K - x^\star\rangle \\ &= \langle \tilde{\nabla} f(x^\star), \bar{x}_K - \Pi_{\mathcal{X}}(\bar{x}_K)\rangle + \langle \tilde{\nabla} f(x^\star), \Pi_{\mathcal{X}}(\bar{x}_K) - x^\star\rangle \\ &\geq \langle \tilde{\nabla} f(x^\star), \bar{x}_K - \Pi_{\mathcal{X}}(\bar{x}_K)\rangle,\end{aligned}$$

where the first step follows from convexity, and the second from optimality of $x^\star$ on $\mathcal{X}$. Cauchy-Schwarz now implies

$$f(\bar{x}_K) - f(x^\star) \geq -\|\tilde{\nabla} f(x^\star)\| \operatorname{dist}(\bar{x}_K, \mathcal{X}).$$

Combining with the high-probability bound (3.25), we have, for any $\delta \in (0, \infty)$,

$$f(\bar{x}_K) - f(x^\star) \geq -\tau^{-1}\|\tilde{\nabla} f(x^\star)\| \delta \tau^{-1} \mathcal{O}\Big(\frac{M\tilde{R}\log^{(2+3e)/2}(K)}{\sqrt{K}}\Big)$$

with probability at least $1 - \exp(-\delta^2/2)$. Note that we can choose $\tilde{R} \leq R$ without loss of generality. Hence, by combining with the upper bound (3.24), we have for any $\delta \in (0, \infty)$ and all $K \in \mathbb{N}$,

$$|f(\bar{x}_K) - f(x^\star)| \leq \tau^{-1} \mathcal{O}\Big(\frac{MR\log^{(2+3e)/2}(K)}{\sqrt{K}}\Big)$$

with probability at least $1 - \exp(-\delta^2/2)$, as desired. □

With theorem 3.22 in hand, we can now immediately answer the question we posed at the beginning of this subsection.

Corollary 3.23 (Time until (ϵ, δ)-solution)**.** Let the assumptions of theorem 3.22 hold. Then there exists a constant $c \in (0, \infty)$ such that, for any $\epsilon \in (0, \infty)$, $\delta \in (0, 1)$, and all $K \in \mathbb{N}$ with

$$\frac{K}{\log^{2+3e}(K)} \geq c\frac{2\log(\delta^{-1})}{\epsilon^2\tau^2},$$

the output $\bar{x}_K$ of algorithm 3 is an (ϵ, δ)-solution of (P).

4

Numerical Experiments

In this chapter, we will apply algorithm 3 to two concrete settings. First, in section 4.1, we will consider the primal hard-margin support vector machine problem, which is a popular method in machine learning for classification. Then, in section 4.2, we will consider the standard control problem of inventory management subject to uncertainty in demand projections. The experiments are implemented in PyTorch [30], and the source code can be found on github. Code development was partially assisted by OpenAI Codex [31].

4.1. Classification with support vector machines

A common problem in machine learning is binary classification, where we are given a dataset $D := \{ (a_i, y_i) \mid i \in \{ 1, \dots, N \} \}$ of size $N \in \mathbb{N}$, where each $a_i \in \mathbb{R}^n$ is called *feature* and each $y_i \in \{-1, 1\}$ is called *label*. As a specific example, one can think of the problem of classifying images (the features) of pets as either showing a dog (label 1) or a cat (label -1). The features and labels can be seen as samples from an underlying probability distribution $\mathbb{P}$ on the space $\mathbb{R}^n \times \{-1, 1\}$. Given this dataset, we aim to learn a rule to classify new samples $a_{n+1} \in \mathbb{R}^n$ from the distribution of the features as either 0 or 1. Ideally this rule would choose the class that maximizes $\mathbb{P}(a_{n+1}, \cdot)$. One approach to find such a rule is called *support vector machine* (SVM), which is an algorithm for constructing a hyperplane that seperates the two subsets $\{ a \in \mathbb{R}^n \mid (a, -1) \in D \}$ and $\{ a \in \mathbb{R}^n \mid (a, 1) \in D \}$. New samples are then classified based on which side of the hyperplane they lie in. Whether or not such a hyperplane even exists depends on the nature of the dataset. Nevertheless, we can always write down the problem of finding such a hyperplane as a convex optimization problem

with affine constraints:

$$\begin{aligned} &\min_{x\in\mathbb{R}^n} \frac{1}{2}\|x\|^2 \\ &\text{s.t.} \quad y_i\langle a_i, x\rangle \geq 1 \quad \text{for all } i \in \{1, \dots, N\}. \end{aligned} \tag{SVM}$$

If a solution to the above problem exists, we say that the dataset is *linearly seperable*. This is often not the case, which is why the above problem is usually relaxed to the unconstrained problem

$$\min_{x\in\mathbb{R}^n} \frac{1}{2}\|x\|^2 + C\sum_{i=1}^{N} \max(0, 1 - y_i\langle a_i, x\rangle),$$

where $C \in (0, \infty)$. This problem always has a solution, regardless of whether or not the dataset is actually linearly seperable. However, the quality of its solution for the classification task is highly dependent on the choice of C [32].

In contrast, our algorithm 3 allows us to bypass this issue because we can directly apply it to (SVM). This problem fits our template problem (P) by taking $F_\xi(x) := \frac{1}{2}\|x\|^2$, $r \equiv 0$, and

$$g(x, (a, y)) := -y\langle a, x\rangle + 1$$

for $(a, y) \in D$.

To test algorithm 3 on real world data, we use the Wisconsin Diagnostic Breast Cancer dataset [33] with $n = 30$ features and $N = 569$ samples. Features are standardized, and we use a stratified 70/30 train/test split.[1] To avoid perfect separability, we flip labels independently with probability 0.1. As the penalty function sequence $(h_k)_{k\in\mathbb{N}_0}$, we use the Huber-type penalty from example 3.1. The objective is 1-strongly convex, so we use the parameter schedule suggested by corollary 3.19:

$$\eta_0 = 2, \qquad \eta_k = \frac{2}{k}, \qquad \gamma_k = \ln(k+1), \qquad \delta_0 = 1, \qquad \delta_k = \frac{1}{k}, \qquad w_k = \eta_k^{-1},$$

for $k \in \mathbb{N}$.

Since there does not exist a solution to this problem, we will track the misclassification rates[2] of the classifiers induced by the sequence of iterates $(\bar{x}_k)_{n\in\mathbb{N}}$, generated by algorithm 3, evaluated on the training and test sets, respectively. We repeat the experiment over 10 independent runs and report the mean and standard de-

[1] For background information on machine learning evaluation methods, see the book [34].

[2] The misclassification rate is defined as the percentage of falsely classified examples in a given set.

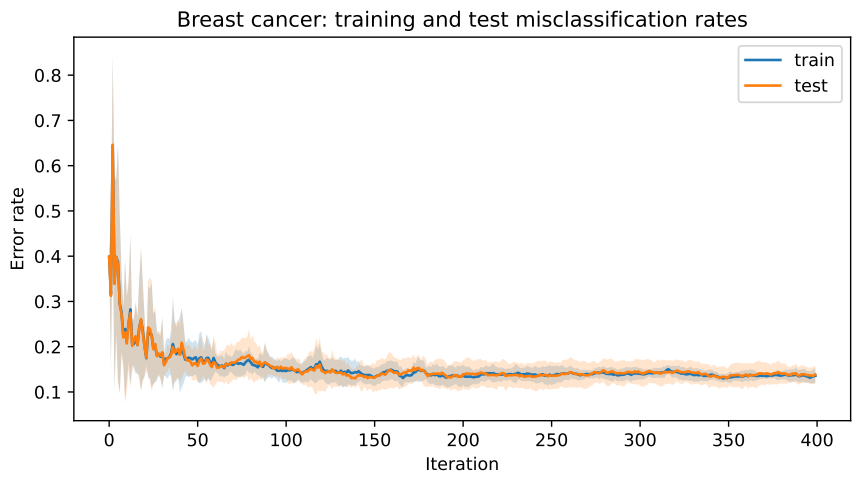

Figure 4.1: Wisconsin Diagnostic Breast Cancer dataset: training and test misclassification rates over iterations. The shaded regions show ±1 standard deviations.

viation of the train and test misclassification rates over the sequence of averaged iterates $(\bar{x}_k)_{k\in\mathbb{N}}$. Figure 4.1 shows the average training and test error across iterations with ±1 standard deviation bands. After 400 iterations, the final training error is 0.1357 ± 0.0158 and the final test error is 0.1386 ± 0.0192.

4.2. Inventory control

This example is adapted from Section 4.8.2 in [35]. We consider a discrete-time inventory problem over a finite horizon $\{0, \ldots, n-1\}$, where $n \in \mathbb{N}$. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $\zeta \colon \Omega \to \mathbb{R}^p$ be a random variable. For each period $k \in \{0, \ldots, n-1\}$, we introduce the following quantities:

- $y_k(\zeta)$: inventory level at the beginning of period k, subject to uncertainty,
- x_k: order quantity placed in period k,
- $d_k(\zeta)$: customer demand in period k, subject to uncertainty,
- γ: per-unit ordering cost,
- β: per-unit holding cost.

The inventory evolves according to

$$y_{k+1}(\zeta) = y_k(\zeta) + x_k - d_k(\zeta) \tag{4.1}$$

for $k \in \{0, \dots, n-1\}$, with a prescribed deterministic initial inventory $y_0 \in (0, \infty)$. We seek to minimize the expected total ordering and holding cost subject to almost-sure nonnegativity of the inventory at every period. Adding a quadratic regulariser $\lambda \|x\|^2$ with $\lambda \in (0, \infty)$ for stability, and imposing a finite ordering capacity $x_{\max} \in (0, \infty)$, the problem reads

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \; & \mathbb{E}\Big(\sum_{k=0}^{n-1} \gamma x_k + \beta \sum_{k=1}^{n} y_k(\zeta) \Big) + \lambda \|x\|^2 \\ \text{s.t.} \quad & y_{k+1}(\zeta) = y_k(\zeta) + x_k - d_k(\zeta) \quad \text{a.s. for all } k \in \{0, \dots, n-1\} \\ & y_k(\zeta) \geq 0 \quad \text{a.s. for all } k \in \{1, \dots, n\} \\ & 0 \leq x_k \leq x_{\max} \quad \text{for all } k \in \{0, \dots, n-1\}. \end{aligned} \tag{4.2}$$

We can eliminate the state variables $(y_k)_{k \in \{0, \dots, n-1\}}$ by recursively solving the dynamics (4.1). This yields

$$y_k(\zeta) = y_0 + \sum_{j=0}^{k-1} \big(x_j - d_j(\zeta) \big),$$

for all $k \in \{0, \dots, n-1\}$. The nonnegativity constraints thus become linear inequalities in x. To write them compactly, let $A = (a_{ij})_{i,j \in \{1, \dots, n\}} \in \mathbb{R}^{n \times n}$ be the lower-triangular matrix with

$$a_{ij} := \begin{cases} 1, & \text{if } i \geq j, \\ 0, & \text{else.} \end{cases}$$

We further define

$$d(\zeta) := \big(d_0(\zeta), \dots, d_{n-1}(\zeta) \big)^\top \in \mathbb{R}^n, \qquad \mathbf{1} := (1, \dots, 1)^\top \in \mathbb{R}^n.$$

Setting

$$v := \mathbf{1}^\top (\beta A + \gamma I_n), \qquad \eta(\zeta) := \beta \, \mathbf{1}^\top A \, d(\zeta),$$

and

$$r(x) := \begin{cases} 0, & \text{if } x \in [0, x_{\max}]^n, \\ \infty, & \text{else,} \end{cases}$$

problem (4.2) fits our template (P) by taking the proximal term r as above,

$$F(x,(i,\zeta)) := \langle v, x\rangle - \eta(\zeta) + \lambda\|x\|^2, \quad \text{and}$$
$$g(x,(i,\zeta)) := -A_i x + A_i d(\zeta) - y_0,$$

where we let $A_i \in \mathbb{R}^{1\times n}$ denote the i-th row of A for $i \in \{0,\dots,39\}$. For data generation, we use a horizon of $n = 40$ periods and generate 2000 demand scenarios. That is, we generate 2000 trajectories $\{d_0(\zeta),\dots,d_{39}(\zeta)\}$. The samples are generated independently, for each period $k \in \{0,\dots,39\}$, according to a lognormal distribution

$$d_k \sim \text{Lognormal}\big(\log(m_k),\, 0.2^2\big), \qquad m_k := 1.5 + 0.30\sin\Big(\frac{2\pi k}{12}\Big).$$

The demand follows a seasonal pattern with noise. For interpretability of the parameter λ, we rescale all inventory quantities by the empirical mean demand

$$s := \frac{1}{2000\cdot 40}\sum_{i=1}^{2000}\sum_{k=0}^{39} d_k^{(i)},$$

where $d_k^{(i)}$ denotes the i-th demand scenario in period $k \in \{0,\dots,39\}$ for $i \in \{1,\dots,2000\}$. The parameters are chosen as

$$\gamma = 0.6, \qquad \beta = 0.35, \qquad \lambda = 0.5, \qquad y_0 = \frac{8}{s}, \qquad x_{\max} = \frac{4}{s}.$$

Since the empirical objective is 2λ-strongly convex, we apply algorithm 3 with the strongly convex schedule from corollary 3.19:

$$\eta_0 = \frac{1}{\lambda}, \quad \eta_k = \frac{1}{\lambda k}, \quad \gamma_k = c_\gamma \ln(k+2), \quad \delta_k = \frac{c_\delta}{k+1}, \quad w_k = \eta_k^{-1},$$

where $k \in \mathbb{N}_0, c_\gamma, c_\delta \in (0,\infty)$, and the penalty function sequence $(h_k)_{k\in\mathbb{N}_0}$ is again the Huber-type penalty sequence from example 3.1. The parameters c_γ and c_δ are tuned by minimizing the relative distance to the true optimum after 1000 iterations[3] over all combinations of

$$c_\gamma \in \{160, 170, 180, 190, 200\}$$

[3]Of course, the distance to the ground-truth minimizer is not known in practice. It is not clear how to tune the parameters in the general case of problem (P). If $r \equiv 0$, however, one could instead consider the norm of the gradient of the penalized objective, $\|\nabla f(\bar{x}_k) + \gamma_k \nabla\pi(\bar{x}_k)\|$, as a scoring criterion. If $\bar{x}_k$ is close to $x^\star$, this value is close to zero. What constitutes a good scoring criterion in the general case remains an open question.

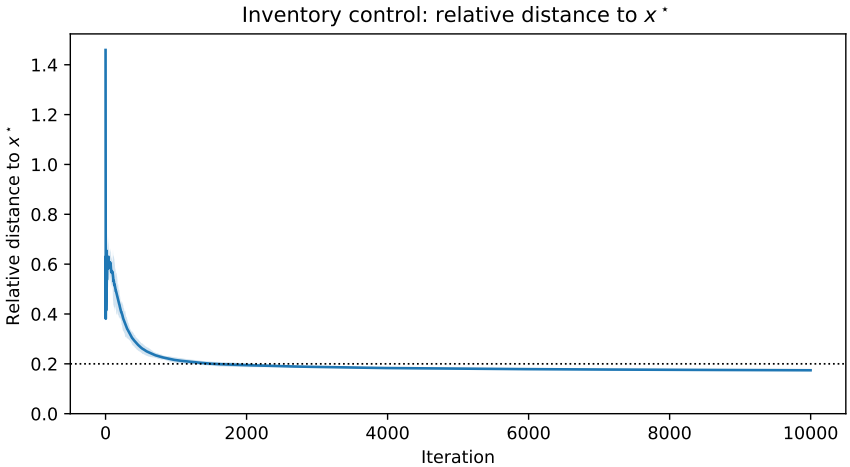

Figure 4.2: Inventory control: relative distance $\|\bar{x}_k - x^\star\|/\|x^\star\|$ over iterations. The shaded region shows ±3 standard deviations, and the dotted horizontal line indicates the level 0.2.

and

$$c_\delta \in \{\, 0.05, 0.075, 0.1, 0.125, 0.15 \,\}.$$

The tuned pair is $(c_\gamma, c_\delta) = (180, 0.125)$, which is then used to rerun the algorithm for 10000 iterations.

To test the performance of our algorithm, we determine the ground-truth solution with CVX [36]. Note that strong convexity of the objective, along with the fact that we sample a finite amount of scenarios for the affine constraints, guarantee the existence of a unique minimizer $x^\star$ (see proposition 2.55). We repeat the experiment over 10 independent runs for 10000 iterations each on the full dataset, and report the mean and standard deviation of the averaged iterates $(\bar{x}_k)_{k\in\mathbb{N}}$. Figure 4.2 shows the relative distance $\|\bar{x}_k - x^\star\|/\|x^\star\|$, where $x^\star$ denotes the unique CVX minimizer, together with ±3 standard deviation bands and a dotted reference line at level 0.2. After 10000 iterations, the final relative distance is $1.74 \times 10^{-1} \pm 1.2 \times 10^{-3}$.

5

Summary and Outlook

We developed a method for stochastic optimization problems subject to almost sure constraints:

$$\min_{x \in \mathbb{R}^n} \{ f(x) := \mathbb{E}(F_\xi(x)) + r(x) \}$$
$$\text{subject to} \quad g(x, \xi) \leq 0 \quad \text{almost surely.}$$

Our method provably handles the case of infinitely many nonlinear constraints, which is more general than the settings which have been considered in related works [7, 11, 15, 18]. Further, because our algorithm uses penalty functions to softly enforce constraints, it is applicable even if the constraints are not fully satisfiable. Empirically, we found that the iterates still converge to a reasonable solution in that case (see section 4.1).

We analyzed the convergence properties of the method in the convex and strongly convex regimes. In the convex setting, one can achieve a convergence rate of $\mathcal{O}(\log^{1/2+\epsilon} K/\sqrt{K})$ for any $\epsilon \in (0, \infty)$. The convergence rate can be improved to $\mathcal{O}(\log^{\epsilon} K/K)$ for any $\epsilon \in (0, \infty)$, in the strongly convex case. It is known that stochastic projected gradient descent achieves the minimax optimal convergence rates of $\mathcal{O}(1/\sqrt{K})$ in the convex setting, and $\mathcal{O}(1/K)$ in the strongly convex setting [37, 38]. Thus, the convergence rates of our method match the optimal convergence rates up to an arbitrarily small logarithmic factor, while being projection-free. Besides the typical convergence guarantees which hold in expectation, we also provided convergence guarantees that hold with high-probability in the general convex setting. Notably, we did not rely on any of the typical global boundedness assumptions regarding the stochastic gradients of the objective f.

There are a lot of directions for future research. One interesting avenue is to adapt the algorithm to the setting of chance constraints, where constraints only need to hold with some high probability, instead of almost surely. A straight-forward

way to approximate a chance constrained problem by an almost sure one is through the sampling-and-discarding approach [39]. However, this is only an approximation and, furthermore, requires the ability to sample a possibly very large amount of times before the optimization begins. This is not always feasible, for example, in online optimization settings, where on-demand sampling is not possible [40]. On a related note, the current analysis of our algorithm does not hold in the general online optimization setting, as we require an i. i. d. assumption on the sampling process, which is unrealistic in the online context. An extension to online settings with a practical stopping rule for handling chance-constraints could greatly enhance the applicability of our method.

Another possible avenue for future research is the use of methods for accelerating convergence. To speed up the convergence of first-order methods, a popular method in deterministic optimization is Nesterov's accelerated gradient method (AGD) [41, 42]. In stochastic optimization, however, a naive application of AGD does typically not lead to faster (asymptotic) convergence rates, due to an increase in variance in the gradient estimates. Instead, acceleration is achieved via variance reduction methods, such as SAGA [43] and SVRG [19]. These algorithms achieve acceleration, while keeping variance managable by periodically calculating full gradients of the objective. Unfortunately, a straight-forward application of these methods to our setting is not possible, unless we assume a finite-sum structure of the objective and a finite number of constraints. Moreover, the number of summands involved in the objective, as well as the number of constraints, could be extremely large. This would make even an occasional calculation of full gradients infeasible for us. A promising recent method that circumvents these issues was proposed in [44], where the authors manage to greatly improve the convergence of stochastic gradient methods, while avoiding the calculation of full gradients. Incorporating their techniques into our algorithm is a promising future research direction to improve upon the convergence rate of our current method.

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